

Quadratic Orthant Certificates

Maximal parallelism, timed irreversible races
and a fixed Waterfall universal machine

Degree-two Diophantine certificates, nonnegative on the real orthant, with negative and selection conditions

AI-assisted research report prepared for Vladimir Reshetnikov from three manuscripts
(batch 78 of ProveIt's incoming reports, manuscripts 08, 12 and 14), October 2, 2026

Abstract

Each Part compiles a fixed finite system into one integer polynomial of total degree two that is nonnegative on the whole real nonnegative orthant and whose natural zeros are in bijection with the intended semantic objects, including a condition that ordinary resource-balance or trace equations cannot express. Part I (source 08) certifies one *maximal* parallel round of a multiset-rewriting system with $d + 2m + 4H$ auxiliaries, proves that the resource-cover rank $\rho(A)$ gives the optimal one-round counting exponent $m - \rho(A)$, shows that one cooperative deadlock already rules out convex quadratic certificates, and proves that numerical stoichiometry as an input needs degree four. Part II (source 12) gives a *total* quadratic semantics for irreversible exclusive-site systems with positive integer delays: one polynomial per catalogue has exactly one natural zero for every delay vector, recording the earliest-completion outcome and certified permanent absence, with $m(2q + 3) + 3D + 3(L + R_0) + R$ witnesses. Part III (source 14) proves the exact macrostep behaviour of a fixed 46-clock Waterfall program, due to Iijil, that simulates the Neary–Woods universal machine $U_{15,2}$, and compiles k source steps into one quadratic with $35k$ natural witnesses whose fibre is empty or a singleton.

All three constructions sit at the level $D_2^+ = \text{SL}$ of the positivity-restricted degree classification proved in Part XI of the collection's report *Canonical Diophantine certificates*: they are fixed-horizon or fixed-catalogue certificates, not fixed-arity universal equations, and every Part says so. The report is AI-assisted and unrefereed; nothing in it is formalized in Lean or Rocq, and priority is not certified for any Part.

Status. Proofs and executable finite checks; not kernel-checked. Section 1 states the common spine, the conventions and what is not claimed; Appendix H gives the provenance.

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1 Scope, common spine and conventions

[write] This section was written when the three manuscripts were merged. Parts I, II and III print the three sources; text there without a [write] marker is the source’s own.

1.1 The three Parts

This report is built from three AI-assisted research manuscripts delivered as batch 78 of ProveIt’s incoming reports (manuscripts 08, 12 and 14; Appendix H gives the full provenance). They are called *source 08*, *source 12* and *source 14* after their batch-78 manuscript numbers, which are also the file prefixes of their shipped programs and data. All three are dated 2 October 2026. None cites another. Sources 08 and 12 consulted the repository at the same commit, `db3b377f0`; source 14 names no repository commit. They are not versions of one manuscript and share no theorem: their common ground is one *format* of certificate, described in Section 1.2.

- **Part I (source 08), *Quadratic certificates for maximal parallelism*.** A fixed multiset-rewriting system fires, in one round, a *maximal* multiset of rule copies: every unused rule must lack a resource. Shared threshold comparators encode this negative condition at degree two with $d + 2m + 4H$ auxiliaries (Theorem 5.1); histories and exact first halting follow (Theorem 6.1, Corollary 6.2). The resource-cover rank $\rho(A)$ gives the optimal uniform one-round counting exponent (Theorems 8.2 and 8.3); fixed-horizon counts are quasipolynomial (Theorem 9.1); root counting is $\#P$ -complete (Corollary 10.2); the deadlock of $A + B \rightarrow C$ has no convex quadratic certificate (Theorem 11.2); and with stoichiometry as an input, degree four is sufficient and necessary (Theorem 12.3).
- **Part II (source 12), *Total quadratic semantics for irreversible computation*.** Sites acquire at most one of several competing labels; a rule completes a positive integer delay after its typed prerequisites appear; the earliest completion wins, ties by label. The capped causal equations have exactly one solution (Theorem 19.2), and one quadratic per catalogue has exactly one natural zero for every delay vector, recording the complete outcome including permanent absence (Theorem 21.2). Applications: Horn closure (Theorem 22.1), fixed-candidate tile assembly (Corollary 22.3), seeded self-assembly in a finite box, timing regions (Theorem 24.1), and the permanent-absence obstruction (Theorem 25.3).
- **Part III (source 14), *Quadratic Diophantine certificates for a fixed Waterfall universal machine*.** Iijil’s fixed 46-clock Waterfall matrix simulates the Neary–Woods machine $U_{15,2}$ with both half tapes as raw integers. Every source step is one exactly described macrostep (Theorem 32.1), proved for all tape values by a finite affine certificate; k steps compile into one quadratic with $35k$ natural witnesses and an empty or singleton fibre (Theorem 33.1). A decidable common-column class has horizon-free certificates (Theorem 36.1).

Each Part keeps its own definitions, letters and numbering of statements. The letters that differ between Parts, and the words that also occur elsewhere in the collection with another meaning, are listed in Section 1.3.

1.2 The common spine and its place in the degree classification

In all three Parts the main certificate has the form

$$P = \sum_i A_i^2 + \sum_j B_j C_j, \quad (1)$$

where every A_i is an integer affine form and every factor B_j, C_j is an affine form with nonnegative coefficients and nonnegative constant term in the natural witnesses. Hence $P \geq 0$ on the whole real nonnegative orthant, P has total degree at most two, and $P = 0$ on that domain exactly when every summand vanishes. Part II states this form explicitly (equation (41)); in Part I it is the shape of the comparator (3) and of the round polynomial (9); in Part III it is the shape of F_k in (78) and of the horizon-free certificate (84). The deliberate exceptions are of degree four: Part I’s program-uniform quartic (Theorem 12.3) and MRDP quartic (Section 13), Part II’s four-square conversion to integer witnesses, and Part III’s common-column certificate with its coefficients promoted to inputs. The unsquared products are what lets each Part express, at degree two and without a population or time cutoff, a condition that equations alone cannot: *maximality* (every unused rule lacks a resource) in Part I; *earliest completion with label priority*, and certified *absence*, in Part II; and the inactive direction group and exact first halt of a *strict-minimum* event process, compressed into macrosteps, in Part III. In each Part, once the semantic object is fixed (a firing multiset or extent history, a delay vector, a halting run), every auxiliary coordinate is forced.

Write $D_n^+(k)$ for the subsets of $\mathbb{N}^k$ that are projections of natural zero sets of integer polynomials of degree at most n , nonnegative on the real nonnegative orthant, and $\text{SL}(k)$, $\text{CE}(k)$ for the semilinear and the computably enumerable subsets. Part XI of the collection’s report *Canonical Diophantine certificates* proves the positivity-restricted degree classification

$$D_2^+(k) = D_3^+(k) = \text{SL}(k), \quad D_4^+(k) = \text{CE}(k),$$

(its Theorem `cdc:of:thm:classification`), and that every semilinear set has a single-fold degree-two representation in this class (`cdc:of:thm:singlefoldsl`). Every degree-two certificate of this report is of the form (1), so every relation it projects to is *semilinear*. This is consistent with each Part, and each Part says so in its own terms:

- Part I’s certificates are for a fixed system and a fixed horizon T ; their projected relations are Presburger (Section 9). Where a non-semilinear relation appears — multiplication $z = kn$ when stoichiometry is an input — Part I moves to degree four and proves that four is necessary (Theorem 12.3).
- Part II’s polynomial is quadratic jointly in the delay parameters and the witnesses; its projection to the delays is all of $\mathbb{N}^R$. The delay regions giving a prescribed outcome are semilinear (Theorem 24.1). The unbounded direction is a family of polynomials indexed by a grounding size b , with “ $\exists b$ ” outside every polynomial (equation (54)).
- Part III’s horizon k is compiler data, not a variable; for each k the parameter set is semilinear (Proposition 35.1).

The tempting false reading. “A degree-two certificate for a fixed *universal* machine (Part III), or for a Turing-complete Horn program (Part II), gives a universal degree-two Diophantine equation.” This is false. The polynomial changes with the horizon k or the grounding size b ; a single orthant-nonnegative polynomial of degree at most three cannot have an undecidable projection (`cdc:of:cor:no-cubic-universal`, re-proved as Proposition 13.1). Without the positivity restriction, MRDP gives universal equations of degree four; that is a different class.

1.3 Notions and letters

No symbol of any source is renamed, and no normalization is changed. Each letter keeps its source’s meaning inside its Part. Table 1 lists words that carry different meanings in different Parts or elsewhere in the collection; Table 2 lists the letters used differently by the three Parts.

Table 1: Notions. “CDC” is the collection’s report *Canonical Diophantine certificates*; “LBH” is *Liveness beyond halting*.

Word	Meaning in this report	Other meanings it must not be confused with
maximal	Part I: a feasible extent vector with no feasible one-copy extension (Definition 3.1); <i>not</i> maximum cardinality. Part II: a maximal finite-box assembly, one admitting no further legal attachment in the box (Corollary 23.3).	CDC Part II’s maximal commutation relation; CDC Part XII’s maximal number of complete repetitions.
flat	Part I: flat maximality, each rule used at most once per round (Proposition 7.2).	CDC Part III’s flat schemes (cdc:cs:thm:flat) and CDC Part XII’s flat schemas of fixed words (cdc:pf:cor:flat).
threshold	Part I: a consumption coefficient a_{ij} read by a comparator; Part II: the tile temperature τ and irreversible threshold growth.	CDC Part XI’s temperature, Part XII’s pumping threshold, Part XIII’s reservoir threshold.
flag	Part I: the threshold flag u_h of a comparator; Part II: the three-witness presence flag b (Section 25).	CDC Part II’s exact flag recurrence (cdc:prop:flags).
rank	Part I: $\rho(A)$, the least rational row rank of a resource cover (Definition 8.1). In the proof of Part II’s Corollary 22.3, “finite rank” means a finite activation time.	CDC Part XI’s first-arrival ranks (cdc:of:thm:ranks); CDC Part XIV’s rank function.
history-free	Part II (source 12’s subtitle, “without execution histories”): the certificate records the outcome, not an execution order.	The <i>same</i> sense as CDC Part XIV (“history-free routing certificates”).
single-fold	Part II: at most one complete natural witness per parameter tuple, for a <i>fixed</i> catalogue; Part III: an empty or singleton fibre for a <i>fixed</i> horizon k .	The single-fold (finite-fold) representation problem for all c.e. sets, with one fixed-arity polynomial, which no Part addresses.
phase	Part II: a region of the delay orthant on which the event signature is fixed (Theorem 24.1; “critical-chain phase”, “label phase”).	A commuting factor of a subgroup product in the report <i>Groups as Diophantine substrates</i> ; CDC Part XIII’s four-phase zero test.

Word	Meaning in this report	Other meanings it must not be confused with
clock	Part III: a Waterfall counter, one coordinate of the deadline vector.	LBH's computable clocks and clock-faithful simulations (time functions).
canonical	Part III: the canonical deadline vector $\kappa(q, s, L, R)$ and canonical cut of Section 32.	CDC's canonical representative of a trace class.
guard	Part I: a pre-state promoter or inhibitor bit e_j (Proposition 7.1).	CDC Part II's interval guards (related, not identical).

Table 2: Letters with different meanings in the three Parts.

Letter	Part I (source 08)	Part II (source 12)	Part III (source 14)
A, B	consumption and production matrices	A, B, C : affine residuals and factors in (41); A, C also the counts of squares and products in Theorem 21.2; $B = q + 1$ the key base, and B a finite box	A_j, B_j : current and next state expressions; $B(p)$ the common shift; $A, \dots, O$ the machine states
C	C_a, C_h comparators; $C_T(n)$ trace counts; $\mathcal{C}(A)$ the resource covers	number of product summands; C_v a set of rules	number of nonhalting Waterfall firings
D	retention diagonal $\text{diag}(\delta_i)$	number of distinct nonseed prerequisite literals	direction L or R
H	number $ \mathcal{H}(A) $ of species-threshold pairs; a Hessian in Lemma 11.1; an undecidable set in Proposition 13.1	—	H_j the one-hot residual; H_{file} a serialization bound; H an input of the common-column class
L, R	L a signed affine form, and a constant in the proof of Theorem 9.1	L the number of prerequisite incidences; R, R_0 the numbers of rules and of empty-body rules	the left and right half tapes, and the two directions
M	largest consumption coefficient	the sentinel $M = S + 1$ standing for ∞	the Waterfall increment matrix
Q, q	$Q_{A,B}$ the round polynomial; q a period, a rank, and $q_{\pm}$ signed quotients	q the number of labels; $Q_{\min}, Q_{\max}$ gates; Q a hypothetical polynomial in Theorem 25.3	Q a binary quotient; q a machine state
T	the horizon	T_e a rule body	$T_1, \dots, T_4$ terminal residuals; T the halt time of the common-column class
W	—	the number of witnesses	W_{jD} the written bit; $W = H + 1$

Letter	Part I (source 08)	Part II (source 12)	Part III (source 14)
d, k, m	numbers of species (d) and rules (m); $k_j = S_j $, and k a stoichiometric coefficient	d_e a delay; m the number of nonseed sites; k a fold size	d the deadline vector; k the horizon; m an exponent
ρ, τ	$\rho(A)$ the cover rank; ρ a remainder in (7)	τ the tile temperature	τ the halt timestamp

1.4 Shared devices, printed in each Part

The three sources use a few small degree-two devices that are also proved in the collection. Their statements differ in the inputs they accept and in the number of witnesses, so each Part keeps its own statement and proof; this subsection only connects them.

- **Minimum and conjunction.** Part I's Boolean gate (6) is Part II's minimum gate (42) (Lemma 20.1) applied to two bits, with a complement bit added; Part II also has the mirror maximum gate (43). Boolean conjunction with unique affine slacks is `cdc:lem:and` and `cdc:of:lem:and` in CDC.
- **Threshold and presence tests.** Part I's canonical comparator (Lemma 4.1, four witnesses, fixed threshold, no bound on the tested quantity) and Part II's three-witness presence flag (Section 25; one slack fewer because $0 \leq t \leq M$ is known) are the same device. CDC Part XI has the unbounded comparison penalty `cdc:of:lem:unboundedcompare` (five witnesses, two free inputs) and the bounded comparator `cdc:of:lem:compare`. Part II's typed-prerequisite selector (Lemma 20.2) and Part III's direction-group products (74) use the same complementary-slack idea.
- **Euclidean remainder.** Part I's congruence test (7) is CDC's penalty E_a (`cdc:of:eq:euclidean`) up to renaming; Part III's horizon-free certificate (84) uses a ceiling-division variant.
- **Unsquarred products.** Sources 08 and 12, and CDC after `cdc:of:lem:unboundedcompare`, observe that the complementarity products must not be squared: squaring would raise the degree to four. Source 14 likewise leaves its products unsquared.

1.5 Results of the collection re-proved or generalized here

Sources 08 and 12 read only the README of *Canonical Diophantine certificates*, not its article, and source 14 did not consult the repository. Several of their statements are therefore second proofs or generalizations of results already in that report. They are printed as their sources wrote them, marked at the place by a [write] note; no novelty is claimed for them, and sources 08 and 12 themselves disclaim novelty for the inherited facts.

This report	Canonical Diophantine certificates	Relation
Lemma 12.1 (natural zeros of positive cubics are semilinear)	cdc:of:lem:affinezero, cdc:of:prop:faces, cdc:of:thm:cubicsemilinear: the inclusion $D_3^+ \subseteq \text{SL}$ of cdc:of:thm:classification	second proof, same Hessian-on-faces idea
Corollary 12.2 (multiplication needs degree four)	cdc:of:cor:squares	same content and argument
Proposition 13.1	cdc:of:cor:no-cubic-universal	same statement
The MRDP-to-quartic paragraph of Section 13	cdc:of:prop:quartic	same circuit argument
Equation (7) and the Presburger guards after it	cdc:of:eq:euclidean and cdc:of:thm:singlefoldsl	identical up to renaming; credited by source 08
Theorem 19.2 (unique capped causal fixed point)	cdc:of:thm:ranks (two-cut characterization of first-arrival ranks)	generalization: positive delays, exclusive labels, absence
Lemma 23.2 (compatible union)	the union-closure step of cdc:of:prop:antimatroid	same argument
The batch serialization of Section 23	cdc:of:lem:closure	same argument
Theorem 25.3, second half	cdc:bd:prop:nobound, cdc:of:prop:nocutoff	a case
Theorem 24.1, Presburger conclusion for integer delays	cdc:of:thm:classification applied to Part II’s own polynomial with a label query	second route
Proposition 35.1	cdc:of:thm:classification	second route

Relative to the collection (not to the literature, which no source searched exhaustively), the other main results are new here: in Part I the shared-threshold round quadratic, the resource-cover exponent and its sharpness, the trace-degree bound, the counting reduction, the convexity obstruction and the program-uniform quartic; in Part II the total compiler with its exact allocation, the Horn and fixed-candidate tile allocations, the locking counterexample, the timing decomposition and its exponential lower bound, and the nonmonotone timing family; in Part III everything except Proposition 35.1.

1.6 Relation to neighbouring reports and to the research programme

- **Canonical Diophantine certificates.** Besides Part XI (above): its Part XIII expressly does not impose a maximal-parallel interpretation on its reaction networks and cites the same universal maximal-parallel system as source 08 (Alhazov–Verlan); Part I of this report supplies certificates for that other convention. Part XIV’s “history-free” is Part II’s sense. Part III bears on CDC’s questions cdc:q:acceleration and cdc:q:restricted ([write] note in Section 35). Parts I and II add data points, not answers, to cdc:q:degree: a program-uniform quartic proved necessary, and a parameter-uniform quadratic with the delays as free inputs.
- **Liveness beyond halting.** Part I’s guarded register-machine embedding bears on that

report’s question on other unconventional substrates with timing certificates ([write] note in Section 7).

- **Signal-machine collision certificates** (batch 78, cluster H2) uses the same family of degree-two certificates (sums of squares of affine residuals in natural witnesses) for rational signal machines. No theorem is shared.
- **Groups as Diophantine substrates** and **Probabilistic, quantum and continuous computation**: no overlap.
- **The Hilbert-tenth-problem research programme** (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/`, read only for this report) reviewed all three archives before this write. Its reviews confirm the mathematics of sources 08 and 12 and record implementation input-contract defects with patches (`incoming_substrate_review_808b53ed8.md`, `incoming_parallel_order_review_808b.md`, `incoming_convergence_review_808b.md`; commit 126028588). Its review of source 14 (`waterfall_report_review_24a743255.md`, commit f19aaa092) confirms the theorems, records two Python interface defects with a patch, and projects the seven-step polynomial from 245 to 107 witnesses (`waterfall_forced_boundary_projection.md`, commit 3b8187ea9); see the dated notes in Part III. Its note `neary_woods_explicit_universal_tm.md` (1 October 2026) transcribes the same machine table as source 14 and found the same erratum. The shipped programs are the delivered bytes, unpatched.
- **The formal project.** The report sits in the collection, not in `Computability/HilbertTenthProblem`, and placement beside a Lean development confers no formal status. Sources 08 and 12 cite MRDP only as a classical theorem; the project’s formal endpoint is `Diophantine.mrdp`, `Diophantine.mrdp_iff` and `Diophantine.mrdp_dioph_iff` in `Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean`, whose blob is unchanged from source 08’s pin to this write. None of this report’s statements is formalized.

1.7 What is claimed, and what is not

Each Part claims conventional mathematical proofs of the results listed in Section 1.1, with exact-integer finite checks of its implementation and examples. The report does *not* claim, for any Part:

- historical priority (all three sources say their literature search was targeted, not exhaustive), peer review, or proof-assistant verification;
- a fixed-arity universal Diophantine equation, a resolution of the single-fold or finite-fold representation problem, or an improvement of the repository’s universal arithmetic-operation bounds (75 and 87 operations at the sources’ pins); source 14 says so explicitly for its fixed universal machine;
- that finite tests prove the all-input theorems: the executable checks are regression checks of the implementations;
- minimality of any witness count or of the published universal systems used as context (the 23-rule maximal-parallel system, the Neary–Woods machine, Iijil’s matrix);
- for Part I, efficient computation of $\rho(A)$ or of root counts, and convex certificates in general; for Part II, correctness for zero delays, negative glues, inhibition, detachment or stochastic

kinetics, or an enumeration of all nondeterministic outcomes; for Part III, an optimality claim for $35k$ or for the four-operation loader, or any cost for encoding ordinary programs as half tapes.

Part I

Quadratic certificates for maximal parallelism

[write] Part I prints source 08 (batch-78 manuscript 08). Its Section n is Section $n+1$ here (Sections 2–16), and its Theorem $n.m$ is Theorem $(n+1).m$; its appendices A–C are Appendices A–C. Its title page follows, without the page layout.

Quadratic Certificates for Maximal Parallelism

Exact witnesses, sharp growth bounds, and a convexity obstruction

Research manuscript prepared for Vladimir Reshetnikov. Developed with ChatGPT. 2 October 2026.

For a fixed multiset-rewriting system, a single polynomial of degree two certifies a maximal parallel round with exactly one auxiliary assignment per firing multiset. Its size is independent of the population magnitude. A resource-cover rank gives the optimal uniform one-round counting exponent. Cooperative deadlock already obstructs convex quadratic certificates. Allowing stoichiometric coefficients to vary raises the sharp orthant-nonnegative degree threshold from two to four.

Status. This manuscript supplies conventional proofs and an executable exact-integer compiler with finite validation. It is not a peer-reviewed publication or a proof-assistant-checked formalization. The constructions and quantitative deductions are developed here; global priority is not asserted. The classical universality, Presburger counting, and MRDP ingredients, and the repository’s existing low-degree positivity result, are explicitly credited.

Package: this article in `.tex` and `.pdf`, a standard-library Python implementation, an exported example polynomial, validation results, a source audit, and reproduction instructions.

[write] Of the package named above, this report ships the Python implementation, the exported example, the validation results and the source audit; the reproduction instructions are in the report’s README. Source 08’s own `.tex` is this Part, and its PDF is not shipped.

Abstract

A maximal parallel multiset-rewriting step is not merely a solution of resource-balance equations: its unused reactants must disable every rule. We encode this negative condition by shared, uniquely determined threshold flags. For fixed consumption and production matrices with d species, m rules, and H distinct positive species–threshold pairs, we construct an integer polynomial nonnegative on the entire real nonnegative orthant, of total degree at most two, using $d + 2m + 4H$ natural auxiliaries when both endpoint states are supplied. Its natural zeros are in bijection with the legal firing multisets. Fixed-horizon histories, exact first halting, pre-state guards, flat maximality, and selective retention admit compatible extensions.

For the consumption matrix A , define $\rho(A)$ to be the least rational row rank of a set of species meeting every reactant support. We prove that the number of maximal firing multisets at population x is $O((1 + \|x\|_1)^{m - \rho(A)})$, and that for every A the exponent is attained along a suitable integral input ray. At fixed horizon T , counts along affine input rays are eventually quasipolynomial, of degree at most $T(m - \rho(A))$; the bound is sharp across systems. We also

obtain a parsimonious graph-to-network counting reduction. Finally, no convex quadratic with finitely many natural auxiliaries can represent the deadlock set of $A + B \rightarrow C$, although the quadratic xy does. A program-uniform quartic construction and a matching lower bound clarify why fixed-system quadratics do not solve fixed-arity unbounded halting or the finite-fold problem.

2 The questions answered and their provenance

2.1 Why maximal parallelism is a substantive test case

Multiset rewriting uses vectors of object counts rather than an ordered tape. The same objects can be required by several rules; one round chooses how many copies of each rule fire using only the objects present at the start of that round. A maximal choice leaves no room for an additional rule copy. The distinction between this semantics and arbitrary feasible parallel firing is decisive.

Maximally parallel multiset rewriting is an established universal computational substrate. Alhazov and Verlan construct a universal system with 23 rules [1]. We use that published universality result as context, not as a new contribution and not as a claim that 23 is currently minimal. Flat maximality, in which each rule is used at most once per round, is also an established semantic variant [2].

[write] Two further credits that source 08 does not give. Maximal parallelism as the semantics of multiset rewriting goes back to Păun’s membrane (P) systems [23]; firing a maximal step in a Petri net (the “maximum firing strategy”) already makes Petri nets Turing-complete, by Burkhard [24]. The MRDP theorem used in Section 13 is due to Matiyasevich [25], completing Davis, Putnam and Robinson [26]; see Davis [27]. Source 08 credits MRDP only through the repository’s Lean statement [6].

The present task is more specific than proving that a halting set is Diophantine. We ask whether maximality can be encoded *explicitly*, at low degree, without a population cutoff and without multiplying the number of witnesses. We then ask what the resulting geometry says about counting and about stronger requirements such as convexity.

2.2 The repository baseline

The consulted ProveIt source exposes the finite-polynomial MRDP theorem in `Computability/HilbertTenthProblem/Lean/Diophantine/MRDP.lean` [6]. Its research-report catalogue already includes witness-faithful certificates for Petri traces, priority systems, queues, heaps, self-assembly, and routing. It also records that orthant-nonnegative polynomials of degree at most three have semilinear projected natural zero sets, and that semilinear relations admit single-fold quadratic representations [7].

Consequently, neither abstract Diophantine representability nor the low-degree semilinearity boundary is advertised here as new. Our focus is a direct shared-threshold implementation of maximal parallelism, explicit size and uniqueness proofs, the optimal resource-cover exponent, and the convexity obstruction. The inspected repository sources were not rebuilt in this session. Appendix C records the snapshot and the scope of the inspection.

[write] The results credited here through the README are proved in Part XI of *Canonical Diophantine certificates*: the classification $D_2^+ = D_3^+ = \text{SL}$, $D_4^+ = \text{CE}$ is its Theorem `cdc:of:thm:classification`, and the single-fold quadratic representation of semilinear sets is `cdc:of:thm:singlefoldsl`. Where this Part re-proves one of its results, a [write] note says so (Lemma 12.1, Corollary 12.2, Proposition 13.1, equation (7)); Section 1.5 lists them.

2.3 A precise result ledger

Constructive answer. At fixed system shape and fixed numerical stoichiometry, exact maximal-parallel rounds admit population-uniform, orthant-nonnegative quadratic certificates with $d + 2m + 4H$ auxiliaries.

Quantitative answer. The optimal exponent valid uniformly over all initial populations is exactly $m - \rho(A)$, in the sense of Theorems 8.2 and 8.3.

Geometric answer. Convexity cannot be imposed in general: one cooperative reaction already prevents it, regardless of auxiliary dimension (Theorem 11.2).

Uniformity answer. With numerical stoichiometry supplied as input on a fixed support, degree four suffices and is necessary in the class of polynomials nonnegative on the whole real orthant (Theorem 12.3).

These are answers to the explicit questions formulated in this manuscript. The literature search does not establish that these exact formulations were previously published open problems, or that no equivalent theorem exists elsewhere. The useful mathematical content is the stated constructions, proofs, and sharpness results, rather than an unverified priority label.

2.4 Three quantifier orders that must not be confused

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. A fixed-system, fixed-horizon theorem has logical form

$$\forall(A, B, T) \quad \exists Q_{A,B,T} \quad \forall x \quad \{\text{natural zeros of } Q_{A,B,T}(x, \cdot)\} \simeq \{\text{length-}T \text{ histories from } x\}.$$

Here the number of variables may grow with T . A program-uniform theorem instead places matrix entries among the input coordinates while keeping d, m, T fixed. An unbounded-halting theorem requires one polynomial and one finite witness dimension before the running time is known. None of these quantifier orders may be substituted for another.

A second distinction concerns uniqueness. The certificate is unique *after a firing multiset, or an entire extent history, has been specified*. Distinct histories may reach the same endpoint. We do not claim one root per reachable endpoint, one root per accepting input of a nondeterministic system, or one root per computably enumerable membership instance.

3 Operational semantics

Let

$$A = (a_{ij}), \quad B = (b_{ij}) \in \mathbb{N}^{d \times m}, \quad d, m \geq 1.$$

Column j of A is the reactant multiset of rule j , and column j of B is its product multiset. Every column of A is assumed nonzero. The assumption excludes spontaneous rules and guarantees finite branching at each finite population.

We write $x \in \mathbb{N}^d$ for the current population and $f \in \mathbb{N}^m$ for the *extent vector*: f_j copies of rule j fire. All vector inequalities are coordinatewise. Define the reactant support

$$S_j = \{i : a_{ij} > 0\}, \quad k_j = |S_j|.$$

Definition 3.1 (Feasibility and maximality). The extent f is feasible at x if $Af \leq x$. Its unused reactants are $r = x - Af$. It is maximal if it is feasible and

$$\forall j \in \{1, \dots, m\} \quad \exists i \in S_j \quad r_i < a_{ij}. \quad (2)$$

The next state is $y = r + Bf$.

Lemma 3.2 (Local test for global maximality). *A feasible f is maximal in the coordinatewise order among feasible extent vectors if and only if (2) holds.*

Proof. The vector $f + e_j$ is feasible exactly when $A_{\cdot j} \leq r$. If some larger feasible vector $g \geq f$, $g \neq f$, exists, choose j with $g_j > f_j$. Then $f + e_j \leq g$, so $f + e_j$ is feasible. Thus excluding all one-copy extensions excludes every strict extension. $\square$

Rule labels are retained: two different labels with identical input and output columns still give different extent choices. Individual molecule identities and permutations of identical firings are not counted. An *extent history* of length T is a sequence $(f^0, \dots, f^{T-1})$ with the corresponding states obtained inductively. The state sequence alone need not determine the extents.

3.1 Three short semantic counterexamples

For $A \rightarrow \emptyset$ and $2A \rightarrow \emptyset$ at population $2A$, both $(2, 0)$ and $(0, 1)$ are maximal. Their total numbers of firings differ. Thus maximality is not maximum-cardinality optimization.

For $A \rightarrow B$ and $B \rightarrow C$ at population A , the vector $(1, 1)$ is illegal. It could occur as two sequential steps but not as one round: the B produced in the round cannot be reused immediately. Feasibility must be tested using $Af \leq x$, not a net-change inequality involving $(A - B)f$.

For $A \rightarrow A$ at population $3A$, ordinary maximality requires three copies of the rule, although the next state is unchanged. The unused reactants, not the next state, are the correct place to test maximality.

3.2 Deadlocks and stuttering

A state is deadlocked if $A_{\cdot j} \not\leq x$ for every j . At such a state, the zero extent is the unique maximal extent. Hence the base semantics has one stuttering step at every deadlock. This makes all fixed lengths well-defined. Section 6 separately excludes early stuttering when counting *first* halting times.

4 A unique threshold gadget at degree two

The following gadget is the arithmetic core. It does not use a guessed large constant bounding the population.

Lemma 4.1 (Canonical threshold comparator). *For a fixed integer $a \geq 1$, define*

$$C_a(r; b, c, v, w) = (b + c - 1)^2 + (r + v - (a - 1) - b - w)^2 + bv + cw. \quad (3)$$

This is nonnegative on the entire real nonnegative orthant and has total degree two. For each $r \in \mathbb{N}$ it has exactly one natural zero in (b, c, v, w) , namely

$$(b, c, v, w) = \begin{cases} (0, 1, a - 1 - r, 0), & r < a, \\ (1, 0, 0, r - a), & r \geq a. \end{cases} \quad (4)$$

In particular, b is the truth value of $r \geq a$ and c its complement.

Proof. Every summand in (3) is nonnegative on the real orthant. At a natural zero, $b + c = 1$, so (b, c) is either $(0, 1)$ or $(1, 0)$. If $(b, c) = (0, 1)$, then $cw = 0$ forces $w = 0$ and the remaining equality is $r + v = a - 1$. Thus $r < a$ and $v = a - 1 - r$ uniquely. If $(b, c) = (1, 0)$, then $bv = 0$ forces $v = 0$ and the equality becomes $r = a + w$. Thus $r \geq a$ and $w = r - a$ uniquely. The displayed tuples also give zeros directly. $\square$

Remark 4.2 (Why inactive slacks matter). A disjunction with a slack left unconstrained on the inactive branch usually produces infinitely many auxiliary witnesses for one fact. The products bv and cw eliminate exactly that defect. They are not squared: their nonnegativity on the allowed domain is already enough. Squaring them would unnecessarily raise the degree to four.

4.1 Signed affine tests and Boolean combination

If L is any integer affine form in natural input coordinates, the polynomial

$$(b + c - 1)^2 + (L + v - b - w)^2 + bv + cw \quad (5)$$

has a unique auxiliary zero. The bit b says $L \geq 1$, while c says $L \leq 0$. Although L itself may be negative, it occurs only inside a square, so orthant nonnegativity is preserved.

Two already certified bits a, b can be combined canonically. Introduce $e, p, q, d \in \mathbb{N}$ and use

$$(a - e - p)^2 + (b - e - q)^2 + pq + (e + d - 1)^2. \quad (6)$$

At a natural zero, $e \leq \min(a, b)$ and $pq = 0$. If e were strictly smaller than both bits, both p and q would be positive. Therefore $e = \min(a, b)$, $p = a - e$, $q = b - e$, and $d = 1 - e$, all uniquely. Complementation and De Morgan's law then give unique AND, OR, and NOT circuits of degree two. This is a concrete implementation of elementary Boolean arithmetic, not a new abstract expressibility theorem.

[write] The gate (6) is the minimum gate (42) of Part II (Lemma 20.1) applied to two bits, with the complement bit d added. Lemma 4.1 is the same device as Part II's three-witness presence flag (Section 25); *Canonical Diophantine certificates* has Boolean conjunction with unique affine slacks (cdc:lem:and, cdc:of:lem:and), an unbounded comparison penalty with five witnesses (cdc:of:lem:unboundedcompare) and the remark that its products must not be squared. See Section 1.4.

4.2 Fixed congruence tests

For completeness, a signed affine form L has a canonical Euclidean remainder modulo a fixed $a \geq 1$. Introduce $q_+, q_-, \rho, \sigma \in \mathbb{N}$ and impose

$$(L - a(q_+ - q_-) - \rho)^2 + (\rho + \sigma - (a - 1))^2 + q_+q_- = 0. \quad (7)$$

The signed quotient and its positive/negative split, the remainder $0 \leq \rho < a$, and the slack σ are all unique. Equality of ρ with a selected residue is read by the affine threshold gadgets. Combining these constructions with quantifier elimination for fixed Presburger formulas gives canonical degree-two guard implementations; the general Presburger and semilinear background is inherited [3, 7]. Our executable implementation covers conjunctions of ordinary threshold and absence tests, not a general Presburger quantifier-elimination engine.

[write] Equation (7) is the penalty E_a of *Canonical Diophantine certificates* (cdc:of:eq:euclidean), up to renaming its coordinates (u, v, s) as (q_+, q_-, σ) ; the canonical degree-two implementation of fixed Presburger guards is that report's Theorem cdc:of:thm:singlefoldsl.

5 The shared-threshold compiler

Define the set of distinct positive species–threshold pairs

$$\mathcal{H}(A) = \{(i, a_{ij}) : 1 \leq j \leq m, a_{ij} > 0\}, \quad H = |\mathcal{H}(A)|. \quad (8)$$

If several rules require the same amount of the same species, their comparison flag is shared. For each $h = (i, a) \in \mathcal{H}(A)$ introduce $\eta_h = (u_h, \bar{u}_h, v_h, w_h)$ and set

$$C_h = C_a(r_i; u_h, \bar{u}_h, v_h, w_h).$$

The new notation u avoids confusing these Boolean flags with entries of the production matrix B .

Given endpoint coordinates x, y , introduce $f \in \mathbb{N}^m$, $r \in \mathbb{N}^d$, the comparator variables η , and $s \in \mathbb{N}^m$. Define

$$\begin{aligned} Q_{A,B}(x, y; f, r, \eta, s) = & \sum_{i=1}^d \left(x_i - \sum_{j=1}^m a_{ij} f_j - r_i \right)^2 \\ & + \sum_{i=1}^d \left(y_i - \sum_{j=1}^m b_{ij} f_j - r_i \right)^2 \\ & + \sum_{h \in \mathcal{H}(A)} C_h \\ & + \sum_{j=1}^m \left(\sum_{i \in S_j} u_{(i, a_{ij})} + s_j - (k_j - 1) \right)^2. \end{aligned} \quad (9)$$

Theorem 5.1 (Exact one-round quadratic). *For fixed A, B , the polynomial (9) has integer coefficients, total degree at most two, and is nonnegative on the entire real nonnegative orthant. For every $x, y \in \mathbb{N}^d$, projection to f is a bijection*

$$\{(f, r, \eta, s) \in \mathbb{N}^{d+2m+4H} : Q_{A,B}(x, y; f, r, \eta, s) = 0\} \longrightarrow \{f : f \text{ is maximal at } x, y = x - Af + Bf\}.$$

The certificate uses exactly $d + 2m + 4H$ auxiliary coordinates, $2d + 2H + m$ affine squares, and $2H$ nonnegative coordinate products.

Proof. The degree and nonnegativity follow term by term. A natural zero forces $r = x - Af \geq 0$ and $y = r + Bf$. Lemma 4.1 makes each flag $u_{(i,a)}$ equal to 1 exactly when $r_i \geq a$. The last equality in (9) is solvable in $s_j \geq 0$ exactly when

$$\sum_{i \in S_j} u_{(i,a_{ij})} \leq k_j - 1,$$

which says that at least one resource required by rule j is insufficient. Lemma 3.2 now proves maximality.

Conversely, start from a maximal f with the stated endpoint. Set $r = x - Af$, use the unique tuples in (4), and set

$$s_j = k_j - 1 - \sum_{i \in S_j} u_{(i,a_{ij})}. \quad (10)$$

All these coordinates are natural and every summand vanishes. The same formulas show uniqueness after f has been specified. There are m extents, d residuals, $4H$ comparator coordinates, and m maximality slacks. Counting the summands gives the remaining assertions. $\square$

5.1 Population uniformity, sparsity, and height

No coefficient in (9) depends on a population cutoff. Population size changes witness values, not the number of coordinates. If the polynomial is stored as the displayed sum, its description uses $O(\text{nnz}(A) + \text{nnz}(B) + H + d + m)$ affine-form entries, up to the fixed bookkeeping for signs and constants. Full expansion remains polynomial-size, but is less sparse and unnecessary for checking.

For a legal step, $\|f\|_1 \leq \|x\|_1$, since every firing consumes at least one object. Also $0 \leq r_i \leq \|x\|_1$. Let

$$M = \max_{i,j} a_{ij} \geq 1, \quad \beta = \max_j \|B_{\cdot j}\|_1.$$

Comparator slacks are at most $M + \|x\|_1$, flags are bits, and $0 \leq s_j \leq k_j - 1$. The output satisfies $\|y\|_1 \leq (1 + \beta)\|x\|_1$. Thus witnesses have bit length controlled by the bit lengths of the population and coefficients, even when their numerical values are large.

5.2 A fully worked cooperative example

Take species (A, B, C) and rules

$$R_1 : A + B \longrightarrow C, \quad R_2 : A \longrightarrow B.$$

Then $d = 3$, $m = 2$, and $\mathcal{H} = \{(A, 1), (B, 1)\}$, so $H = 2$. The endpoint-supplied certificate has $3 + 4 + 8 = 15$ auxiliary coordinates. For $x = (2, 1, 0)$, the maximal extents are

$$f = (1, 1), \quad y = (0, 1, 1), \quad \text{and} \quad f = (0, 2), \quad y = (0, 3, 0).$$

For the first extent, $r = (0, 0, 0)$. Both threshold tuples equal $(u, \bar{u}, v, w) = (0, 1, 0, 0)$, and the maximality slacks are $(s_1, s_2) = (1, 0)$. For the second extent, $r = (0, 1, 0)$; the A flag is 0, the B flag is 1, and both maximality slacks are zero. The feasible extent $(1, 0)$ is not maximal because the remaining A permits R_2 .

The file `data/example_certificate.json` supplies both a structured and an expanded polynomial for this example, together with the first witness. Appendix A gives its compact mathematical form.

[write] Shipped as `data/08-maximal-parallel-example_certificate.json`.

6 Histories and exact first halting

6.1 Fixed-horizon extent histories

Fix $x^0 = x$ and $T \in \mathbb{N}$. For every $0 \leq t < T$, take a fresh copy of the one-round variables and identify its endpoints with x^t, x^{t+1} . Define

$$Q_{A,B}^{[T]}(x; \mathbf{z}) = \sum_{t=0}^{T-1} Q_{A,B}(x^t, x^{t+1}; f^t, r^t, \eta^t, s^t). \quad (11)$$

The states $x^1, \dots, x^T$ are now among the witnesses.

Theorem 6.1 (Bounded histories). *The natural zeros of (11) are in bijection with all length- T extent histories from x . The polynomial is orthant-nonnegative and quadratic, and has*

$$N_T = T(2d + 2m + 4H) \quad (12)$$

witness coordinates. The statement includes the unique empty history when $T = 0$.

Proof. Every summand is nonnegative. Their simultaneous vanishing is equivalent to a legal maximal step at each time. Once the extent sequence is given, the states are determined recursively; Theorem 5.1 uniquely supplies every remaining coordinate. Each step contributes d new state coordinates in addition to its $d + 2m + 4H$ one-round auxiliaries. $\square$

Endpoint constraints such as $x_i^T = c$ or $x_i^T \geq c$ can be imposed by additional affine squares with unique slacks. A fixed Presburger terminal condition can instead use Section 4. The base count (12) excludes such optional conditions.

6.2 No earlier deadlock

To require first halting after exactly T rounds, add, for $0 \leq t < T$,

$$\left(\sum_j f_j^t - 1 - h_t \right)^2, \quad h_t \in \mathbb{N}. \quad (13)$$

At the last state, use a fresh bank of H threshold comparators on x^T , together with m slacks enforcing

$$\sum_{i \in S_j} u_{(i, a_{ij})}^T + s_j^T = k_j - 1.$$

This terminal certificate introduces no last-round extent or residual.

Corollary 6.2 (First-halting certificate). *First-halting extent histories of exactly T rounds have a root-faithful orthant-nonnegative quadratic certificate with*

$$N_T^{\text{halt}} = T(2d + 2m + 4H + 1) + 4H + m \quad (14)$$

natural witness coordinates. This also holds for $T = 0$.

Proof. The terminal bank says precisely that x^T is deadlocked. At any earlier state, a zero maximal extent is possible exactly when that state is already deadlocked. Equation (13) therefore excludes exactly the histories that halted earlier. Its slack is uniquely $h_t = \sum_j f_j^t - 1$. All other uniqueness claims follow from the previous theorem and the comparator lemma. $\square$

6.3 Witness height at a fixed horizon

Induction gives

$$\|x^t\|_1 \leq (1 + \beta)^t \|x^0\|_1. \quad (15)$$

Consequently every coordinate in a base trace certificate is at most a constant depending on A, B, T times $1 + \|x^0\|_1$. This establishes finite fibers and a linear-in-population height bound at fixed horizon. The factor depending on T need not be small; no polynomial-time algorithm for unbounded simulation is implied.

7 Guards, flat maximality, and retention

The same arithmetic separates three semantic choices that should not be silently conflated: which rules are eligible, how often an eligible rule may fire, and what happens to objects not consumed in the round.

7.1 Promoters and inhibitors read the pre-state

Suppose rule j has a fixed eligibility predicate $E_j(x)$, read from the *pre-state*. A promoter may require $x_i \geq a$; an inhibitor may require $x_i < a$, including absence as $x_i < 1$. Let a canonical guard certificate provide complementary bits

$$e_j = \mathbf{1}_{E_j(x)}, \quad d_j = 1 - e_j.$$

Keep the balance equations and residual threshold flags. Add the nonnegative term

$$d_j f_j \quad (16)$$

to forbid firings of disabled rules. Replace the unguarded maximality square by

$$\left(\sum_{i \in S_j} u_{(i, a_{ij})} + e_j + s_j - k_j \right)^2. \quad (17)$$

When $e_j = 1$, this enforces a missing residual resource. When $e_j = 0$, it is automatically satisfiable, with a uniquely determined nonnegative slack. Eligibility is not retested after some firings consume a promoter; that would be a different semantics.

Proposition 7.1 (Guarded maximality). *If every E_j has a unique auxiliary degree-two orthant-nonnegative certificate for its complementary truth bits, then guarded maximal rounds have a witness-faithful degree-two certificate. Its extra size is the size of the guard circuits and their bits; it is independent of the population magnitude.*

Proof. Term (16) forces $f_j = 0$ whenever $e_j = 0$. The only possible one-copy extensions are enabled rules. For exactly those rules, (17) enforces an insufficient residual resource. Conversely, every legal guarded extent gives unique guard bits, unique threshold flags, and the unique slack $s_j = k_j - e_j - \sum_{i \in S_j} u_{(i, a_{ij})}$. $\square$

7.2 Flat maximality

In flat semantics, each rule can be used at most once. Introduce $g_j \in \mathbb{N}$ and impose

$$(f_j + g_j - 1)^2 = 0. \quad (18)$$

Then f_j, g_j are complementary bits. A used rule cannot be appended again, even when its reactants remain available. The correct maximality condition is therefore

$$\left(\sum_{i \in S_j} u_{(i, a_{ij})} + e_j + g_j + s_j - (k_j + 1) \right)^2 = 0. \quad (19)$$

If $e_j = g_j = 1$, the rule is both enabled and unused, so one residual resource must be insufficient. In every other case the inequality is vacuous. Term (16) still excludes used-but-disabled rules.

Proposition 7.2 (Flat certificate). *Equations (18)–(19), together with guarded feasibility, give an exact witness-faithful quadratic for flat maximal rounds. In the unguarded case the endpoint-supplied auxiliary count is $d + 3m + 4H$.*

Proof. The preceding truth-table analysis proves the maximality equivalence. The complement $g_j = 1 - f_j$ is unique, and the last slack is

$$s_j = k_j + 1 - e_j - g_j - \sum_{i \in S_j} u_{(i, a_{ij})}.$$

Every other auxiliary is unchanged. The unguarded construction adds exactly one g_j per rule to Theorem 5.1. $\square$

For example, $A \rightarrow A$ at $3A$ has extent 3 in ordinary maximal semantics and extent 1 in flat maximal semantics. Both return the same state, so state observations alone can conceal the semantic difference. The published flat-maximal literature motivates this variant [2]; the certificate above is for the explicitly stated local semantics.

7.3 Discarding or selectively retaining unused objects

Fix a diagonal matrix $D = \text{diag}(\delta_1, \dots, \delta_d)$ with $\delta_i \in \{0, 1\}$. Replace the output equality by

$$y = Dr + Bf. \quad (20)$$

The case $D = I$ is ordinary retention and $D = 0$ discards all unused objects. Feasibility and maximality are still tested before this discard, against $r = x - Af$. Equation (20) is linear for fixed D , so all degree and uniqueness results survive unchanged. This statement concerns the defined transition relation; it does not assert that all retention variants have the same computational power.

7.4 Fixed membrane topology

For a finite fixed membrane topology, replace each object type by a pair (region, symbol). A fixed rule that consumes and produces specified multisets in these regions becomes one column of A and one of B . Pre-state promoters and inhibitors fit the guard mechanism above. The conversion applies when that membrane model's parallel semantics agrees with the resource semantics used here. Unbounded membrane creation, division, dissolution, dynamically generated names, or model-specific priority rules require additional definitions and are not covered merely by changing notation.

7.5 An explicit guarded register-machine embedding

The arithmetic compiler does not require universality, but a small direct simulation makes the computational connection transparent. Represent each machine label ℓ by one control species L_ℓ and each register by its object count. Exactly one control object is present.

An instruction “increment register A and go to k ” becomes

$$L_\ell \longrightarrow L_k + A.$$

A zero/decrement instruction becomes the two rules

$$L_\ell + A \longrightarrow L_k, \quad L_\ell \longrightarrow L_h \quad \text{inhibited by the presence of } A.$$

If $A > 0$, only the first rule is eligible and feasible; if $A = 0$, only the second is. Each instruction uses the single control object, and the next control object is unavailable until the next round. Maximality forces the unique applicable instruction. A halt label has no consuming rules.

Induction on steps proves a one-to-one correspondence between machine instructions and parallel rounds, with the same register contents and halting time. This is an elementary simulation of the familiar register model, whose universality is classical [5]; it is not an improvement on the unguarded universality results of [1]. Composing it with our certificate preserves individual deterministic histories at every fixed horizon.

[write] This bears on the question “Other unconventional substrates with timing certificates” of the report *Liveness beyond halting*, which asks for clock-faithful translations into, among others, reaction systems with an explicit effective zero-test mechanism, together with a local polynomial compiler for the target’s finite steps. Source 08 does not draw the connection. The embedding above uses one round per instruction, maximality forces the unique applicable rule, and there are no internal steps; so it appears to be a computable signal-faithful block simulation with block length one in the sense of that report’s Theorem lbh:thm:transfer, and Proposition 7.1 supplies a degree-two local step. That would be a partial answer for register machines simulated by guarded maximal-parallel multiset rewriting. The transfer hypotheses have not been checked in detail here, and guarded histories are not implemented in the shipped code.

8 The optimal one-round counting exponent

We now turn from certificates to a quantitative invariant of the consumption matrix. This section concerns *unguarded ordinary* maximality. Productions play no role in the number of choices available in a single round.

Definition 8.1 (Resource covers and their rank). A set of species $J \subseteq \{1, \dots, d\}$ is a *resource cover* if $J \cap S_j \neq \emptyset$ for every rule j . Let A_J denote the submatrix formed by its rows. Define

$$\rho(A) = \min_{J \text{ a resource cover}} \text{rank}_{\mathbb{Q}}(A_J). \quad (21)$$

All columns are nonzero, so covers exist and $1 \leq \rho(A) \leq \text{rank } A \leq m$. A resource cover is not necessarily a minimum-cardinality hitting set: linear dependence of its rows matters. The number $\rho(A)$ is a property of numerical stoichiometry as well as support.

Let

$$F_A(x) = \#\{f \in \mathbb{N}^m : f \text{ is maximal at } x\}.$$

Theorem 8.2 (Uniform resource-cover bound). *Put $M = \max_{i,j} a_{ij} \geq 1$ and let $\mathcal{C}(A)$ be the finite collection of resource covers. Then for every $x \in \mathbb{N}^d$,*

$$F_A(x) \leq \left(\sum_{J \in \mathcal{C}(A)} M^{|J|} \right) (1 + \|x\|_1)^{m-\rho(A)}. \quad (22)$$

In particular, the leading constant can be replaced by $2^d M^d$.

Proof. For every maximal f , choose for each rule one deficient species as in (2), and let J be the union of these choices. This J is a resource cover. Every coordinate r_i , $i \in J$, was chosen because $r_i < a_{ij}$ for at least one rule, so $0 \leq r_i < M$. Consequently f satisfies, for at least one such pair (J, r_J) ,

$$A_J f = x_J - r_J. \quad (23)$$

There are at most $M^{|J|}$ possible r_J .

Fix J, r_J and write $q = \text{rank } A_J$. Choose q linearly independent columns of A_J . Once the other $m - q$ entries of f are fixed, (23) determines the chosen entries uniquely over $\mathbb{Q}$, if they exist at all. Feasibility and nonzero consumption give $\|f\|_1 \leq \|x\|_1$, so each unchosen coordinate has at most $1 + \|x\|_1$ possibilities. The number of solutions for this pair is at most $(1 + \|x\|_1)^{m-q}$, and $q \geq \rho(A)$. Sum over all covers and all allowed residual tuples. The sum may overcount a maximal extent, which is harmless for an upper bound. $\square$

The bound is not just sharp for one convenient family. For every consumption matrix its exponent is attained by an integral input ray.

Theorem 8.3 (Sharpness for every consumption matrix). *For every A satisfying the standing assumptions, there exists $v \in \mathbb{N}^d$ such that*

$$F_A(nv) = \Theta(n^{m-\rho(A)}) \quad (n \rightarrow \infty). \quad (24)$$

One may choose a rank-minimizing resource cover J and set

$$v = A\mathbf{1} + \mathbf{1}_{J^c}. \quad (25)$$

Thus $m - \rho(A)$ is the smallest exponent that can bound $F_A(x)$ by a constant times $(1 + \|x\|_1)$ to that exponent uniformly over all x .

Proof. Let $q = \rho(A)$ and $k = m - q$. The rational kernel of A_J has dimension k . Choose linearly independent integer vectors $z_1, \dots, z_k \in \mathbb{Z}^m$ spanning it over $\mathbb{Q}$. Such vectors are obtained from a rational basis by clearing denominators. If $k = 0$, the vector $f = n\mathbf{1}$ already gives one maximal extent for every n , which is the required lower bound.

Suppose $k > 0$. For integers $t_1, \dots, t_k$, set

$$f = n\mathbf{1} + \sum_{h=1}^k t_h z_h.$$

For sufficiently small fixed $\varepsilon > 0$, all choices $|t_h| \leq \varepsilon n$ make every component of f at least $n/2$. Indeed, it suffices to choose ε smaller than the reciprocal of twice the maximum, over coordinates, of $\sum_h |(z_h)_j|$. Also choose it small enough that

$$\left| \sum_h t_h (Az_h)_i \right| \leq n/2 \quad \text{for every } i \notin J.$$

Zero maxima impose no restriction on this choice.

At $x = nv$, the residual is

$$r = nv - Af = n\mathbf{1}_{J^c} - \sum_h t_h Az_h.$$

It vanishes on J because $A_J z_h = 0$, and is nonnegative outside J by the choice of ε . Thus f is feasible. Every rule consumes a positive amount of some species in J , whose residual is zero; hence f is maximal. The integer basis vectors are linearly independent, so different tuples $(t_1, \dots, t_k)$ give different extents. Their number is $(2\lfloor \varepsilon n \rfloor + 1)^k = \Omega(n^k)$. The upper bound is Theorem 8.2, since $\|nv\|_1 = n\|v\|_1$.

If a uniform exponent smaller than k existed, its restriction to this ray would contradict the established lower bound. $\square$

8.1 Examples and interpretation

If all m rules consume one object of the same species, a single row covers every rule and $\rho(A) = 1$. There can be $m - 1$ independent large-scale allocation choices. If each rule has its own private consumed species and no overlap, the relevant rows have rank m , $\rho(A) = m$, and the number of maximal choices is uniformly bounded. For independent pools with $m_1, \dots, m_p$ competitors, $\rho(A) = p$ and the maximal exponent is $\sum_i (m_i - 1) = m - p$.

The rank theorem should not be used with all columns of a guarded system without further work. Disabled rules do not need a deficient resource; the effective cover condition then involves an enabled submatrix, which may vary with the pre-state. This is a genuine extension question, not an automatic corollary of the unguarded theorem.

9 Trace counts, quasipolynomials, and sharp degree bounds

Fix A, B, T and an affine input ray

$$x(n) = nv + w, \quad v, w \in \mathbb{N}^d.$$

Let $C_T(n)$ count length- T extent histories from $x(n)$, optionally filtered by a fixed Presburger condition on the final state and n . First-halting counts are included by imposing the conditions of Section 6.

Theorem 9.1 (Quasipolynomial trace counts). *The function $C_T(n)$ is eventually quasipolynomial: there are $q \geq 1$, $N \geq 0$, and rational polynomials $p_0, \dots, p_{q-1}$ such that*

$$C_T(n) = p_r(n) \quad \text{for } n \geq N, \quad n \equiv r \pmod{q}.$$

Every nonzero p_r has degree at most

$$T(m - \rho(A)). \tag{26}$$

The generating function $\sum_{n \geq 0} C_T(n)z^n$ is rational, with possible finite poles only at roots of unity.

Proof. At a fixed horizon, all state recurrences have constant integer coefficients. Feasibility is a conjunction of linear inequalities and maximality is the finite Boolean combination

$$\bigwedge_{t,j} \bigvee_{i \in S_j} \left(x_i^t - \sum_h a_{ih} f_h^t < a_{ij} \right).$$

The initial state is affine in n . Thus the set of pairs consisting of n and a valid extent history is Presburger definable. The fibers are finite by (15). The Presburger counting theorem of Woods gives piecewise quasipolynomiality; in one nonnegative parameter a finite polyhedral partition is eventually a single ray, giving the asserted eventual form [3]. The rational generating-function conclusion also follows directly by summing polynomial sequences on residue classes.

For the degree bound, set $k = m - \rho(A)$ and $K = \sum_{J \in \mathcal{C}(A)} M^{|J|}$. Along the input ray there is a constant L with $1 + \|x(n)\|_1 \leq L(n + 1)$. At time t , (15) gives

$$1 + \|x^t\|_1 \leq (1 + \beta)^t L(n + 1).$$

Theorem 8.2 bounds the branching at that state by $K((1 + \beta)^t L(n + 1))^k$. Multiplying these bounds over $t < T$ gives

$$C_T(n) \leq K^T L^{kT} (1 + \beta)^{kT(T-1)/2} (n + 1)^{kT}. \quad (27)$$

A terminal filter only removes histories. An eventual residue polynomial of degree larger than kT would violate this estimate along its residue class. Therefore all degrees satisfy (26). $\square$

The quasipolynomial conclusion is an application of established Presburger counting theory, not a new replacement for it. The explicit resource-cover degree bound and its sharpness are the quantitative input proved here. The argument is about fixed T ; it is not uniform in an unbounded time parameter.

9.1 A family attaining the horizon bound

For $m \geq 1$, take a shared pool species A and inert output species $B_1, \dots, B_m$, with rules



Start with nA and no output objects. Every maximal round distributes exactly n firings among the m labels. The pool size remains n , and the inert outputs do not constrain subsequent choices. Therefore

$$C_T(n) = \binom{n + m - 1}{m - 1}^T. \quad (29)$$

Here $\rho(A) = 1$, and the degree is exactly $T(m - 1)$. For $m \geq 2$ and fixed T , the leading asymptotic is

$$C_T(n) = \frac{n^{T(m-1)}}{((m-1)!)^T} \left(1 + \frac{Tm(m-1)}{2n} + O(n^{-2}) \right). \quad (30)$$

In fact (29) is a polynomial, so all further coefficients are obtained by multiplying the finite factors $\prod_{h=1}^{m-1} (1 + h/n)^T$. This is a transparent extremal example, not an assertion of an unresolved OEIS asymptotic.

With p independent pools and m_i competitors for pool i , the count is

$$\prod_{i=1}^p \binom{n + m_i - 1}{m_i - 1}^T,$$

and the degree is $T(m - p) = T(m - \rho(A))$.

9.2 Genuine periodicity

Two labelled rules both consuming $2A$ and producing nothing have $\rho(A) = 1$. Maximality forces $f_1 + f_2 = \lfloor n/2 \rfloor$ at input nA , so

$$C_1(n) = \lfloor n/2 \rfloor + 1, \quad \sum_{n \geq 0} C_1(n) z^n = \frac{1+z}{(1-z^2)^2}. \quad (31)$$

The two eventual residue polynomials differ. Thus “eventually polynomial” is too strong even for a one-species example.

9.3 Flat maximality stabilizes at fixed horizon

Proposition 9.2 (Eventual constancy in the unguarded flat case). *For a fixed unguarded flat system and fixed T , the number of extent histories from $nv + w$ is eventually constant in n . The conclusion also holds with fixed Boolean combinations of affine threshold guards and terminal tests, but need not hold with modular tests.*

Proof. There are only 2^{mT} candidate flat extent sequences, independently of n . For each candidate, state recurrence makes every state affine in n . Feasibility, eligibility, maximality, and the stated terminal tests are finite Boolean combinations of affine inequalities in n . Every one of these inequalities has a stable truth value for sufficiently large n . Hence each fixed candidate is eventually always accepted or always rejected. Take the maximum of the finitely many stabilization thresholds and sum the resulting constant indicators. $\square$

A guard such as “the initial count is even” yields periodic behavior instead, explaining the exclusion of modular tests from the constancy statement.

10 Counting roots can remain hard at one round

Population-uniform low degree does not imply efficient exact counting when the network itself is part of the input. The next reduction preserves the count exactly and uses only Boolean initial populations and Boolean consumption coefficients.

Theorem 10.1 (Graph-to-network root bijection). *Given a finite simple graph G with at least one vertex, one can construct in polynomial time an unguarded multiset-rewriting system and a Boolean initial population such that its maximal one-round extents are in bijection with the independent sets of G . Every reactant species occurs in exactly two rules, every consumption coefficient is 0 or 1, and every feasible extent is Boolean. The natural roots of the one-round quadratic, with the endpoint allowed to vary, preserve the same count.*

Proof. Form a graph H by adding a private leaf v' adjacent only to each original vertex v . An independent set S of G gives the maximal independent set

$$S \cup \{v' : v \notin S\} \quad (32)$$

of H . Conversely, a maximal independent set of H must contain exactly one of v, v' for every v : it cannot contain both, and containing neither would allow the leaf to be added. Its original vertices form an independent set of G . These maps are inverse.

For each edge e of H , create one species X_e initially present once. For each vertex u of H , create a rule consuming all X_e on incident edges and producing nothing. There are no isolated vertices in H , so every rule consumes something and every feasible multiplicity is at most one. Two vertex rules can be used together exactly when their vertices are nonadjacent. Hence feasible extents are independent sets of H , and a further rule can be appended exactly when another vertex can be adjoined. Maximal extents are therefore precisely its maximal independent sets. Each edge species belongs to its two endpoint rules. Compose this bijection with (32) and Theorem 5.1. $\square$

Corollary 10.2 (Complexity of one-round root counting). *Counting these roots is #P-complete under polynomial-time Turing reductions, even under the restrictions in Theorem 10.1. The displayed map from independent-set counting to root counting is parsimonious.*

Proof. Counting independent sets is #P-complete under the stated reduction convention: it is equivalent, by complementation, to counting vertex covers, which is counting assignments to a monotone 2-CNF with one clause per edge. This is among Valiant’s counting-completeness results [4]. Theorem 10.1 gives hardness and preserves each count exactly.

For membership, guess the Boolean extent, check feasibility and maximality, and reconstruct the unique remaining coordinates in polynomial time. More generally, for arbitrary binary-encoded numerical populations, feasible extents and all canonical auxiliaries have polynomial bit length by the bounds following Theorem 5.1. Fixed-length binary guesses count each witness once. $\square$

There is no claim here of parsimonious completeness for arbitrary #P functions. In particular, every finite resource system has at least one maximal feasible extent, whereas a satisfiability instance can have zero satisfying assignments. The graph-to-network bijection and the Turing-reduction completeness statement are different assertions. Nor does this reduction bound the number of reactants in a rule: vertex degree can be large, even though each species occurs in only two rules.

11 Why convex quadratic certificates fail

The certificates above are quadratic but usually not convex. This is not merely an implementation artifact.

11.1 A structural lemma for convex zero sets

Lemma 11.1 (Rational polyhedral zeros). *Let $q(z) = z^T H z + \ell^T z + c$, with H rational symmetric positive semidefinite, be nonnegative on $\mathbb{R}_{\geq 0}^N$. Suppose $q(z_0) = 0$ for some $z_0 \in \mathbb{Q}_{\geq 0}^N$. Then its orthant zero set is a rational polyhedron. More precisely, with $g = 2H z_0 + \ell$,*

$$\{z \geq 0 : q(z) = 0\} = \{z \geq 0 : H(z - z_0) = 0, g^T z = 0\}. \quad (33)$$

Proof. At the constrained minimum z_0 , the directional derivative in a positive coordinate direction is nonnegative. If $(z_0)_i > 0$, both signs of that direction are allowed locally, so $g_i = 0$. If $(z_0)_i = 0$, we have $g_i \geq 0$. Thus $g \geq 0$ and $g^T z_0 = 0$.

Taylor’s identity gives

$$q(z) = g^T(z - z_0) + (z - z_0)^T H(z - z_0) = g^T z + (z - z_0)^T H(z - z_0).$$

Both terms are nonnegative for $z \geq 0$. They vanish together exactly when $g^\top z = 0$ and $H(z - z_0) = 0$, using positive semidefiniteness. All displayed coefficients are rational. $\square$

We also use the elementary recession fact that, if a linear functional is unbounded above on a nonempty rational polyhedron, some rational recession direction makes that functional strictly positive. One proof uses a finite rational vertex-and-ray description: a polyhedron with no such ray would give a finite upper bound from its finitely many vertices and zero or negative ray contributions. In the nonnegative orthant there is no lineality. Clearing denominators gives an integral direction.

11.2 The cooperative deadlock obstruction

Theorem 11.2 (No convex quadratic for a two-reactant deadlock). *There are no $k \in \mathbb{N}$ and rational quadratic polynomial $Q(x, y, z_1, \dots, z_k)$ that is convex and nonnegative on $\mathbb{R}_{\geq 0}^{k+2}$ and satisfies*

$$\forall x, y \in \mathbb{N} \quad (\exists z \in \mathbb{N}^k : Q(x, y, z) = 0) \iff (x = 0 \text{ or } y = 0). \quad (34)$$

Nevertheless xy is an orthant-nonnegative quadratic with exactly the required natural zero set and no auxiliaries.

Proof. Suppose such a convex quadratic exists. Its constant Hessian is positive semidefinite: convexity on the orthant already implies this by considering small line segments through an interior point. Choose a natural zero z_0 of the full polynomial projecting to $(0, 0)$. By Lemma 11.1, the real orthant zero set Z is a rational polyhedron.

The slice $Z \cap \{y = 0\}$ contains a natural point with $x = n$ for every $n \in \mathbb{N}$, so the coordinate x is unbounded on it. The recession fact gives a rational recession vector u with $u_x > 0$ and $u_y = 0$. Every recession direction of a subset of the nonnegative orthant is nonnegative. Scaling u by a positive integer makes it integral. Similarly, the slice $Z \cap \{x = 0\}$ supplies an integral nonnegative recession vector v with $v_x = 0$ and $v_y > 0$.

Both u and v satisfy the homogeneous constraints defining $\text{rec } Z$. Hence $z_0 + u + v$ is an integral nonnegative point of Z . Its projected coordinates are both positive, contradicting (34). The final assertion is immediate from $xy \geq 0$ on the orthant. $\square$

For the sole reaction $A + B \rightarrow C$, let x, y be the counts of its two reactants. Deadlock is exactly the union of the coordinate axes. Thus arbitrary finite existential integer lifting cannot turn this particular deadlock condition into a convex quadratic zero problem. The obstruction does not concern a prescribed finite population box, and it does not exclude nonconvex quadratic certificates such as ours.

11.3 A convex subclass that really works

Proposition 11.3 (Single-species reactant supports). *Suppose each S_j has size one. Let I be the species consumed by at least one rule, and for each $i \in I$ put*

$$c_i = \min\{a_{ij} : S_j = \{i\}\}.$$

Then maximal rounds have a convex quadratic certificate with $d + m + |I|$ natural auxiliaries when both endpoint states are supplied.

Proof. The no-extension condition is now precisely $0 \leq r_i < c_i$ for every $i \in I$. Introduce one slack t_i and impose $r_i + t_i = c_i - 1$. Together with $x = Af + r$ and $y = Bf + r$, these are affine equalities. The sum of their squares is convex and nonnegative on all of $\mathbb{R}$. Its variables are f (m), r (d), and t ($|I|$), and all coordinates other than the chosen extent are uniquely determined. $\square$

This is a sufficient structural condition, not a complete classification. Some systems with cooperative rules may have redundant deadlock constraints that simplify to convexly representable relations. Theorem 11.2 proves that a general compiler cannot assume such a simplification is always available.

12 Program-uniform quartics and a sharp degree threshold

The fixed numerical matrices are coefficients of our quadratic. If those numbers become inputs, multiplication of a rule multiplicity by a stoichiometric coefficient becomes a genuine quadratic expression, and its square has degree four. This difference is mathematically necessary in the orthant-nonnegative setting.

12.1 A positivity lemma inherited from the repository

We first give a self-contained proof of the needed lower-degree fact. The semilinearity conclusion is already recorded in the repository's order-free certificate report [7]; it is not a new contribution of the present paper.

[write] Second proof (source 08). Lemma 12.1 is a second proof of the inclusion $D_3^+ \subseteq \text{SL}$ in Theorem cdc:of:thm:classification of *Canonical Diophantine certificates*; the first proof is cdc:of:lem:affinezero, cdc:of:prop:faces and cdc:of:thm:cubicsemilinear there, by the same Hessian-on-faces idea. Corollary 12.2 is cdc:of:cor:squares, and Proposition 13.1 is cdc:of:cor:no-cubic-universal. They are printed as source 08 wrote them.

Lemma 12.1 (Natural zeros of positive cubics). *Let $p \in \mathbb{Q}[z_1, \dots, z_N]$ have total degree at most three and satisfy $p \geq 0$ on $\mathbb{R}_{\geq 0}^N$. Then its natural zero set, and every coordinate projection of that set, is semilinear.*

Proof. Partition the orthant by exact coordinate support. Fix a support $I \subseteq \{1, \dots, N\}$ whose face contains a natural zero a in its relative interior. Work only in the coordinates of I ; all others are zero. The point a is an interior minimum of the restricted polynomial, so its gradient is zero and its Hessian H_a is positive semidefinite. These derivatives have rational entries because a is integral.

If b is any other zero in the relative interior of the same face, consider $q(t) = p(a + t(b - a))$. The line segment lies in the relative interior, as do sufficiently small extensions beyond its endpoints. The polynomial q is nonnegative on an open interval containing $[0, 1]$ and has zeros at 0 and 1. If $a \neq b$, both zeros have even multiplicity. Since $\deg q \leq 3$, q must vanish identically. Thus $(b - a)^\top H_a (b - a) = 0$, so $b - a \in \ker H_a$. The same kernel conclusion is immediate if $a = b$.

Conversely, let $v \in \ker H_a$. Taylor expansion, using degree at most three, has the form

$$p(a + tv) = t^3 R(v)$$

for a homogeneous cubic expression R , with $R = 0$ in the lower-degree case. For sufficiently small positive and negative t , the point $a + tv$ is still in the relative interior. Nonnegativity for both signs forces $R(v) = 0$. Therefore every point of $a + \ker H_a$ that stays in that relative interior is a zero.

The zeros in this face are consequently exactly $(a + \ker H_a) \cap \text{relint}(\mathbb{R}_{\geq 0}^I)$. Its natural points are described by rational affine equations, coordinates at least one on I , and zeros outside I . Clearing denominators gives a Presburger description. Faces without any natural zero contribute nothing, and there are only finitely many supports. A finite union of these descriptions is Presburger, as are its coordinate projections; equivalently, these sets are semilinear [3]. $\square$

This proof is a structural argument. Choosing an integer zero in every nonempty face is not presented as an algorithm for deciding which faces have zeros.

Corollary 12.2 (Multiplication requires degree at least four). *The relation $z = kn$ on $\mathbb{N}^3$ cannot be represented as the projection of the natural zero set of any finite-variable polynomial of degree at most three that is nonnegative on its entire real orthant.*

Proof. Such a representation would make the multiplication graph semilinear by Lemma 12.1. Intersecting with $k = n$ and projecting to z would make the set of squares Presburger. Every one-dimensional Presburger set is eventually periodic. If positive period q worked for the squares, choose n large enough that n^2 lies past the eventual threshold and $q < 2n + 1$. Periodicity would make $n^2 + q$ a square strictly between n^2 and $(n + 1)^2$, a contradiction. $\square$

12.2 Uniform construction on a fixed consumption support

Fix d, m and a support mask $E \subseteq \{1, \dots, d\} \times \{1, \dots, m\}$ meeting each column. At a mask entry, write

$$a_{ij} = 1 + \alpha_{ij}, \quad \alpha_{ij} \in \mathbb{N},$$

and put $a_{ij} = 0$ off the mask. Treat all α_{ij} and all production entries $b_{ij} \in \mathbb{N}$ as input coordinates. Let $K = |E|$.

Use the same one-round formula, but give each mask entry its own comparator. Its threshold is now $1 + \alpha_{ij}$. Numerical thresholds that happen to coincide on a particular input are not merged, since this compiler must work before the input values are known.

Theorem 12.3 (Sharp program-uniform degree). *On every fixed nonempty-column support mask, maximal one-round execution has an orthant-nonnegative quartic certificate, uniform in all numerical consumption and production entries, with exactly $d + 2m + 4K$ natural auxiliaries when the endpoint states are supplied. Its zeros remain in bijection with legal extent vectors. Across this class of systems, degree four is optimal among polynomials nonnegative on the entire real orthant, even with an arbitrary finite number of auxiliaries.*

Proof. The resource and output residuals now contain products $\alpha_{ij}f_j$ and $b_{ij}f_j$, so they have degree at most two and their squares have degree at most four. The threshold residual is still affine:

$$r_i + v_{ij} - \alpha_{ij} - u_{ij} - w_{ij}.$$

All complementarity terms are nonnegative coordinate products. Therefore the resulting polynomial is orthant-nonnegative of degree at most four. For each natural parameter assignment, Theorem 5.1's

proof applies without change, using the specified positive thresholds. The coordinate count replaces H by K .

For necessity, consider just one rule $A \rightarrow kB$. At the initial state nA with no B , maximality forces extent $f = n$ and output zB with $z = kn$. A hypothetical program-uniform orthant-nonnegative representation of degree at most three would, after fixing the other matrix and state coordinates to their constant values, represent multiplication. This contradicts Corollary 12.2. $\square$

The optimality claim includes the positivity hypothesis. It is not a claim about arbitrary cubic Diophantine equations with unrestricted sign, nor about an undecidability threshold for all degree-three polynomials. Also, program uniformity here fixes the matrix dimensions and support mask; it does not encode an arbitrarily large rule table in a fixed number of coordinates.

[write] Theorem 12.3 is not in *Canonical Diophantine certificates*: it applies $D_3^+ = \text{SL}$ to a program-uniform family. It is a data point for that report’s question `cdc:q:degree`, not an answer to it.

13 Unbounded computation: what follows and what does not

13.1 The ordinary MRDP consequence

Finite histories of any fixed effective system can be enumerated and checked. Its accepting-input set is therefore computably enumerable. The MRDP theorem supplies a single ordinary integer polynomial with a fixed finite witness dimension for membership in that set. The repository exposes precisely such a finite-polynomial statement in `Diophantine.mrdp` [6].

The passage from an arbitrary MRDP polynomial to degree four can be made explicit at the level of arithmetic circuits. Write its equation as $P_+(n, w) = P_-(n, w)$, where both sides have nonnegative coefficients. Compute them using addition and multiplication gates, each with a fresh natural output coordinate. For addition impose $v - u - w = 0$; for multiplication impose $v - uw = 0$; impose the final equality of the two outputs. The sum of the squares of all gate residuals is a globally nonnegative quartic. For a given original witness, every gate coordinate is uniquely determined. This preserves the number of original witnesses, whatever that number was; it does not make those witnesses unique.

[write] This paragraph is the circuit argument of `cdc:of:prop:quartic` (“Quartic normalization”) in *Canonical Diophantine certificates*.

13.2 Why our quadratics cannot compress all time

Proposition 13.1 (No positive cubic uniform halting certificate). *Let $H \subseteq \mathbb{N}$ be an undecidable accepting-input set of a fixed effective universal computational device. There is no finite-variable polynomial of degree at most three, nonnegative on the entire real orthant, whose projected natural zero set is H .*

Proof. Lemma 12.1 would make H semilinear and hence decidable, contrary to the hypothesis. $\square$

[write] This is cdc:of:cor:no-cubic-universal of *Canonical Diophantine certificates*. Here H denotes a set, not the number of species–threshold pairs of (8).

Even a nonuniversal growth example already shows a barrier to the simplest kind of horizon compression. The single rule $A \rightarrow 2A$, starting at one A , has 2^T objects after T maximal rounds. For each fixed T , our history certificate is quadratic. But a single orthant-nonnegative cubic with free input T , free output y , and fixed witness dimension cannot represent $y = 2^T$: projecting would make the powers of two eventually periodic. Thus the difficulty is not only nondeterministic computation.

13.3 Finite-fold and single-fold caution

A bounded-horizon certificate may have exactly one root for a specified deterministic history and still have a number of variables growing with the horizon. An unbounded Diophantine representation obtained by MRDP may have many or infinitely many witnesses. Neither property implies a fixed-arity finite-fold or single-fold representation for every accepting input.

Accordingly, this manuscript does not settle the finite-fold or single-fold representation questions studied in the repository. It gives no elimination of the horizon that preserves our root bijection. Its uniform quartic theorem concerns fixed-shape numerical rule parameters, not an unbounded history compressed with unique witnesses into a fixed tuple. These limitations are essential parts of the theorem statements, not exceptions added after the construction.

13.4 A formalization route

The most direct Lean or Rocq development can be divided into three independently reusable interfaces. First, formalize a finite multiset system as natural matrices and define a maximal extent by the failure of all one-copy extensions. Prove Lemma 3.2 without any arithmetization. Second, prove the threshold comparator’s natural-zero classification and its uniqueness, then assemble the one-round and history bijections by finite sums of nonnegative terms. Third, formalize the resource-cover bound using finite rank and injectivity after a choice of pivot columns. The sharpness proof adds an integral kernel basis and an explicit coefficient box.

The convexity obstruction and cubic-face lemma require a different library layer: rational quadratic forms, positive semidefiniteness, polyhedral recession, and Presburger projections. The first comparator and compiler layer is a useful initial target because it isolates all semantic errors before invoking any substantial geometry. No claim is made that these new interfaces have already been checked by a proof assistant.

14 Executable validation and reproducibility

14.1 What is implemented

The package contains `code/parallel_certificates.py`, a Python 3.10-or-later implementation using only the standard library. Its `RoundCertificate` class emits the quadratic as a sum of affine squares and nonnegative coordinate products, and can export its expanded integer monomials. The same implementation supports flat semantics, conjunctions of pre-state threshold/absence guards, and selective retention. Its constructor returns the canonical witness for a supplied legal extent; it is not an integer-root search engine.

HistoryCertificate assembles fixed-horizon and exact-first-halting quadratics for the base unguarded, ordinary, retaining semantics, including horizon zero. The code also provides an independent extent enumerator, exact rational rank, an exhaustive resource-cover search, and fixed-horizon trace counting. The cover search is exponential in the number of species; no efficient optimization algorithm for $\rho(A)$ is claimed.

The program-uniform quartic, general Presburger guards, geometric lower bounds, and formal proof-assistant development are mathematical results or proposed work in the article, not features silently claimed for the implemented compiler.

14.2 Actual exact-arithmetic checks

The supplied run used Python 3.13.5. All tests passed, with the following counts taken from `data/verification.txt`:

Check	Cases
Threshold-gadget assignments	160,524
Exhaustive full-polynomial assignments for $A \rightarrow 2A$	32,805
Distinct two-species, two-rule consumption matrices	64
Candidate one-round extents across those matrices	7,542
Constructed legal roots across those matrices	2,488
Single-coordinate witness mutations rejected	38,496
Guarded, flat, and retention round checks	72
Exact trace-count and rank-family identities	124
History and exact-first-halting checks	328
All labelled simple graphs on 1–4 vertices	75
Roots constructed for the graph reductions	599

The comparator check exhausts thresholds $1 \leq a \leq 7$, inputs $0 \leq r \leq 12$, flags $0 \leq b, c \leq 2$, and slacks $0 \leq v, w \leq 13$. It finds exactly the tuple in Lemma 4.1 in every case. The full-polynomial test exhausts all seven auxiliary coordinates in $\{0, 1, 2\}$ for $A \rightarrow 2A$, with $0 \leq x \leq 2$ and $0 \leq y \leq 4$; exactly three roots appear, the expected ones.

The network tests include nonmaximal feasible choices, illegal same-round reuse of products, and the difference between maximality and maximum-cardinality firing. Graph tests enumerate every labelled graph in the stated range, build the private-leaf construction, and compare independent-set counts with exact polynomial roots. Count identities include the sharp shared-pool family, independent pools, and the periodic example (31).

These finite checks cannot establish the universal uniqueness, rank bound, convexity obstruction, or degree lower bound. Those claims rest on the proofs in the corresponding sections. Tests are a reproducible check on the arithmetic implementation and on small instances, not a substitute for those proofs.

14.3 Reproduction

From the package root, run

```
python code/verify.py
```

```
python code/parallel_certificates.py > data/example_certificate.json
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The first command rewrites `data/verification_counts.json` and prints the checks. Redirect its output to refresh the text log. The TeX source has an embedded bibliography and requires no BibTeX run. The PDF was rendered and visually inspected after compilation.

[write] These are the delivered paths. In this report the programs are `code/08-maximal-parallel-parallel_certificates.py` and `code/08-maximal-parallel-verify.py`, and the data are `data/08-maximal-parallel-example_certificate.json`, `data/08-maximal-parallel-verification.txt` and `data/08-maximal-parallel-verification_counts.json`. The first command rewrites a shipped file and the second, as printed, overwrites one; run them only on a copy in the delivered layout, as the report’s README describes. The `article.tex` and PDF named here are source 08’s own; this Part replaces them and the delivered PDF is not shipped. The package was rerun at placement with Python 3.14.4: all checks passed, and the outputs equal the shipped files apart from line endings, the Python version line and the elapsed time. The research programme’s review found two input-contract defects in the compiler (mutable network inputs, unchecked guard polarity) and supplies a patch, `maximal_parallel_immutable_guards.patch`; no polynomial changes, and the shipped file is unpatched.

15 Further research questions

The most useful continuations are questions whose proof obligations can be separated cleanly from a new assertion of universality.

- Q1. Classify convexly representable maximality relations.** Single-species reactant supports suffice; one two-reactant deadlock can fail. Find necessary and sufficient conditions on A for the deadlock set, and then for the full round relation, to admit a convex quadratic with finitely many natural auxiliaries. A useful first subproblem is whether a characterization can be read from the minimal resource covers and redundant inequalities.
- Q2. Determine ray-specific counting degrees.** The invariant $m - \rho(A)$ is the optimal exponent uniformly over all populations. A fixed ray $nv + w$ can have smaller degree. Describe its exact eventual quasipolynomial degrees using the faces $\{f \geq 0 : Af \leq v, A_J f = v_J\}$, together with the bounded residual shifts and integral congruence obstructions. Any proposed formula must handle zero-dimensional and arithmetically empty slices.
- Q3. Measure the complexity of computing $\rho(A)$.** The definition minimizes rational rank over hitting sets of rows. The included exact algorithm enumerates 2^d row subsets. Determine the complexity of this optimization, and whether useful structural classes admit efficient algorithms or certificates of optimality. Minimum row count and minimum row rank need not be conflated.
- Q4. Extend the rank theory to guards.** For pre-state guards, maximality sees only the enabled columns. Seek piecewise bounds involving enabled-submatrix ranks, with careful treatment of states where no rules are enabled. For fixed Presburger guards, the parameter regions are accessible to arithmetic partitioning, but one must show that their bounds remain stable under the transition relation.

- Q5. Optimize canonical comparison circuits.** Four coordinates per distinct threshold are sufficient here. Determine whether the count can be reduced while keeping global orthant nonnegativity, degree two, no population bound, and unique auxiliaries. Shared ordered thresholds on one species may admit further reuse beyond identical-threshold sharing. A smaller existential encoding without uniqueness would not answer this question.
- Q6. Locate bounded-arity counting frontiers.** Our hard family has Boolean stoichiometry, Boolean initial populations, and reactant occurrence exactly two, but unbounded rule arity. Determine sharp complexity boundaries when every rule consumes at most two, three, or four species, perhaps with planar or bounded-treewidth incidence structure. State the counting reduction convention explicitly.
- Q7. Compute periods and chamber data effectively.** Theorem 9.1 supplies eventual quasipolynomiality and an explicit degree bound. Develop a practical algorithm that returns the actual period, residue polynomials, and validity threshold for small fixed systems. Multiple initial population parameters should lead to a chamber version with a resource-rank bound on total degree.
- Q8. Compile a published small universal system end to end.** Instantiate the exact matrices of an independently checked universal maximal-parallel system, record $d, m, H, \rho(A)$, and export its concrete fixed-horizon polynomial. The published 23-rule construction is an interesting target, but its rule count should not be mistaken for a certificate-size optimum or a new minimality claim.
- Q9. Add dynamic membranes without witness multiplicity.** Fresh membrane identifiers and symmetries can multiply certificates for one physical computation. A canonical birth-order treatment might combine with the repository’s heap and fresh-name work, but division and simultaneous object transport require a precise, model-specific semantics. A result should distinguish histories modulo renaming from histories with persistent labels.
- Q10. Find restricted history-compression theorems.** For systems with ultimately periodic, affine, or otherwise explicitly summarizable evolution, determine when a variable horizon can be compressed with unique witnesses. The positive-cubic obstruction for 2^T is a useful negative test. A theorem for a special class must not be promoted to a fixed-arity single-fold theorem for universal computation.
- Q11. Prove the compiler in Lean or Rocq.** Start with the exact comparator, then the shared-threshold one-round bijection and its size formula, then history composition. The rank bound and its integer-kernel sharpness should be separate theorems rather than properties inferred from experiments. The code and exported example in this package provide small regression cases for such a formalization.

16 Conclusion

Maximal parallelism introduces a negative local condition: every unused rule must lack a resource. Shared canonical threshold comparisons encode that condition at degree two without a population bound. This gives a concrete, witness-faithful continuation of the repository’s Diophantine compilation program rather than another abstract invocation of MRDP.

The same semantics has a sharp quantitative structure. The resource-cover rank determines the

optimal worst-case one-round exponent for every consumption matrix. At fixed horizon, established Presburger counting combined with that rank bound yields controlled quasipolynomial growth, with sharp examples. Low degree does not make counting easy, and it does not guarantee convexity: the cooperative deadlock obstruction is already present in one reaction. Finally, numerical program parameters and unbounded time expose distinct uniformity barriers. The quartic threshold and the explicit limitations on witness compression keep those barriers visible.

Part II

Total quadratic semantics for irreversible computation

[write] Part II prints source 12 (batch-78 manuscript 12). Its Section n is Section $n+16$ here (Sections 17–29), and its Theorem $n.m$ is Theorem $(n+16).m$; its appendices A–B are Appendices D–E.

Total Quadratic Semantics for Irreversible Computation

Exclusive sites, timed self-assembly, and Horn closure without execution histories

Research prepared for Vladimir Reshetnikov. Developed with ChatGPT. 2 October 2026.

Abstract

We construct an explicit, history-free Diophantine semantics for finite irreversible systems in which each site acquires at most one of several competing labels. A rule becomes available after all its typed prerequisites have appeared; its positive integer delay determines its completion time. The earliest completion fills the site, with a fixed label order resolving ties. Cycles, incompatible prerequisites, and sites that never activate are allowed.

For every fixed rule catalogue we emit one polynomial, nonnegative on the entire real nonnegative orthant and of total degree at most two, whose natural-number zero is unique for every assignment of the delay parameters. That zero records the complete outcome, including permanent nonactivation within the finite system. The number of witnesses is

$$m(2q + 3) + 3D + 3(L + R_0) + R,$$

where m counts nonseed sites, q labels, D distinct nonseed prerequisite literals, L prerequisite incidences, R_0 empty-body rules, and R all rules. Delays are parameters, not time-unrolling bounds. A single-label specialization uses only $m + 3(L + R_0) + R$ witnesses.

We prove causal uniqueness, exact allocation and height bounds, a finite polyhedral timing decomposition, and an exponential lower bound on explicit outcome-region lists. A four-tile counterexample shows that directedness does not justify replacing exclusive occupancy by independent typed facts. Applications cover seeded singleton-tile assembly, irreversible threshold growth, and finite groundings of Turing-complete Horn programs. The unbounded boundary is identified precisely: positive finite certificates persist, whereas universal permanent absence cannot have an existential Diophantine definition.

Status. Mathematical proofs and executable finite checks are included. The contribution is the specified compiler and its guarantees, not a claimed resolution of the general finite-fold problem or a new smallest universal polynomial. Historical priority has not been certified; no Lean build or independent peer review is claimed.

17 Research target and relationship to prior work

17.1 The question addressed

A computation can have many histories but only one completed outcome. In self-assembly, the histories differ in attachment order. In an irreversible inference system, they differ in the order of

discovering consequences. When labels compete for the same site, the outcome also depends on timing. A useful arithmetic representation should not pay for an entire tableau when a small causal object already determines the answer.

The precise question studied here is the following.

Can an exclusive-site, positively timed irreversible system be represented by one low-degree polynomial whose *complete* natural witness is unique, even when the result contains blocked sites, and with a number of coordinates independent of the numerical delays?

The answer is yes for the model defined below. The polynomial is quadratic, not quartic, because complementarity products can be added directly to squares of affine residuals. The difficult semantic point is not the minimum/maximum gate. It is making a cyclic system of such gates describe exactly the actual causal outcome, while preserving exclusivity and preventing self-supported events.

17.2 What the repository already contains

The repository was inspected at commit

db3b377f0ae3ab8cb585df4eb4b859cbee5ba3c5.

Its Hilbert-tenth README reports a 75-operation complete universal certificate and a separate 87-operation universal polynomial [8, 9]. Those figures concern fixed universal arithmetic constructions. They are not the size metric used in this paper.

More importantly, the merged report *Canonical Diophantine certificates* already includes manuscript 09, *Order-Free Diophantine Self-Assembly and a Sharp Positivity-Restricted Degree Threshold* [10]. Its README reports a convex quadratic with one natural zero per producible labelled assembly, first-arrival ranks, and a whole-grid terminality extension. It also reports degree-classification results for orthant-nonnegative representations. Thus neither “self-assembly is Diophantine” nor “self-assembly admits quadratic certificates” is presented here as a new result.

The extension pursued here is a *total, parameter-uniform outcome compiler*: competing labels are exclusive, all positive integer rule delays are inputs to the same polynomial, and every finite instance has exactly one complete zero whether or not a queried site activates. Exact formulas include the costs of label selection, inactive prerequisites, and the variable sentinel. The smaller single-label construction is provided as a transparent specialization and an independently proved allocation, not as a claim of priority over the earlier first-arrival construction.

The audit used the repository READMEs and source listings. The large merged article.tex could not be retrieved in full through the available connector. Consequently, this paper does not assert an exhaustive nonduplication audit or a numerical improvement over manuscript 09’s allocation, whose interface differs from ours.

[write] “Manuscript 09” here is source 09 of *Canonical Diophantine certificates* (batch-62 manuscript 02), printed there as Part XI; it is not batch-78 manuscript 09. Its article, which source 12 could not retrieve, gives the following relations. Theorem 19.2 generalizes that Part’s two-cut characterization of first-arrival ranks (cdc:of:thm:ranks) to positive delays, exclusive labels and certified absence: with one label and unit delays the Horn times (53) are Part XI’s first-arrival ranks of a fixed candidate target. The mechanism differs (a min/max fold fixed point here, two support cuts there), and so does the object: here one

deterministic eager outcome per delay vector, with a nonconvex polynomial; there every producible labelled target, with a convex one. Lemma 23.2 is the union-closure step of `cdc:of:prop:antimatroid`; the batch serialization in Section 23 is `cdc:of:lem:closure`; the second half of Theorem 25.3 is a case of `cdc:bd:prop:nobound` and `cdc:of:prop:nocutoff`. The allocations remain incomparable, as the paragraph above says.

17.3 Classical ingredients and genuine scope

Algorithmic self-assembly’s computational universality is classical [11]. Least-fixed-point semantics for positive logic programs is also classical [12]. Minimum/maximum/plus descriptions of timed discrete-event systems have an extensive literature [14]. We do not rebrand these facts as new.

[write] Further credits that source 12 does not give. The one-label equation (53), $t_v = \min_e (d_e + \max_{u \in T_e} t_u)$, is an instance of Knuth’s grammar problem with superior functions, solved by his generalization of Dijkstra’s algorithm [28], equivalently a shortest-path problem in an AND–OR directed hypergraph [29]; the event algorithm of the subsection “Canonical event semantics” is such a label-setting algorithm, with the exclusive-site competition added. Linear-time forward chaining for propositional Horn formulas is due to Dowling and Gallier [30]. Max-plus models of timed event graphs are standard [31].

Our results combine these ingredients with an explicit natural-number compiler, a proof of a unique *total* causal fixed point, a symbolic cap that does not raise degree, and exact allocation ledgers. The polyhedral timing and succinctness results describe this particular compiler’s semantic family. They do not settle general min–max dynamical systems, arbitrary reaction networks, or self-assembly with detachment.

The term *single-fold* will be used only relative to stated parameters and a fixed finite catalogue: each allowed parameter tuple has at most one complete natural witness, and the total compiler has exactly one. Varying the catalogue changes the polynomial and usually its arity. This distinction is essential when comparing with the general single-fold Diophantine representation question [17].

18 Exclusive-site irreversible systems

18.1 Sites, labels, rules, and delays

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. Fix a finite set V of sites and a label alphabet $[q] = \{1, \dots, q\}$, with $q \geq 1$. A site is initially either unoccupied or seeded with one label. Write $S_0 \subseteq V$ for the seeded sites and $m = |V \setminus S_0|$.

A *literal* is a pair (u, b) asserting that site u has label b . A rule e consists of a head (v, a) , a finite set T_e of prerequisite literals, and a delay $d_e \in \mathbb{Z}_{>0}$. Rule heads are nonseed sites. The same site can occur in several different literals in a body, including incompatible labels; such a body can never become satisfied. A body may also contain a literal at its own head site. No acyclicity assumption is made. Duplicate identical literals are removed during input validation.

[write] The shipped compiler rejects duplicate body literals instead of removing them, as Section 26 states; the research programme’s review noted the inconsistency. No theorem is affected.

The catalogue comprises the sites, alphabet, seeds, and rule heads and bodies. It is fixed when compiling the polynomial. In the timed version,

$$d_e = 1 + \delta_e, \quad \delta_e \in \mathbb{N} \quad (35)$$

are its numerical parameters. Glue tables, topology, and the rule catalogue are not silently treated as free uniform inputs.

18.2 Canonical event semantics

Seed labels appear at time zero and never change. If all literals in T_e have appeared, the rule offers its head label at time

$$c_e = d_e + \max\{t_u : (u, b) \in T_e\}, \quad \max \emptyset = 0. \quad (36)$$

An already occupied head ignores every offer. At an unoccupied site, the smallest offered completion time wins; among equal times, the smallest label wins. Offers to different sites at the same time are applied together.

Equivalently, repeatedly inspect all rules whose bodies are satisfied and whose heads remain vacant. Choose the minimum of their completion times, fill every head having an offer at that time, and choose its minimum label. Stop when there is no such rule. At most m sites are filled. Newly enabled rules have strictly later completion times than the events that enabled them, because $d_e > 0$. Thus simultaneous events cannot depend on each other, and the algorithm is well defined.

The output is a label $\lambda_v \in \{0, 1, \dots, q\}$ and a time $t_v \in \mathbb{N} \cup \{\infty\}$ for every site. Label zero means that the site never becomes occupied. This is a total function of the delay vector. It is not an enumeration of all asynchronous resolutions of a nondirected system.

Example 18.1 (A tie and a blocked cycle). Two empty-body rules for $(v, 1)$ and $(v, 2)$ with delays 2 and 2 fill v with label 1 at time 2. Rules $(u, 1) \rightarrow (w, 1)$ and $(w, 1) \rightarrow (u, 1)$ with positive delays and no external source activate neither site. Combining the two examples does not create any ambiguity: the tie is resolved and the cycle remains blocked.

18.3 Which physical assumptions are being made?

This model describes irreversible, one-time acquisition of labels. Prerequisites persist once true. It has no consumption, detachment, inhibition by a newly arriving neighbor, negative glue contribution, stochastic attachment time, or repeated firing at the same site. A pending offer need not be represented separately because a satisfied body stays satisfied. The only cancellation is occupation of its head.

Integer delays are a mathematical timing interface, not a claim that molecular waiting times are deterministic integers. For ordinary aTAM applications we will use unit delays and explicitly identify the eager schedule being represented. The distinction from models permitting prebuilt multi-tile attachments is substantive, not cosmetic; their verification problems can be harder [16].

19 Finite causality and a unique capped fixed point

19.1 A bound that does not unroll time

Let E be the rule set, $R = |E|$, and define

$$S = \sum_{e \in E} d_e, \quad M = S + 1. \quad (37)$$

If all delays equal one, the smaller choice $M = m + 1$ is sufficient.

Lemma 19.1 (Critical-chain bound). *Every finite nonseed activation time is at most S . With unit delays it is at most m . More precisely, it is a sum of the delays of distinct rules whose heads are distinct sites.*

Proof. Choose a winning rule at the activated site. If its body is empty, its activation time is that rule's delay. Otherwise choose a prerequisite with maximum activation time. If that prerequisite is a seed, stop; otherwise choose one of its winning rules and continue. Each step strictly decreases activation time. Moreover, a site cannot recur: it has one activation time, so recurrence would contradict strict decrease. Hence the chain contains at most m nonseed sites, and its rules are distinct. Unwinding the equalities in (36) expresses the original time as the sum of the chain's delays, with a zero seed or empty-body base. The asserted bounds follow. $\square$

The integer M is an arithmetic sentinel for infinity. It is not an external stopping time and does not cause M copies of a transition relation to be emitted. Its bit length, rather than its value, will control witness height.

19.2 Effective prerequisite times

Use M in place of ∞ . A proposed complete output consists of $\lambda_v \in \{0, \dots, q\}$ and $t_v \in \{0, \dots, M\}$, with seeds fixed, subject to $t_v = M$ precisely when $\lambda_v = 0$. For a literal (u, b) put

$$f_{u,b} = \begin{cases} t_u, & \lambda_u = b, \\ M, & \lambda_u \neq b. \end{cases} \quad (38)$$

An incompatible final label is therefore not allowed to act as a prerequisite, even if the site was occupied early.

Define the capped offered time for a label by

$$\theta_{v,a} = \min \left(M, \min_{e: \text{head}(e)=(v,a)} \left[d_e + \max_{(u,b) \in T_e} f_{u,b} \right] \right). \quad (39)$$

An inner minimum over no rules contributes no candidate, so $\theta_{v,a} = M$ if no rule has that head. The completed site is selected by lexicographically minimizing the pairs

$$(M, 0), \quad (\theta_{v,1}, 1), \dots, (\theta_{v,q}, q). \quad (40)$$

Thus absence wins when every candidate is at the sentinel; otherwise the smallest genuine time wins, followed by label order.

Theorem 19.2 (Total causal fixed point). *With M as in (37), or $M = m + 1$ for unit delays, (38)–(40) have exactly one proposed complete output. It is the canonical event outcome, with every infinite time replaced by M .*

Proof. First consider the actual event outcome. Every actual finite time is strictly below M by Lemma 19.1. A winning rule at a site therefore contributes its actual winning time to (39). Any competing rule whose final body is satisfied and whose offer is strictly earlier would already have been enabled before the winning activation; the event algorithm would have chosen it. The same argument excludes an equal-time offer with a smaller label. A rule whose body is not satisfied has an effective prerequisite equal to M and therefore cannot compete below M . If a site is absent at termination, no rule with that head has its body satisfied: otherwise the algorithm would not have stopped. Consequently every capped candidate there equals M . This proves existence.

For uniqueness, let a proposed output satisfy the equations. Every nonseed finite time is positive: a finite candidate is the sum of a positive delay and nonnegative prerequisite times. We compare its events with the actual events by induction on integer time $r < M$.

Assume agreement on all events before r . For any site not already occupied, an offer at time at most r must have every prerequisite at a time strictly smaller than r , since its delay is positive. A wrong-label prerequisite contributes M and cannot occur in such an offer. By the induction hypothesis, the offers at times at most r are therefore identical in the proposed and actual descriptions. Minimizing first time and then label makes their new events at r identical. The base $r = 0$ contains only the fixed seeds. This establishes agreement at every finite time below M . Both descriptions have no other finite times: the proposed one by definition of its capped candidates, and the actual one by Lemma 19.1. All remaining sites are absent in both. $\square$

Remark 19.3 (Why positive delay is structural). Allow delay zero and take one unseeded site with the rule $(v, 1) \rightarrow (v, 1)$. With cap $M = 1$, both absence and the unsupported event $(\lambda_v, t_v) = (1, 0)$ satisfy the capped equations. The actual empty execution only produces absence. Positivity is exactly what makes “my prerequisites happened before me” a well-founded argument.

20 Quadratic arithmetic without ambiguous auxiliaries

20.1 Nonnegative penalties

Our emitted polynomial has the form

$$P = \sum_{A \in \mathcal{E}} A^2 + \sum_{(B, C) \in \mathcal{C}} BC, \quad (41)$$

where all A, B, C are affine integer expressions and every factor B, C has nonnegative coefficients and a nonnegative constant. Hence $P \geq 0$ on the entire real nonnegative orthant. On that domain, $P = 0$ is equivalent to the vanishing of every summand.

The affine residuals need not be nonnegative. Products in the second sum are *not squared*. Squaring a quadratic complementarity residual would unnecessarily increase the total degree to four. The natural domain is part of the theorem, not an implicit convenience.

20.2 Minimum and maximum gates

For natural inputs x, y , output z , and slacks a, b , define

$$Q_{\min}(x, y; z, a, b) = (x - z - a)^2 + (y - z - b)^2 + ab, \quad (42)$$

$$Q_{\max}(x, y; z, a, b) = (z - x - a)^2 + (z - y - b)^2 + ab. \quad (43)$$

Lemma 20.1 (Unique gates). *For each $x, y \in \mathbb{N}$, (42) has exactly one natural zero in (z, a, b) , with $z = \min(x, y)$. Equation (43) has exactly one, with $z = \max(x, y)$.*

Proof. For the minimum, zero implies $x = z + a$, $y = z + b$, and $ab = 0$. Thus $z \leq x, y$ and equals at least one input; it is their minimum. The two slacks are then forced. Conversely these values satisfy all three conditions. The maximum proof reverses the two inequalities. Equal inputs force both slacks to zero, so ties introduce no witness multiplicity. $\square$

[write] Part I's Boolean gate (6) is the minimum gate applied to two bits, with a complement bit added (Section 1.4).

A fold of $k \geq 1$ inputs uses $k - 1$ gates. An empty maximum is the constant zero. These gates accept affine input expressions; when embedded in the compiler, their actual values are nonnegative.

20.3 A symbolic-cap typed-prerequisite gate

Let $\ell_0, \dots, \ell_q \in \mathbb{N}$ satisfy

$$\sum_{j=0}^q \ell_j = 1. \quad (44)$$

Exactly one label is selected; separate Boolean equations are unnecessary. For a requested label $b \geq 1$, introduce natural variables f, a, h and add the penalty

$$(f - t - a)^2 + (M - f - h)^2 + \ell_b a + \left(\sum_{j \neq b} \ell_j \right) h. \quad (45)$$

Lemma 20.2 (Unique effective-time selector). *Assume $0 \leq t \leq M$. Under (44), the unique natural zero of (45) is*

$$f = t, \quad a = 0, \quad h = M - t \quad \text{if } \ell_b = 1,$$

and

$$f = M, \quad a = M - t, \quad h = 0 \quad \text{otherwise.}$$

Proof. The first two squares impose $t \leq f \leq M$. If b is selected, the product $\ell_b a$ forces $a = 0$, hence $f = t$. If another label is selected, the other product forces $h = 0$, hence $f = M$. All remaining values follow from the affine equations. $\square$

The point of (45) is that M may itself be a variable. Writing $f = \ell_b t + (1 - \ell_b)M$ and squaring its residual would be quartic. The two affine bounds plus complementary slacks retain total degree two uniformly in the delay parameters. Writing $(1 - \ell_b)h$ would be nonnegative only after imposing one-hotness; the sum over the other labels is syntactically nonnegative even away from the zero set.

20.4 Selecting a winner without multiplying time by a selector

Set $B = q + 1$. For a site define the offered integer costs

$$K_0 = BM, \quad K_a = B\theta_a + a \quad (1 \leq a \leq q). \quad (46)$$

For output time t and one-hot labels, define an affine selected cost

$$K = Bt + \sum_{a=0}^q a\ell_a. \quad (47)$$

Introduce $q + 1$ natural gaps $g_0, \dots, g_q$ and add

$$\sum_{a=0}^q (K_a - K - g_a)^2 + \sum_{a=0}^q \ell_a g_a. \quad (48)$$

Lemma 20.3 (Unique winner). *Suppose $M, \theta_1, \dots, \theta_q \in \mathbb{N}$ and $\theta_a \leq M$. Together with (44), the zero condition (48) uniquely determines $t, \ell_0, \dots, \ell_q$ and all gaps. It gives exactly (40). In particular, $t \leq M$, and $\ell_0 = 1$ if and only if $t = M$.*

Proof. Let j be the selected label. The nonnegative product $\ell_j g_j$ forces $g_j = 0$, and the squares give $K = K_j \leq K_a$ for every a . The costs have pairwise different residues modulo B , so their minimum is unique. For $j \neq 0$, the equality $Bt + j = B\theta_j + j$ gives $t = \theta_j$. If $\theta_j = M$, its cost exceeds K_0 , which is impossible; thus $t < M$. For $j = 0$, $Bt = BM$ gives $t = M$. Conversely the smallest cost gives a one-hot choice and natural gaps $g_a = K_a - K$. The integer key order agrees with lexicographic order: a one-unit increase of time costs B , exceeding every label difference. $\square$

This last integrality observation matters. A key $B\theta + a$ would not encode lexicographic priority correctly for arbitrary real times whose differences could be smaller than $1/B$. Our separate real timing theorem will use the actual lexicographic event semantics, not a claim that the same integer key represents arbitrary real delays.

21 The total quadratic compiler

21.1 Construction and exact allocation

Let D be the number of distinct prerequisite literals at nonseed sites that occur anywhere in the catalogue. Let

$$L = \sum_{e \in E} |T_e|, \quad R_0 = |\{e : T_e = \emptyset\}|, \quad G = L + R_0.$$

Every site uses the common alphabet of q labels; unused labels are allowed.

The compiler first creates a sparse cap circuit. Order the R rules and introduce natural prefix variables

$$c_0 = 1, \quad c_i = c_{i-1} + 1 + \delta_i \quad (1 \leq i \leq R), \quad M = c_R. \quad (49)$$

Only $c_1, \dots, c_R$ are witnesses. If $R = 0$, $M = 1$ is a constant. The alternative unit-delay compiler uses $M = m + 1$ directly and has no prefix variables or delay parameters.

At each nonseed site allocate its time and $q + 1$ one-hot labels. For each of the D distinct nonseed prerequisite literals allocate the three variables of (45); reuse that effective time at all occurrences.

A matching seed literal has effective time zero and a mismatching seed literal has effective time M , with no auxiliary variables.

For each rule, fold a maximum over its effective body times and add its delay. For each head label with rules, fold the minimum over those offers and cap the result at M . Head labels with no rules use M directly. Finally, allocate $q + 1$ winner gaps at each nonseed site and use (48). Add every component into (41).

Lemma 21.1 (Exact gate count). *The specified folds use exactly $G = L + R_0$ minimum/maximum gates.*

Proof. Maximum folds use $\sum_e \max(|T_e| - 1, 0) = L - R + R_0$ gates. If r rules have a given head label, their minimum uses $r - 1$ gates and its cap uses one, for a total of r . Summing over the nonempty head-label groups gives R more gates. Their sum is $L + R_0$. $\square$

Theorem 21.2 (Exact total quadratic representation). *For every fixed catalogue, the above algorithm emits an integer polynomial $P(\delta; \mathbf{w})$ of total degree at most two, nonnegative on the entire real nonnegative orthant, with the following properties.*

For every $\delta \in \mathbb{N}^R$ there is exactly one $\mathbf{w} \in \mathbb{N}^W$ such that $P(\delta; \mathbf{w}) = 0$. Its semantic coordinates are precisely the complete canonical outcome. The exact number of natural witness coordinates is

$$W = m(2q + 3) + 3D + 3(L + R_0) + R. \quad (50)$$

The penalty description contains

$$A = R + m(q + 2) + 2D + 2(L + R_0) \quad (51)$$

affine squared residuals and

$$C = 2D + (L + R_0) + m(q + 1) \quad (52)$$

nonnegative product summands. The unit-delay compiler deletes the final R from (50) and the initial R from (51).

Proof. Every summand is of the form allowed by (41), including the prefix equations. Thus the polynomial is orthant-nonnegative and has total degree at most two, even when the delay parameters are counted as variables for the degree calculation.

At a natural zero the prefix variables have their unique intended values. All site labels are one-hot. The min/max gates force their exact intended values by Lemma 20.1. In particular each offered time is capped at M . Lemma 20.3 forces every site time to be at most M , so the hypothesis of Lemma 20.2 follows from the winner equations; it is not an extra unproved bound. The effective-time gates then have exactly the meaning of (38). The resulting semantic coordinates satisfy the capped equations and hence are the unique outcome by Theorem 19.2.

Conversely that outcome determines one-hot labels and times. It has $t_v \leq M$, so each effective-time gate has its unique natural extension. Every min/max fold has its unique extension, as does every winner gap. Together with the prefix values this gives a natural zero. There are no unused branch variables, arbitrary support choices, or unnormalized rank witnesses. Thus uniqueness holds for the entire witness tuple.

The site time, one-hot labels, and winner gaps contribute $m[1 + (q + 1) + (q + 1)] = m(2q + 3)$ variables. Typed effective times contribute $3D$, gates contribute $3G$, and prefixes contribute R . The affine residuals are the R prefix equations, m one-hot equations, $m(q + 1)$ winner equations,

$2D$ effective-time equations, and $2G$ gate equations. The product counts follow from two per effective-time gate, one per min/max gate, and $q + 1$ per site. $\square$

[write] Added 2 October 2026. The research programme’s review (`incoming_convergence_review_808b.md`) proves a local reduction that is not implemented in the shipped compiler: in each min gate set $z = x - a$ and keep $y - x + a - b = 0$ with $ab = 0$ (for max, $z = x + a$), and in each typed selector set $f = t + a$. This removes $G + D$ witnesses and $G + D$ affine squares, giving $W' = m(2q + 3) + 2D + 2G + R$ and, for Horn closure, $m + 2G + R$, with degree and orthant nonnegativity unchanged. No operation saving is claimed. The counts printed in Theorem 21.2 are source 12’s, for its compiler as delivered.

21.2 Height, bit size, and sparse expansion

Theorem 21.3 (Height and description bounds). *At the unique zero, every witness is at most $(q + 1)(M + 1)$. Consequently each coordinate has bit length*

$$O(\log(q + 1) + \log(2 + \sum_e d_e)).$$

For bounded q , the emitted polynomial has a sparse expanded description of size $O(m + L + R)$; its arity and number of monomials do not depend on the values of the delays. For general q , an upper bound on the monomial count is $O(mq^2 + qD + L + R)$.

Proof. Site times, effective times, and their slacks are at most M . Each rule delay is at most $S = M - 1$. Hence a raw rule offer is at most $2M - 1$; all min/max outputs and gap slacks are bounded by that quantity. Prefixes are at most M . Every offered key is at most $(q + 1)M + q$, the selected key is nonnegative, and winner gaps are their differences. Since $q + 1 \geq 2$, all these bounds are below $(q + 1)(M + 1)$.

Each prefix and min/max residual has bounded affine support. The same is true of effective-time squares; their nonnegative label-sum products have $O(q)$ terms. The one-hot and winner residuals have $O(q)$ affine support, so squaring them costs $O(q^2)$ monomials per site after sharing like terms. All $q + 1$ winner residuals share the same selected-key square, and each other offered key has bounded support. Thus their union of monomials is $O(q^2)$ per site, not $O(q^3)$. Distinct coefficients may aggregate but do not create new monomials. The cap prefixes avoid a dense square of $\sum_e \delta_e$. Finally, $R \leq L + R_0$, except trivially when both are zero, so the claimed bounded-alphabet description is linear in the catalogue size. $\square$

Remark 21.4 (What the bound measures). A huge delay can make a coordinate huge in value, but costs only its binary bit length. The result is not an arithmetic-circuit lower bound, a universal-polynomial record, or a claim that arbitrary Diophantine feasibility is easy. For these particular polynomials the event algorithm constructs the unique zero. Catalogue compilation and event evaluation are polynomial-time operations in the explicit catalogue and binary delay input; a simple implementation scans every rule after each batch.

21.3 Queries and changes of domain

To certify that site v has label a , add $(\ell_{v,a} - 1)^2$ to P . To certify that it does not have that label, add $\ell_{v,a}^2$. Each query polynomial has zero or one natural zero. An “at least one of these sites has

the halt label” flag is the maximum of the corresponding Boolean coordinates, implemented by unique max gates. Fixing the flag to zero or one again preserves degree two and unique extension.

Both positive and negative finite queries are therefore represented. This is possible because the finite closure problem is decidable. It says nothing about the complement of unbounded halting.

Natural witnesses may be converted bijectively to strictly positive ones by replacing w_i with $z_i - 1$, $z_i \geq 1$. Degree and uniqueness are preserved. Over arbitrary integers the nonnegative-domain proof is not valid: for instance (42) with $x = 1, y = 2, z = 2, a = -1, b = 0$ vanishes but does not describe the minimum. Replacing each natural witness by a sum of four integer squares gives an integer-domain polynomial of degree at most four. It has finitely many integer witnesses for each fixed parameter vector, but generally not one. This elementary domain conversion is not a uniform finite-fold theorem for all computably enumerable sets.

22 Horn closure and irreversible threshold growth

22.1 A smaller one-label compiler

When $q = 1$, exclusivity between different labels disappears. Each site is just a fact, and rules are Horn implications $\bigwedge_{u \in T_e} u \Rightarrow v$. Seed facts are initially true. The ordinary forward closure is the set of all eventually activated facts; with delays, each fact also has its first derivation time.

For a nonseed fact introduce only t_v . Seed times are constant zero. Define

$$t_v = \min \left(M, \min_{e: \text{head}(e)=v} \left[d_e + \max_{u \in T_e} t_u \right] \right). \quad (53)$$

No one-hot variables, typed effective-time gates, or winner keys are needed. Represent the right side with the same folds, and equate it to t_v .

Theorem 22.1 (Total Horn certificate). *The complete timed Horn closure has a unique quadratic natural certificate with exactly*

$$W_H = m + 3(L + R_0) + R$$

witnesses. It has $R + m + 2(L + R_0)$ affine squared residuals and $L + R_0$ product summands. For unit delays, the R prefix witnesses and residuals are omitted. The unique value $t_v = M$ certifies that the fact is not in the closure.

Proof. A false prerequisite has time M , making any offer that uses it at least $M + 1$ before capping. Thus (53) is exactly the one-label specialization of the causal fixed point. Its uniqueness follows from the same time induction. There are m time variables, $3G$ gate variables, and optionally R cap-prefix variables. The remaining counts follow directly from the m defining equalities, $2G$ gate residuals, and the prefix equations. $\square$

The proof does not enumerate derivation trees. A fact supported by many proofs still has one time, and a minimum tie has one pair of zero-normalized slacks. Cycles with no seeded or empty-body entrance remain at the sentinel.

22.2 Fixed labelled assemblies and minimal sufficient supports

Suppose a finite set of lattice positions has already been assigned its candidate tile labels, including a fixed seed. Retain only attachments of those candidate labels. At each nonseed position, the relevant data are the nonnegative bond strengths to at most four candidate neighbors. A tile is enabled when the sum of contributions from already present neighbors reaches the positive temperature τ .

Create one Horn rule for every inclusion-minimal sufficient neighbor set. A set is sufficient if its bond sum is at least τ . Minimality removes redundant supports but does not change when the tile becomes enabled. The resulting closure contains exactly the candidate tiles that can be constructed while staying within the candidate assembly.

Lemma 22.2 (Four-neighbor incidence bound). *An antichain $\mathcal{A}$ of subsets of a four-element set satisfies $\sum_{A \in \mathcal{A}} |A| \leq 12$.*

Proof. Choose a uniformly random permutation of the four elements and consider its chain of initial segments. An antichain meets this chain in at most one member. A fixed k -element subset occurs as its k th initial segment with probability $1/\binom{4}{k}$. Taking expectations gives $\sum_{A \in \mathcal{A}} \binom{4}{|A|}^{-1} \leq 1$. Therefore

$$\sum_{A \in \mathcal{A}} |A| \leq \max_{0 \leq k \leq 4} \left(k \binom{4}{k} \right) \sum_{A \in \mathcal{A}} \binom{4}{|A|}^{-1} \leq 12.$$

The bound is attained by all six pairs, and also by all four triples. $\square$

Corollary 22.3 (A $38m$ candidate-productibility allocation). *For a fixed labelled candidate assembly with m nonseed positions, nonnegative glues, and $\tau > 0$, its complete restricted closure has a unique quadratic certificate with at most $37m$ witnesses. Requiring the entire candidate to be producible uses at most $38m$ witnesses, still with zero or one natural zero.*

Proof. Minimal sufficient supports form an antichain, so $L \leq 12m$. Positive temperature excludes empty sufficient supports; hence $R_0 = 0$. The unit-delay Horn compiler has $m + 3L \leq 37m$ variables. Set $M = m + 1$ and add one natural slack s_v and equation $t_v + s_v = M - 1$ at each nonseed position. This is feasible exactly when $t_v < M$, and determines s_v uniquely. All sites have finite rank exactly when the candidate has a legal attachment sequence, obtained by increasing rank. $\square$

Here the candidate labels and coordinates are fixed in the polynomial's construction. These are not the variable-label parameters counted in the repository's reported assembly compiler. Neither an improvement over its allocation nor a new general degree threshold is inferred.

23 Self-assembly with competing tile types

23.1 Compilation of a finite box

Consider a seeded singleton-tile assembly system with nonnegative matching glue strengths and a positive integer temperature. Fix a finite box B containing the seed. Each unseeded location is a site, and every tile type is a possible label.

For a head tile type a at v , enumerate subsets of at most four neighboring positions and assign a tile type to each selected position. Retain the typed subset if its matching bond sum to a is at least τ , and remove nonminimal sufficient subsets. This produces rules with at most four literals.

Literal choices at the same neighboring position are not placed together in a body. Seed literals incompatible with the actual seed can be deleted together with their impossible rules, or retained as inactive literals; both are sound.

With unit delays, the event semantics is an eager parallel attachment schedule with fixed tile-type priority. Every batch has a legal sequential serialization, because its members depend only on earlier batches and additional tiles cannot reduce nonnegative bond strength. Its final assembly is terminal *within the box*: any legal attachment at a remaining position would have a satisfied sufficient-support rule.

For a fixed tile set, catalogue size is linear in the box area. More explicitly, each head label has at most $\sum_{k=1}^4 \binom{4}{k} q^k = (q+1)^4 - 1$ typed support candidates before minimality pruning, a constant when q is fixed. The complete outcome then has a quadratic certificate of linear size in $|B|$.

23.2 A directed system that defeats occupancy-free relaxation

Proposition 23.1 (Directedness does not make labels independent). *There is a directed, temperature-two, four-tile seeded assembly system with a unique three-tile terminal assembly for which independent typed-fact closure derives a tile label that cannot occur at its proposed position.*

Proof. Use two matching glue names g, h , both of strength two. All unspecified glues have strength zero; horizontal glues are all zero. Tile types have north/south glues as follows:

Tile	North	South
S	g	0
A	h	g
C	0	h
B	h	0

Seed the single tile S at $(0,0)$. Initially only A can attach, at $p = (0,1)$. Then only C can attach, at $(0,2)$. There is no exposed positive glue supporting a further vacant attachment. Thus the system has one legal sequence and the unique terminal assembly S, A, C .

In an occupancy-free typed-fact relaxation, S implies A at p , that fact implies C above it, and C implies B below it through their matching h glues. The last implication derives B at the already occupied position p . It is a correct local glue condition but an impossible attachment. Directedness of the original system did not remove the need to track that occupancy. $\square$

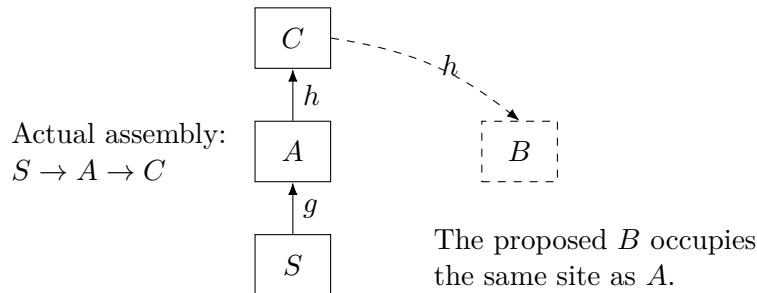


Figure 1: A later compatible bond is not a later available site. The dashed step is valid only in the unsound relaxation.

This is a counterexample to a particular proposed simplification, not a claimed defect in the existing repository’s occupancy-aware compiler. The package includes both a reduced three-rule instance and the literal four-tile rule catalogue restricted to the three vertical sites.

23.3 What directedness does permit

Call a seeded system *compatible* when every two finite producible assemblies agree wherever both are defined. This standard confluence property holds for directed aTAM systems in the usual setting.

Lemma 23.2 (Compatible union). *For nonnegative singleton attachments, the union of two compatible finite producible assemblies is producible. The statement also holds when all positions are restricted to a fixed box.*

Proof. Build the first assembly. Follow a legal sequence for the second, skipping tiles already present. A skipped tile has the same type by compatibility. At each unskipped step all supporting tiles from the second sequence are present, and extra tiles cannot decrease the bond sum. Thus every remaining step is legal. No new position outside the two assemblies is used. $\square$

[write] This is the union-closure step of `cdc:of:prop:antimatroid` in *Canonical Diophantine certificates*, Part XI, with the same skip-already-present argument; the batch serialization of Section 23 is its `cdc:of:lem:closure`.

Corollary 23.3 (Boxwise uniqueness and monotone exhaustion). *In a compatible system, every maximal finite-box assembly is the same. The canonical unit-delay compiler produces it. For nested boxes $B_1 \subseteq B_2$ containing the seed, these maximal assemblies are nested. Every finite producible assembly is contained in a sufficiently large box outcome.*

Proof. Two maximal assemblies in the same box have a producible union there; maximality forces equality. A maximal assembly in B_1 and one in B_2 are compatible, and their union is producible in B_2 , so maximality in B_2 gives inclusion. Finally any finite producer fits in some box; union with that box’s maximal outcome and maximality give containment. $\square$

The directedness claim here is not *intrinsic universality*. There is no implication that one directed two-dimensional system simulates the full dynamics of every directed system. The impossibility result for directed intrinsic universality is a different theorem [15].

24 Timing regions, succinctness, and nonmonotonicity

24.1 Finite polyhedral timing semantics

The event model also makes sense for strictly positive real delays. Its integer polynomial representation is not being extended to all reals; we study the semantic map separately in this subsection.

Theorem 24.1 (Finite timing decomposition). *For a fixed catalogue and seed, the positive real delay orthant has a finite partition into relatively open rational polyhedral cones such that on each*

part the output labels and absent sites are constant. Every finite activation time on a part is a homogeneous integer linear form

$$t_v = \sum_{e \in C_v} d_e$$

with C_v a set of distinct rules. In particular, the set of positive integer delay vectors giving any prescribed label outcome is effectively Presburger-definable and semilinear.

Proof. There are finitely many possible critical chains with distinct heads. Each contributes a linear form with coefficients zero or one. Include the zero form. By Lemma 19.1, whose strict-decrease proof also works for positive real delays, every finite event time is one of these forms. A currently enabled offer extends a critical chain to its vacant head, again with distinct heads, so it too belongs to this finite family.

Take the central hyperplane arrangement obtained from equality of every pair of these forms, and intersect all its relatively open faces with the positive orthant. This is a finite rational polyhedral partition. On each part every comparison between potential event times has a fixed sign, including equality. Start with the fixed seed. The earliest enabled time and the label tie rule consequently select the same sites and labels throughout the part. Induction over at most m successful site activations shows that the whole event signature is fixed. Choosing a fixed critical chain on each comparison face supplies the stated linear form for every finite time.

All faces can be enumerated effectively from the catalogue, although there can be many. Their integer points are described by integer linear inequalities and equalities; strict inequalities can be replaced by the corresponding one-unit inequalities on integer points. Taking the finite union for a specified output yields an effective Presburger formula. The classical equivalence between Presburger sets and semilinear sets [13] gives the last terminology; the explicit linear description is already enough for our applications. $\square$

[write] Second route for integer delays. Adding the query squares $(\ell_{v,a} - 1)^2$ and $\ell_{v,a}^2$ (subsection “Queries and changes of domain”) to the polynomial of Theorem 21.2 keeps the form (41); by Theorem cdc:of:thm:classification of *Canonical Diophantine certificates* ($D_2^+ = \text{SL}$), the set of integer delay vectors giving a prescribed label outcome is therefore semilinear. This recovers only the Presburger conclusion, not the real polyhedral partition or the linear forms of the theorem.

Remark 24.2 (Where affine terms reappear). Finite event times are homogeneous in the positive delay variables d_e . The sentinel $1 + \sum_e d_e$ is affine, not homogeneous. Rewriting $d_e = 1 + \delta_e$ also turns the event-time forms into affine expressions in δ_e . There is no contradiction between these statements.

Corollary 24.3 (Common scaling). *Multiplying every delay by the same positive scalar preserves all output labels and absent sites and scales every finite activation time by that scalar. For positive integer scaling, the semantic part of the quadratic certificate has this property, although its sentinel and auxiliary slacks need not scale homogeneously.*

24.2 Why listing all regions can be exponentially wasteful

Theorem 24.4 (An exponential explicit-outcome lower bound). *There is a family of catalogues with m sites and $2m$ empty-body rules whose timing map has 2^m distinct label outcomes. Any representation that explicitly lists separate constant-outcome regions needs at least 2^m nonempty*

entries. The total quadratic compiler for this family uses exactly $15m$ witnesses and $O(m)$ sparse monomials.

Proof. At site i , put two empty-body rules with labels 1 and 2 and delays x_i, y_i . Label 1 wins exactly when $x_i \leq y_i$; otherwise label 2 wins. The independent choices $(x_i, y_i) = (1, 2)$ or $(2, 1)$ realize every label vector. A region on which the outcome is constant cannot contain points with two different vectors, so at least 2^m entries are necessary for an explicit constant-outcome list. Here $q = 2$, $D = L = 0$, $R_0 = R = 2m$. Equation (50) gives $7m + 6m + 2m = 15m$. The sparse bound follows from Theorem 21.3. $\square$

This is an output-list lower bound, not a lower bound against every symbolic representation. A decision diagram, a program, or the polynomial can encode many outcomes without listing them. The theorem explains why a compact Diophantine formulation may be more useful than an expanded Presburger disjunction even when both define the same parameter set.

24.3 Slowing one rule can accelerate a downstream event

Proposition 24.5 (Unbounded nonmonotone timing effect). *For a fixed exclusive-site catalogue, increasing one delay can make a particular downstream activation arbitrarily many times faster, while keeping that downstream label unchanged.*

Proof. At site p , let empty-body rules offer labels A and B at times 2 and 3. A downstream site x acquires its sole label from $p : A$ with delay K , or from $p : B$ with delay 1. Initially p chooses A , so x activates at time $K + 2$. Increase only the A offer delay from 2 to 4. Now p chooses B at time 3, and x activates at time 4. The ratio $(K + 2)/4$ is unbounded as K grows. $\square$

Within a fixed critical-chain phase, finite times have nonnegative linear coefficients. Globally, changing the winning label changes the available rules, and monotonicity fails. Therefore a generic monotone min–max sensitivity argument cannot be imported into the exclusive-label model without checking the label phase. The example also distinguishes the model from ordinary positive Horn closure, which has no such label race.

25 Turing-complete substrates and the unbounded boundary

25.1 An explicit positive Horn encoding of machine computation

To make the computational boundary independent of an uninstantiated tile set, we give a direct positive Horn encoding. Fix a deterministic Turing machine with alphabet containing a blank symbol $\square$. Represent a configuration by

$$C_q(L, a, R),$$

where q is the state, a is the current symbol, and L, R are finite lists containing the tape symbols to the left and right, nearest first. An empty list denotes an all-blank unrepresented tail. Lists use constructors `nil` and `cons`.

For a right-moving transition $(q, a) \mapsto (q', b, \text{right})$, include

$$\begin{aligned} \text{Reach}(C_q(L, a, \text{cons}(c, R))) &\Rightarrow \text{Reach}(C_{q'}(\text{cons}(b, L), c, R)), \\ \text{Reach}(C_q(L, a, \text{nil})) &\Rightarrow \text{Reach}(C_{q'}(\text{cons}(b, L), \square, \text{nil})). \end{aligned}$$

For a left-moving transition include

$$\begin{aligned} \text{Reach}(C_q(\text{cons}(c, L), a, R)) &\Rightarrow \text{Reach}(C_{q'}(L, c, \text{cons}(b, R))), \\ \text{Reach}(C_q(\text{nil}, a, R)) &\Rightarrow \text{Reach}(C_{q'}(\text{nil}, \square, \text{cons}(b, R))). \end{aligned}$$

Stationary transitions, if used, simply replace q, a by q', b . For a halting state h , include

$$\text{Reach}(C_h(L, a, R)) \Rightarrow \text{Halt}.$$

The seed fact is the input configuration. Variables range over the indicated constructor terms. All clauses are positive and every rule has one body atom.

Proposition 25.1 (Exact Horn simulation). *The derived ground Reach atoms are exactly the machine configurations on its run, and Halt is derivable exactly when that run halts. A fixed universal machine therefore gives one fixed finite Horn program whose halting derivability problem is computably enumerable complete.*

Proof. Every ground transition clause performs exactly the designated tape operation, including the empty-tail blank case. Induction on the length of the run proves that every reached configuration is derivable. Conversely, induction on a finite derivation shows that every newly produced configuration is the prescribed successor of one already on the run. The only clauses with head Halt have a halting configuration as their premise. Fixing a universal machine and encoding a machine/input pair in its initial tape gives the usual halting reduction. $\square$

The representation is a Turing-complete computational substrate consisting only of monotone accumulation of structured facts. No register decrement or destructive rewrite is required; computational power comes from the unbounded term universe.

25.2 Every finite grounding has a total quadratic certificate

Choose a size bound b and enumerate a finite set of well-formed configuration terms, for example all configurations whose two list lengths sum to at most b , together with Halt. Include only ground rule instances all of whose atoms belong to this finite set. For sufficiently large b the initial configuration is included. This is a finite Horn catalogue and Theorem 22.1 applies.

Let $h_b \in \{0, 1\}$ indicate whether Halt belongs to its closure. The groundings may be chosen nested, so h_b is nondecreasing. A finite machine run uses only finitely many configurations of finite size. Thus

$$\text{machine halts} \iff \exists b \, h_b = 1. \tag{54}$$

Each individual closure and its negative answer $h_b = 0$ have unique quadratic natural certificates. Grounding cost is explicit: the finite universe can be exponentially large in its term-size bound. Neither term enumeration nor the changing number of sites is free.

Remark 25.2 (A three-witness presence flag). The optimized Horn compiler records only a time. For a queried fact with $0 \leq t \leq M$, introduce $b, h, s \in \mathbb{N}$ and add

$$(b + h - 1)^2 + (M - t - b - s)^2 + hs.$$

Its unique zero has $b = 1, h = 0, s = M - t - 1$ when $t < M$, and $b = 0, h = 1, s = 0$ when $t = M$. Indeed one-hotness forces b, h to be complementary Boolean values, and $hs = 0$ distinguishes the zero gap from a positive gap. Thus b is precisely the truth flag, with three additional witnesses and still total degree two. Fixing b to zero or one adds one affine square and no further witness. These three query coordinates are additional to the unqueried Horn count.

25.3 Why permanent absence is not existentially compressible

Theorem 25.3 (Permanent-absence obstruction). *For the fixed universal Horn program in Proposition 25.1, there is no single integer polynomial $Q(n, \mathbf{z})$ such that, for every valid coded input n ,*

$$\text{Halt is never derived on } n \iff \exists \mathbf{z} \in \mathbb{N}^k Q(n, \mathbf{z}) = 0,$$

where k is fixed. Nor is there a total computable function bounding a sufficient accepting grounding size in terms of the encoded input length.

Proof. Existential natural zeros of any fixed integer polynomial can be effectively enumerated, so such a Q would make nonhalting computably enumerable. Halting is also computably enumerable. Dovetailing the two enumerations would decide halting, a contradiction.

Suppose a computable function f guaranteed that every halting input of length n had $h_b = 1$ for some $b \leq f(n)$. Given an input, enumerate those finitely many groundings and compute their finite closures. A positive answer detects halting; exhaustion with no positive answer detects nonhalting. This is the same contradiction. The argument works for a bound depending computably on the entire input as well. $\square$

Corollary 25.4 (The size escapes every computable bound). *For every total computable function f , there is a halting input whose least accepting grounding size exceeds f of its input length. In fact there are infinitely many such inputs.*

Proof. If there were only finitely many exceptions, their finitely many minimal accepting sizes would have a maximum C . The computable bound $\max(f(n), C)$ would work for every halting input, contradicting the previous theorem. This is an existence argument; it does not compute C . $\square$

[write] The second half of Theorem 25.3 is a case of cdc:bd:prop:nobound of *Canonical Diophantine certificates*, whose list of specializations already includes tile systems, and of its Part XI form cdc:of:prop:nocutoff.

Equivalently, permanent absence is the universal statement $\forall b \ h_b = 0$, not one more finite closure query. Arithmetic coding can replace a changing-length witness by a finite sequence code, but preserving polynomial degree, fixed arity, and uniqueness through that coding is a separate theorem that has not been supplied here. The general single-fold or finite-fold Diophantine problem is not resolved by having a single-fold polynomial for every fixed finite size [17].

25.4 The same boundary in algorithmic self-assembly

Classical tile constructions simulate computation [11]. On any fixed directed universal computational tile construction with a suitable finite input seed and a designated halting output, Corollary 23.3 gives the geometric counterpart of (54): a halting tile occurs exactly when it occurs in some finite-box outcome. Every box has the total quadratic certificate of Section 23.¹ Nonoccurrence in *every* box is again a universal condition; a fixed existential polynomial for all nonhalting inputs would contradict the same computability argument.

The artifact does not instantiate a new universal numerical tile alphabet. Its literal tile example is the four-type occupancy counterexample. The universal Horn reduction is explicit above, while the aTAM universality transfer uses the classical computational substrate theorem. This keeps semantic correctness separate from a numerical universal compiler claim.

26 Implementation, exported polynomials, and tests

26.1 What the package actually implements

The file `code/quadratic_compiler.py` is a standard-library Python implementation using exact integers. It contains an event simulator, a sparse affine-expression class, the total exclusive-site compiler, an optimized Horn compiler, a canonical witness constructor, and an expanded polynomial exporter. No floating-point arithmetic, numerical optimization, or symbolic-system black box is used.

[write] In this report the programs are `code/12-total-quadratic-quadratic_compiler.py`, `code/12-total-quadratic-test_compiler.py` and `code/12-total-quadratic-verify_exports.py`, and every file of `results/` is `data/12-total-quadratic-*`. The test program rewrites its receipt and the four example files in place: run it only on a copy in the delivered layout (README). It was rerun at placement with Python 3.14.4: all checks passed, the four examples were reproduced apart from line endings, and the receipt differed only in its run time. The research programme’s review found that the compiler keeps references to its mutable inputs, returns aliased export lists and accepts a fractional seed label; its patch `total_quadratic_immutable_inputs.patch` changes no valid polynomial. The shipped file is unpatched.

The polynomial exporter retains both the structured penalty description and a fully expanded sparse monomial list. A monomial is represented by its integer coefficient and zero, one, or two variable names. All variable names, delay parameters, and a complete zero assignment are included in each exported example. The JSON representation is an explicit polynomial, not merely pseudocode for an unspecified arithmetic relation.

A separate `code/verify_exports.py` reconstructs each exported polynomial from its penalty data without importing the compiler, checks exact equality with the sparse monomial list, and verifies the supplied zero and that all of its coordinates are natural numbers.

The executable compiler accepts a fixed site count, label count, seed, and catalogue. It rejects repeated seed sites, invalid literals, rules whose heads are already seeded, duplicate body literals, and nonpositive or wrongly sized delay lists. Rules for seeded heads can be removed semantically before compilation; rejecting them keeps the interface unambiguous.

¹[write] Source 12 wrote “Section 7”, its own number for this section.

26.2 Examples and exact counts

Example	Parameters	Witnesses	Squares	Products	Monomials
Reduced locking example	0	29	18	13	126
Literal four-tile example	0	58	36	27	302
Timed race and blocked cycle	8	85	56	35	435
Four-rule huge-delay chain	4	45	30	18	206

All four expanded polynomials have total degree two. Their complete assignments evaluate to zero in exact arithmetic.

The huge-delay chain has four delays

$$10^{30} + 1, \quad 10^{30} + 3, \quad 10^{30} + 7, \quad 10^{30} + 9.$$

Its four nonseed activation times are respectively

$$10^{30} + 1, \quad 2 \cdot 10^{30} + 4, \quad 3 \cdot 10^{30} + 11, \quad 4 \cdot 10^{30} + 20.$$

The largest witness needs 103 bits, but the compiler still allocates 45 witnesses and 206 expanded monomials. The values of the parameters change; the polynomial does not. This is a direct demonstration of delay-uniform size rather than a benchmark against an optimized time-unrolled compiler.

In the timed-race example the output labels are $[1, 1, 1, 0, 0, 2]$ and the finite times are 0, 2, 5, 9 at sites 0, 1, 2, 5; sites 3 and 4 form an unseeded blocked cycle. Equal-time offers of labels 1 and 2 at site 1 select label 1. All these values are recorded in the exported JSON.

26.3 Finite verification receipts

Running

`python code/test_compiler.py`

regenerates the examples and `results/test_receipt.json`. The recorded run passed the following checks.

Check	Count
Min/max gate assignments examined	15,552
Typed effective-time assignments examined	5,868
Complete semantic assignments enumerated	68,885
Instances with exactly one enumerated semantic fixed point	1,059
General compiled instances checked	1,663
Single-coordinate witness mutations rejected	33,574
Expanded-polynomial evaluations compared with penalties	216
Reproducible random catalogues	300
Distinct outputs of the eight-site timing family	256
Four-neighbor weight/threshold policies checked	3,072
Optimized Horn/general compiler comparisons	320
Malformed inputs rejected	7

The semantic enumeration includes *all* subsets of the eight Horn rules on two unseeded one-label sites with arbitrary subset bodies. It also includes every catalogue of at most two distinct rules on two unseeded two-label sites with empty or singleton bodies, with every assignment of delays from $\{1, 2\}$. It enumerates candidate active time zero as well as positive times, so unsupported zero-time fixed points would not be missed.

Random tests use seed 20261002 and include incompatible labels in a body, self-dependencies, cycles, empty bodies, no-rule heads, and partly or fully seeded systems. A separate deliberate zero-delay self-rule exhibits the predicted two fixed points outside the model. An independent sparse monomial evaluator is compared against the structured penalty evaluator both at the intended zero and at off-zero natural tuples.

The mutation check changes one witness by $+1$ while keeping all other coordinates fixed. Rejecting all such changes is a useful sensitivity test, but is not a proof of global uniqueness. The unrestricted uniqueness claim rests on Theorems 19.2 and 21.2; the finite enumerations and mutations are regression tests. Likewise, the threshold enumeration supports but does not replace the antichain proof.

27 Integration into ProveIt

The main formalization target is small enough to isolate from a universal Diophantine backend. A natural dependency sequence is:

1. Define the finite catalogue, immutable seed, and event semantics. Prove strict causal dependence and the critical-chain bound without any polynomial syntax.
2. Define the capped output equations. Prove existence and uniqueness by the time-induction argument, explicitly distinguishing absence from an unproved high-rank event.
3. Prove the natural-number min/max, typed-prerequisite, and integer-key winner lemmas. Include the hypotheses $q \geq 1$, positive delays, and natural domains in the theorem statements.
4. Build an affine-penalty syntax with a theorem that every emitted product has nonnegative factors. Prove the zero-set equivalence between the assembled polynomial and its semantic equations.
5. Connect the list-based allocator to exact counts and an executable witness constructor. Only then connect to a general Diophantine predicate interface and add finite-box self-assembly adapters.

The Python compiler is not a trusted proof certificate until such a checker theorem is proved. Conversely, no large numerical solver should need to be trusted: the emitted polynomial and the witness can be evaluated by exact arithmetic, while the all-input soundness argument is structural.

A useful interface should expose four different theorems rather than one ambiguous “compiler correctness” statement: exact outcome decoding, existence for every legal delay vector, uniqueness of all auxiliaries, and the domain/size ledger. Optimizations can then specify whether they preserve the same tuples, provide a bijective coordinate change, or only preserve the outer accepted-input relation. The repository already treats such distinctions carefully in its research ledgers [8, 9]; this compiler should preserve that discipline.

28 Further research questions

28.1 Verified allocation and sharing

Can the exact witness formulas be proved together with a verified sparse polynomial generator in Lean? The highest-value first targets are the symbolic-cap selector and the winner lemma, because they protect against accidental quartic degree growth and inactive-branch multiplicity. A subsequent optimization should share identical maxima and repeated head offers while proving a full zero-set bijection, not merely counting fewer printed operations.

28.2 Smaller exclusive-label interfaces

The current allocation spends $2q + 3$ site coordinates plus typed-literal selectors. Can a binary label representation reduce dependence on q without introducing ambiguous comparison witnesses or increasing degree? The integer-key construction avoids one selector per rule, but a binary encoding requires exact equality-to-label tests. A useful answer must report coefficient sizes, monomial counts, and complete witness multiplicity, not only the number of semantic bits.

28.3 Parametric support selection

Here the topology and sufficient-support catalogue are fixed. For self-assembly, glue strengths and temperature are compiled into that catalogue before the polynomial is emitted. Can one make these values uniform numerical parameters, retain one natural root, and avoid exponentially enumerating support choices as neighborhood degree grows? For four neighbors the finite enumeration is harmless; unbounded fan-in requires a genuinely different mechanism.

28.4 All nondeterministic outcomes rather than one eager outcome

The present total compiler chooses a deterministic label-priority timing semantics. Representing all asynchronous terminal outcomes exactly once requires quotienting away attachment order without discarding causal competition. Existing order-free assembly work is the relevant starting point, not an assumption that our eager outcome enumerates all outcomes. A desirable theorem would count one zero per final assembly and specify how timing feasibility interacts with that quotient.

[write] The existing order-free work is Part XI of *Canonical Diophantine certificates*, whose convex compiler has one natural zero per producible labelled target (cdc:of:thm:compiler) and a whole-grid terminality extension (cdc:of:thm:terminal). What this question adds is the interaction with timing.

28.5 The sharp zero-delay and inhibition boundary

Zero delays create unsupported cyclic fixed points. One possible repair is to separate physical time from a strictly increasing causal rank. Can rank normalization be made unique without paying for a history? Negative glues and inhibitors are harder: a body can cease being enabled, so pending offers matter. Identify a maximal nonmonotone extension in which a finite outcome still determines a canonical compact certificate. Counterexamples should distinguish failed abstractions from impossible representations in principle.

28.6 Timing geometry without enumerating every phase

The phase theorem is constructive but its explicit list can be exponential. Can one compute a compact decision structure for a queried output, or certify robustness on an entire delay polytope, without expanding every phase? The slowdown/acceleration example shows that coordinatewise monotonicity cannot be assumed. A precise target is a certificate that a fixed output persists throughout a specified rational box of positive delays, with sharp complexity bounds.

28.7 Canonical small obstructions to finite activation

A false fact in a finite Horn closure is certified here by the entire unique rank vector. Is there a canonical minimal blocking object that can be smaller, while admitting a polynomial-time, unique-witness arithmetic description? Minimal cuts need not be unique, and choosing the lexicographically first cut can hide additional verification work. The useful objective is a provable size–height trade-off with an exact uniqueness theorem.

28.8 From finite boxes to fixed arity

For universal Horn computation, the finite-size hierarchy is explicit, and the least accepting size outruns every computable bound. What structure of a restricted class permits fixed-arity compression of this hierarchy while preserving finite or single fold? Candidate restrictions include bounded treewidth of dependency supports, ultimately periodic space use, and a fixed finite set of critical-chain shapes. Such results would need a new coding theorem; they cannot follow from writing $\exists b$ before a family of different polynomials.

28.9 Literal universal tile compilation

A concrete integration project is to choose a published directed computationally universal tile construction, freeze its exact numerical alphabet and input encoding, and run the finite-box compiler on it. This would replace the current classical aTAM transfer by an audited end-to-end implementation. It should not be confused with a directed intrinsic-universal simulator, whose stronger two-dimensional requirement has a known obstruction [15].

29 Conclusion

The central result is a total quadratic semantics, not merely an existential encoding of successful traces. Positive delays enforce causality; a finite sentinel handles absence; typed effective-time gates preserve exclusive occupancy; and integer winner keys remove tie multiplicity. These ingredients give an exact, delay-uniform allocation and one complete natural zero for every finite input instance.

The applications illustrate two complementary facts. Irreversible substrates can compute universally through an unbounded supply of sites or structured facts, while every fixed finite restriction can have an extremely rigid low-degree certificate. The difference between these statements is where the unbounded computational difficulty resides. The supplied proofs and executable sparse polynomials establish the finite side explicitly and identify, rather than conceal, the obstacle to universal negative or fixed-arity single-fold compression.

Part III

Quadratic Diophantine certificates for a fixed Waterfall universal machine

[write] Part III prints source 14 (batch-78 manuscript 14). Its Section n is Section $n+29$ here (Sections 30–38), and its Theorem $n.m$ is Theorem $(n+29).m$; its appendices A–B are Appendices F–G. Source 14 names no repository commit and did not consult the repository; the repository material it bears on is added in [write] notes (Sections 31, 34, 35 and 37).

Quadratic Diophantine certificates for a fixed Waterfall universal machine

Research note prepared with AI assistance. 2 October 2026.

[write] The delivered author line adds “for private review”; this public report prints it in Appendix H only.

Abstract

A fixed 46-clock Waterfall program due to Iijil simulates the binary 15-state universal Turing machine of Neary and Woods. We give an explicit two-half-tape input interface, prove its complete macrostep behavior by a finite exact affine certificate, and compile an externally fixed number k of Turing-machine steps into one quadratic Diophantine equation. The equation has exactly $35k$ natural witnesses, $5k + 4$ squared linear residuals, and $2k$ nonnegative complementarity terms. Its natural fiber is empty or a singleton and records both the exact prehalt clock-firing count and the physical halt timestamp. The certificate eliminates every repetition internal to a Waterfall macrostep, including repetitions exponential in input bit length. A separate decidable common-column class admits horizon-free certificates. All scope distinctions are explicit: the universal machine is prior work, the witness dimension grows with k , and no fixed-arity representation of unbounded universal computation or universal arithmetic-operation record is claimed. An exact-integer replay package accompanies the note.

30 Results and scope

The Waterfall Model is an esoteric counter substrate in which all counters decrease together and a counter reaching zero applies a fixed nonnegative increment vector. Its lack of an instruction pointer makes its event sequence an interesting arithmetic object. The direct question here is whether its long runs admit explicit Diophantine certificates that exploit the substrate’s structure rather than merely recording every event.

The main result is a *bounded-macrostep* answer. The fixed machine, its Turing-machine source, and the compiler from which it comes are prior work of Iijil [19, 20, 21]; the underlying universal Turing machine is due to Neary and Woods [18]. The contribution of this note is the explicit input contract and its symbolic verification, the compressed arithmetic certificates, and the proofs and replay artifacts tying them together. Priority of this particular arithmetic presentation has not been established by an exhaustive literature search.

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$. State labels $A, \dots, O$ have numeric codes $0, \dots, 14$. The head symbol has code 0 or 1. The finite-support half tapes, read away from the head with least significant digit first, are natural integers L, R ; the current head symbol is stored separately.

Theorem 30.1 (Main result). *There is a fixed 46×46 nonnegative integer Waterfall increment matrix M , with entries at most 12, and an explicitly given canonical initial vector $\kappa(A, 0, L, R)$ with the following properties.*

- (i) *Every $L, R \in \mathbb{N}$ gives a tie-free run up to and including its possible halt. The program exactly simulates the Neary–Woods 15-state binary machine, with both half tapes supplied as raw integers. Computing its two varying clock values takes four binary arithmetic operations.*
- (ii) *A Turing-machine step that pops $X = 2Q + r$, $r \in \{0, 1\}$, and pushes onto half tape Y uses exactly*

$$c = 6 + 3Q + r + 3Y$$

nonhalting Waterfall firings. Consecutive transition-event timestamps differ by $2c + 7$.

- (iii) *For every externally fixed $k \geq 1$, an explicitly constructible integer polynomial*

$$F_k(L_0, R_0, C, \tau; \mathbf{z}), \quad \mathbf{z} \in \mathbb{N}^{35k},$$

of total degree two vanishes exactly when this run first halts after k Turing-machine steps, after C nonhalting Waterfall firings, at timestamp τ . For fixed parameters, there is at most one natural witness tuple.

The numerical size of a tape value does not increase the number of macrostep witnesses. The number of Turing-machine steps does: k is compiler data specifying a polynomial and its arity, not one more quantified variable in a fixed polynomial. This distinction is central. In particular, the theorem does not supply a universal fixed-arity quadratic Diophantine equation.

The note first fixes the semantics and complete machine data, then proves the macrostep and arithmetic theorems. Section 35 explains the finite-horizon and semilinearity boundaries. Section 36 treats a different, decidable family where a genuinely horizon-free certificate is available. Appendix F specifies every matrix entry compactly; Appendix G records the finite affine certificate behind the all-input proof.

31 The fixed substrate and its input contract

31.1 Deadlines and strict minimum semantics

There are n actual clocks, indexed by i , and one distinguished halt clock h . The column $M_{\cdot i}$ is the increment vector for firing clock i . Assume $M_{ji} \geq 0$, $M_{ii} > 0$ for $i \neq h$, and $M_{\cdot h} = 0$. Starting from an integer vector $a > 0$, let p_i count previous nonhalting firings. Define the vector of absolute deadlines by

$$d = a + Mp. \tag{55}$$

The next event is the unique minimum coordinate of d . A minimum tie is undefined. When that coordinate is h , the computation halts *before* any update. Otherwise increment p_i by one, equivalently adding $M_{\cdot i}$ to d .

This is the integer version of the standard simultaneous-decrement semantics [22]. Indeed, at physical time t with current clock values v_i , each deadline is $t + v_i$. Advancing time to the first zero changes neither deadline; adding a trigger vector changes deadlines by that same vector. Induction proves (55). After a legal nonhalting event the selected deadline strictly increases, all others are still larger than the old minimum, and no deadline decreases. Thus successive event times are strictly increasing integers; infinitely many events cannot accumulate in finite time. We keep physical timestamps even though an implementation may accelerate idle intervals.

The source file uses the opposite matrix orientation: each trigger is a *row*. It has 47 rows and columns because the first row and column carry serialization metadata and initial values. The mathematical matrix M is the transpose of its 46×46 trigger submatrix. The metadata coordinate is never treated as a clock here.

31.2 Clock names and the source machine

For each direction $D \in \{L, R\}$ there are eight auxiliary clocks:

$$DTape, DTemp, DTrans, DDiv_0, DDiv_1, DMult, DWrite_0, DWrite_1.$$

There are also thirty control clocks $Trans_{q,s}$, one for each state-symbol pair. These are distinct from the two auxiliary clocks $DTrans$. Their total is $16 + 30 = 46$.

Table 4 gives the complete source transition function. An entry (w, D, q') means write w , move in direction D , and enter state q' . The sole missing instruction is $(J, 1)$; its clock, one-based number 28 in the source order, is the halt clock. The matrix and its complete compact definition in Appendix F agree in all 2,116 entries.

Table 4: The fixed source machine. State codes run from $A = 0$ to $O = 14$.

State	Read 0	Read 1	State	Read 0	Read 1
A	$(0, R, B)$	$(1, R, A)$	I	$(0, R, A)$	$(1, L, J)$
B	$(1, R, C)$	$(1, R, A)$	J	$(1, L, K)$	undefined
C	$(0, L, G)$	$(0, L, E)$	K	$(0, R, L)$	$(1, R, N)$
D	$(0, L, F)$	$(1, L, E)$	L	$(0, R, M)$	$(1, R, L)$
E	$(1, R, A)$	$(1, L, D)$	M	$(0, L, B)$	$(1, R, L)$
F	$(1, L, D)$	$(1, L, D)$	N	$(0, L, C)$	$(0, R, O)$
G	$(0, L, H)$	$(1, L, G)$	O	$(0, R, N)$	$(1, R, N)$
H	$(1, L, I)$	$(1, L, G)$			

In Neary–Woods notation, $u_1, \dots, u_{15}$ correspond to $A, \dots, O$ and c, b to $0, 1$. The table is Table 16 on PDF page 17 of the author-hosted primary paper. Its missing (u_{10}, b) transition agrees with the final displayed halting configuration on PDF page 19. The following prose instead says c , a local inconsistency resolved here in favor of the table and displayed configuration. In particular, the u_{15}, b target is u_{14} , as in the primary table.

[write] The repository’s research note `Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/neary_woods_explicit_universal_tm.md`, added on 1 October 2026, a day before source 14, transcribes the same Table 16 for its own universal polynomial and reports the same inconsistency between Table 16’s missing (u_{10}, b) and the prose of Neary and Woods’s Section 3.5. Table 4 agrees with that transcription entry by entry (states $u_i \mapsto$ the i th letter, $c \mapsto 0$, $b \mapsto 1$). Source 14 did not know the note; the erratum is printed once, here, and the two transcriptions were made independently.

31.3 Canonical states and a four-operation loader

Define a vector $\kappa(q, s, L, R)$ of normalized deadlines by

$$\kappa_{LTape} = 2 + 2L, \quad \kappa_{RTape} = 2 + 2R, \quad (56)$$

$$\kappa_{DDiv_j} = 2 + \mathbf{1}_{\{j \neq s\}}, \quad D \in \{L, R\}, j \in \{0, 1\}, \quad (57)$$

$$\kappa_{Trans_{u,t}} = 1 + \mathbf{1}_{\{u \neq q\}} + \mathbf{1}_{\{t \neq s\}}. \quad (58)$$

Every other coordinate is 2. Exactly one coordinate has value 1, namely the next control clock $Trans_{q,s}$.

The universal initial condition is $q = A, s = 0$. All initial values except the two in (56) are fixed. Loading raw L, R therefore costs two multiplications by the fixed constant 2 and two additions: four operations in the binary $+, -, \times$ model, where constants and copying are free. This is a statement about the *interface*, not the cost of encoding an arbitrary external program convention as L, R .

If a syntactically valid source serialization is desired, set its upper-left metadata bound to

$$H_{file} = 2(L + R) + 47 = (2 + 2L) + (2 + 2R) + 43.$$

It is strictly larger than every actual initial value, every trigger increment, and the metadata clock count 46. Two further additions suffice. Thus the operational loader has four operations and the full serialization loader has six.

The primary paper starts its universal machine in u_1 over the rightmost c of bc , with the encoded program to the left. Allowing *both* half tapes is therefore part of the input contract. Its finite input encoding yields such natural L, R effectively. Combining that universality result with the exact simulation proved below shows that halting on this fixed Waterfall matrix, as a set of pairs (L, R) , is computably enumerable complete. No small arithmetic cost is asserted for the separate program-to-half-tape encoding.

32 An exact macrostep theorem

A *canonical cut* is a deadline vector $\kappa(q, s, L, R) + t\mathbf{1}$ just before the next control-clock event; its event timestamp is $t + 1$. We also use this vector identity immediately after the last trigger of the preceding macrostep, when an idle interval can remain before that event.

Theorem 32.1 (Complete macrostep). *Suppose $(q, s) \mapsto (w, D, q')$ is defined in Table 4. Let O denote the opposite direction, X the half tape in direction D , and Y the other half tape. Write $X = 2Q + r$, with $Q \in \mathbb{N}$ and $r \in \{0, 1\}$. Starting at $\kappa(q, s, L, R)$, the exact firing word before the next control-clock event is*

$$\begin{aligned} & Trans_{q,s} (DDiv_0 \ DDiv_1)^Q (DDiv_0)^r DTape (DTrans)^Q DTemp \\ & (OMult)^Y OTape (OTrans)^{2Y} OTemp OWrite_w. \end{aligned} \quad (59)$$

Every selected clock is the unique minimum. The final deadline vector is

$$\kappa(q', r, L', R') + \Delta \mathbf{1}, \quad \Delta = 19 + 6Q + 2r + 6Y, \quad (60)$$

where the popped half becomes Q and the pushed half becomes $2Y + w$. The word has length

$$c = 6 + 3Q + r + 3Y, \quad \Delta = 2c + 7. \quad (61)$$

Zero exponents in (59) mean that the corresponding block is absent.

32.1 A finite proof that covers unbounded tape values

The only unbounded quantities in one macrostep are Q and Y . Their loop repetitions can be certified by affine inequalities, without enumerating their possible values.

Lemma 32.2 (Every-prefix endpoint criterion). *Let v be the clock-count vector of a fixed finite word w , and let u be the count vector of the prefix before one specified letter i of that word. Before that letter in repetition ℓ of w , the gap between competitor j and selected clock i is*

$$d_j - d_i + (Mu)_j - (Mu)_i + \ell((Mv)_j - (Mv)_i). \quad (62)$$

For $K \geq 1$, the repeated word w^K is a legal strict-minimum firing word if and only if, for every prefix position and every $j \neq i$, expression (62) is positive at $\ell = 0$ and $\ell = K - 1$.

Proof. A prescribed prefix determines its deadline vector by (55), whether or not it is legal. Expression (62) is its exact deadline difference. A real affine function of ℓ attains its minimum on $[0, K - 1]$ at an endpoint. Thus positivity of both endpoint values is equivalent to positivity at every integer iteration. Applying the criterion successively to every letter proves legality. Conversely, a legal repeated word necessarily passes its first- and last-iteration inequalities. $\square$

It is important that the criterion is applied at *every prefix position of each repeated block*. Checking only the beginning and end of a whole computation is insufficient; Section 37 gives an explicit counterexample to that different shortcut.

Proof of Theorem 32.1. Fix one of the 29 defined instructions and one of the two values of r . Substitute $X = 2Q + r$ and the appropriate placement of X, Y into the canonical vector. Process the blocks of (59) in order, adding the appropriate repeated trigger vectors to the deadline vector.

For a one-letter nonrepeated block, check all 45 competing clocks directly. For a block repeated Q times, apply Lemma 32.2 on its active domain $Q \geq 1$; the zero case contributes no event. Similarly, use $Y \geq 1$ for blocks repeated Y or $2Y$ times. Every endpoint gap has the exact form

$$c + aQ + bY.$$

After substituting $Q = Q_{\min} + q$ and $Y = Y_{\min} + y$, where the active-domain lower bounds are 0 or 1, each gap has a positive integer constant and nonnegative coefficients of q, y . Consequently every selected clock is uniquely minimal on the entire relevant natural domain.

For full reproducibility, the verifier checks 58 instruction-residue cases and 43,065 strict-minimum inequalities. There are 64 distinct affine forms when their active domains are retained; Appendix G records this finite certificate. The source code uses coefficient tuples and exact integers, and its final check subtracts the proposed target in (60) from *all 46 coordinates*. Every residual coefficient is zero. This supplies an explicit finite algebraic verification of the stated identity, rather than extrapolation from a finite input grid.

Finally, counting letters gives $1 + 2Q + r + 1 + Q + 1 + Y + 1 + 2Y + 1 + 1 = 6 + 3Q + r + 3Y$. The verified common shift is $19 + 6Q + 2r + 6Y = 2c + 7$. $\square$

A second implementation establishes the loop inequalities by a different parameterization: for an iteration $j < Q$ use $Q = j + 1 + u$; for $j < Y$ use $Y = j + 1 + u$; for $j < 2Y$ split $j = 2v + e$, $e \in \{0, 1\}$, and use $Y = v + 1 + u$. It proves nonnegativity of the resulting free-variable coefficients. This is an independent exact replay, not a proof-assistant formalization.

32.2 Halting time and an independent linear potential

The new canonical vector in (60) has its next control event at $1 + \Delta$. Thus Δ is a difference between *consecutive control-event timestamps*, not necessarily the time elapsed to the immediately post-write state. For instance, from $L = R = 0$ the first control event is at time 1, its final write is at time 10, and the next control event is at time 20.

Corollary 32.3. *If the source machine first halts after k executed instructions and the Waterfall run has made C nonhalting firings, its halt timestamp is*

$$\tau = 1 + 2C + 7k. \quad (63)$$

The final observation of the halt clock is not included in C .

Proof. The initial control event has timestamp 1. Apply Theorem 32.1 successively; each macrostep ends at a canonical cut with exactly the source-machine successor. Summing (61) gives (63). At $(J, 1)$ the next unique minimum is the zero-trigger halt clock, so no additional update occurs. $\square$

There is also a concise invariant checking the timing coefficients independently of the gap inequalities. Give the coordinates integer weights

$$\omega_i = \begin{cases} 288, & i = L\text{Trans or } R\text{Trans}, \\ -373, & i \text{ is a division clock}, \\ 970, & i \text{ is a multiplication clock}, \\ -4, & i \text{ is a write clock}, \\ 0, & \text{otherwise.} \end{cases} \quad (64)$$

Direct substitution in Appendix F gives

$$\omega \cdot \mathbf{1} = 1008, \quad \omega \cdot \kappa(q, s, L, R) = 1270, \quad (65)$$

$$\omega \cdot M_{\cdot i} = 2016 + 7056 \mathbf{1}_{\{i \text{ is a nonhalt control clock}\}} \quad (i \neq h). \quad (66)$$

The halt column has dot product zero. A prefix with C firings and k control firings therefore increases the potential by $2016C + 7056k$. If its endpoints are canonical vectors differing by common shift t , (65) makes that increase $1008t$. Hence $t = 2C + 7k$. This proves the same accounting identity; it does not by itself prove that a proposed event word is legal.

33 One quadratic equation for an exact TM horizon

We now use the macrostep semantics to eliminate all its internal repetition counts from the event-level trace. Fix $k \geq 1$. The four supplied parameters are $L_0, R_0, C, \tau \in \mathbb{N}$. They are not counted as witnesses.

33.1 Variables and linear expressions

Let $\mathcal{I}$ be the set of the 29 defined instructions in Table 4. Write instruction i as $(a_i, s_i) \mapsto (w_i, D_i, b_i)$, using numeric state codes. At each step $j = 0, \dots, k-1$, introduce 29 natural selectors e_{ji} and six natural direction-group variables

$$Q_{jL}, r_{jL}, Y_{jL}, \quad Q_{jR}, r_{jR}, Y_{jR}.$$

There are exactly $35k$ witnesses. Define the following *linear expressions*, which introduce no further variables:

$$A_j = \sum_{i \in \mathcal{I}} a_i e_{ji}, \quad S_j = \sum_{i \in \mathcal{I}} s_i e_{ji}, \quad B_j = \sum_{i \in \mathcal{I}} b_i e_{ji}, \quad (67)$$

$$E_{jD} = \sum_{i: D_i=D} e_{ji}, \quad W_{jD} = \sum_{i: D_i=D} w_i e_{ji}. \quad (68)$$

The input and output half-tape expressions are

$$I_{jL} = 2Q_{jL} + r_{jL} + Y_{jR}, \quad I_{jR} = Y_{jL} + 2Q_{jR} + r_{jR}, \quad (69)$$

$$O_{jL} = Q_{jL} + 2Y_{jR} + W_{jR}, \quad O_{jR} = 2Y_{jL} + W_{jL} + Q_{jR}. \quad (70)$$

When the move is left, the right-group triple is zero, so these equations read $(L, R) = (2Q + r, Y)$ and $(L', R') = (Q, 2Y + w)$. When the move is right, the left-group triple is zero, and they give the opposite placement. The polynomial will force precisely this behavior.

33.2 The polynomial in full

For each j define five linear residuals

$$H_j = \sum_{i \in \mathcal{I}} e_{ji} - 1, \quad (71)$$

$$U_j = A_j - \begin{cases} 0, & j = 0, \\ B_{j-1}, & j > 0, \end{cases} \quad V_j = S_j - \begin{cases} 0, & j = 0, \\ r_{j-1,L} + r_{j-1,R}, & j > 0, \end{cases} \quad (72)$$

$$X_j = I_{jL} - \begin{cases} L_0, & j = 0, \\ O_{j-1,L}, & j > 0, \end{cases} \quad Z_j = I_{jR} - \begin{cases} R_0, & j = 0, \\ O_{j-1,R}, & j > 0. \end{cases} \quad (73)$$

Use two quadratic complementarity terms per step:

$$K_{jL} = E_{jR}(Q_{jL} + r_{jL} + Y_{jL}), \quad K_{jR} = E_{jL}(Q_{jR} + r_{jR} + Y_{jR}). \quad (74)$$

Finally, define four linear terminal residuals

$$T_1 = B_{k-1} - 9, \quad T_2 = r_{k-1,L} + r_{k-1,R} - 1, \quad (75)$$

$$T_3 = C - 6k - \sum_{j=0}^{k-1} (3(Q_{jL} + Q_{jR} + Y_{jL} + Y_{jR}) + r_{jL} + r_{jR}), \quad (76)$$

$$T_4 = \tau - 1 - 2C - 7k. \quad (77)$$

The promised equation is

$$F_k = \sum_{j=0}^{k-1} (H_j^2 + U_j^2 + V_j^2 + X_j^2 + Z_j^2 + K_{jL} + K_{jR}) + T_1^2 + T_2^2 + T_3^2 + T_4^2 = 0. \quad (78)$$

All displayed abbreviations are fixed linear expressions and can be expanded mechanically. Thus (78) is an ordinary integer polynomial, not a formula using a minimum oracle, an execution predicate, variable-length products, or powers with variable exponents. It has $5k + 4$ squared linear residuals and $2k$ unsquared quadratic products. In particular, its total degree is two. The coefficient of τ^2 is 1, so this is exactly degree two.

33.3 Soundness completeness and uniqueness

Theorem 33.1 (Canonical natural certificate). *For every fixed $k \geq 1$ and natural parameters L_0, R_0, C, τ , the natural solutions of (78) are in bijection with the fixed machine's first halt after exactly k executed source instructions, with the stated count C and timestamp τ . The solution set is empty or a singleton.*

Proof. Nonnegativity without auxiliary assumptions. Every square is nonnegative. Every factor in (74) is a sum of natural variables with nonnegative coefficients, so every product is nonnegative on *all* natural assignments, even those failing the selector constraints. Therefore $F_k = 0$ forces every displayed square residual and every complementarity product to vanish separately. No cancellation between invalid constraints is possible.

One selected instruction. The equation $H_j = 0$ and naturalness force exactly one e_{ji} to equal 1 and the others to equal 0. Hence A_j, S_j, B_j are the current state, current bit, and next state of a defined instruction; exactly one of E_{jL}, E_{jR} is 1 and the other is 0. The appropriate W_{jD} is its written bit and the other write expression is zero.

The unused direction vanishes. If the selected instruction moves right, $E_{jR} = 1$, so $K_{jL} = 0$ gives $Q_{jL} + r_{jL} + Y_{jL} = 0$. All three are natural and are therefore zero. The left-move case similarly kills the entire right-group triple. Let Q_j, r_j, Y_j denote the remaining active triple.

The residue is genuinely binary. For $j < k - 1$, $V_{j+1} = 0$ identifies $r_{jL} + r_{jR}$ with the next selected instruction's source bit $S_{j+1} \in \{0, 1\}$. For $j = k - 1$, $T_2 = 0$ gives that sum as 1. Since the inactive residue is zero, the active r_j always belongs to $\{0, 1\}$. This is why no separate equations $r_j(r_j - 1) = 0$ are needed.

The tape update is forced. At step zero, $U_0 = V_0 = 0$ fixes state A and head bit 0. Equations $X_0 = Z_0 = 0$, together with (69), identify the directional half tape as $2Q_0 + r_0$ and the other half tape as Y_0 . Binary Euclidean division uniquely determines Q_0, r_0 from the given directional half tape, and the other half uniquely determines Y_0 . Equation (70) is then exactly the source-machine tape update.

Inductively, $U_j = V_j = 0$ fixes the next source instruction from the preceding selected rule and remainder, while $X_j = Z_j = 0$ identifies its incoming halves with the preceding output expressions. The same division argument forces the complete next active triple. Thus every step is the actual deterministic successor of the preceding one, with no extra tape witnesses to choose.

Exact first halt and counters. Every selected instruction is defined because $\mathcal{I}$ omits $(J, 1)$. An earlier halt cannot be passed over: its state-symbol pair would have no available selector at the next step. Equations $T_1 = T_2 = 0$ make the state and head bit after step $k - 1$ equal to $(J, 1)$. Hence the first halt is after exactly k instructions. By Theorem 32.1, $T_3 = 0$ gives the exact number of nonhalting Waterfall firings. Corollary 32.3 and $T_4 = 0$ give the exact halt timestamp.

Completeness and uniqueness. Conversely, for an actual first halt at step k , select its successive instructions, put each actual quotient, remainder, and other-half value in the active direction triple, and put zeros in the inactive triple. All residuals and complementarity products vanish. The preceding induction shows that every selector and every natural coordinate is forced. The natural witness tuple is therefore unique. $\square$

For $k = 0$ use the constant polynomial $F_0 = 1$, since the prescribed initial control $(A, 0)$ is not halting. If strictly positive witnesses are required, replace each natural witness z by $z^+ - 1$ and quantify $z^+ \geq 1$; the degree remains two, but the domain and evaluation cost must be stated

accordingly. Unrestricted signed integer witnesses do not satisfy the nonnegativity argument above.

34 A complete seven-step halting example

Take $(L_0, R_0) = (6, 0)$ with initial control $(A, 0)$. Its full source trajectory and macrostep accounting are in Table 5. Each row gives the preinstruction control and half tapes, the active quotient/remainder/other-half triple, the number c_j of Waterfall firings in that macrostep, and the timestamp t_j of its first control firing.

Table 5: The actual halting run from $(L_0, R_0) = (6, 0)$.

j	q	s	L	R	D	Q	r	Y	c_j	t_j
0	A	0	6	0	R	0	0	6	24	1
1	B	0	12	0	R	0	0	12	42	56
2	C	0	25	0	L	12	1	0	43	147
3	G	1	12	0	L	6	0	0	24	240
4	G	0	6	1	L	3	0	1	18	295
5	H	0	3	2	L	1	1	2	16	338
6	I	1	1	5	L	0	1	5	22	377
7	J	1	0	11	halt observation					428

The macrostep counts sum to

$$C = 24 + 42 + 43 + 24 + 18 + 16 + 22 = 189, \quad \tau = 1 + 2 \cdot 189 + 7 \cdot 7 = 428.$$

For the quadratic equation, choose the seven one-hot selectors for $A0, B0, C0, G1, G0, H0, I1$. In each row place the table's (Q, r, Y) in the indicated direction group and set the other triple to zero. This is the entire 245-coordinate witness, specified without printing all zero and selector entries individually. The exported fixture provides every coordinate explicitly.

At $k = 7$, equation (78) has 245 natural witnesses, 39 squared linear residuals, and 14 complementarity products. Exact evaluation gives zero at the fixture. Direct event-by-event deadline simulation, and a separately implemented physical countdown simulation, verify the actual 189 prehalt firings and halt at time 428. These finite replay checks corroborate the example; the all-input theorem rests on the preceding proofs and finite symbolic certificate.

[write] Added 2 October 2026. After this manuscript arrived, the repository's research programme reviewed it (`waterfall_report_review_24a743255.md`, commit `f19aaa092`): all eight replay commands pass, the macrostep theorem and the first-halt quadratic are confirmed over natural coordinates, and two Python interface defects of the grouped compiler (floating-point and Boolean inputs accepted; exported name lists aliased) are repaired by `waterfall_grouped_exact_domains.patch`, with no change to valid exports. The programme also improved the certificate (`waterfall_forced_boundary_projection.md`, commit `3b8187ea9`): the first instruction is forced to be $A0$ and the three before halt to be $G0, H0, I1$, so their coordinates can be restored by nonnegative affine expressions. For $k \geq 4$ the projected polynomial has $35k - 138$ natural witnesses, $5k - 15$ squares and $2k - 8$ products, still of exact degree two; at $k = 7$ that is 107 witnesses, 20 squares and 6 products, and a complete evaluation schedule of 720 operations (203 multiplications and 517 additions) instead of 1558 (430 + 1128). The counts printed in this Part are source 14's, for its construction as delivered.

35 What the compression does and does not prove

35.1 Exponential event compression in one macrostep

Take $L_0 = 2^m, R_0 = 0$. The initial instruction is $A0 \mapsto (0, R, B)$, so $Q = r = 0$ and $Y = 2^m$. Its macrostep has

$$c = 6 + 3 \cdot 2^m, \quad \Delta = 19 + 6 \cdot 2^m.$$

The left-half integer has $m + 1$ binary digits. The macrostep description and its direction-group witness count remain constant while the literal Waterfall event count grows exponentially in m . This establishes a genuine event-to-macrostep compression. It does not claim that these particular inputs halt after one source instruction, nor that every universal computation exhibits this separation. For a prefix certificate, supply a terminal state $q_* \in \{0, \dots, 14\}$ and bit $s_* \in \{0, 1\}$, replacing T_1, T_2 by $B_{k-1} - q_*$ and $r_{k-1,L} + r_{k-1,R} - s_*$. The binary terminal-bit constraint must be retained. The exponential example ends at $(B, 0)$. The original F_k specifically asks for first halt.

Witness *number* is not witness *bit length*. Large tape integers still require large binary encodings. Likewise, constructing the appropriate k -step certificate from a halting run presupposes access to that run or to its computation. The result removes the potentially much larger Waterfall event horizon, not the source-machine horizon.

35.2 Why fixed-horizon relations are semilinear

Proposition 35.1. *For each fixed k , the set of natural parameter tuples (L_0, R_0, C, τ) for which F_k has a natural zero is effectively semilinear.*

Proof. There are only finitely many possible one-hot instruction sequences of length k . Fix one. The state and head constraints either reject it or prescribe a binary active residue at every step, using the next instruction's source bit and the terminal bit 1. The direction complementarity terms force the inactive triples to zero. All remaining conditions are linear equalities and natural nonnegativity constraints in the active Q, Y values and parameters. Thus each branch is definable by an existential formula of Presburger arithmetic, equivalently an effectively semilinear set. A finite union over instruction sequences remains effectively semilinear. $\square$

[write] Second route. Every factor of the products (74) is a linear form with nonnegative coefficients, and every other summand of (78) is a square; so F_k is nonnegative on the whole real nonnegative orthant, parameters included, and has degree two. By Theorem cdc:of:thm:classification of *Canonical Diophantine certificates* ($D_2^+ = \text{SL}$), its projected natural zero set is semilinear. Source 14's proof above is direct and also explains the branches.

The same statement can be seen more concretely by repeatedly applying the affine updates $Y \mapsto 2Y + w$ and division by 2 with a fixed residue. The quadratic certificate packages finitely many such branches into one nonnegative natural-domain equation. No convexity property is asserted for the quadratic polynomial; natural-domain nonnegativity of its summands is the only required property.

There is consequently no conflict between the simple finite-horizon certificate and undecidability of unbounded universal halting. The latter asks whether *some* k in a family of growing-arity equations succeeds. Replacing that family by one fixed-arity equation requires a further, fully accounted unbounded-sequence representation. This note does not perform that step.

35.3 What remains open in this direction

The most useful next questions concern genuine unbounded compression rather than further hiding a trace in its notation. One would need a fixed-dimensional representation of the sequence of macrostep branch choices and tape recurrences, with explicit polynomial constraints enforcing every transition and the strict-minimum semantics inherited here. Any packed-history decoder, divisibility device, or universal polynomial used for this purpose must have its variables, degree, constants, and arithmetic evaluation costs included.

The present $35k$ construction leaves several narrower questions: whether one can reduce its witness coefficient while retaining a canonical natural fiber; whether other esoteric substrates yield longer verifiable affine blocks; and whether structural classes between the common-column family below and the full fixed universal matrix admit useful horizon-free certificates. No optimality claim is made for $35k$ or for the four-operation loader under alternative encodings or cost models. In particular, the results do not improve or challenge a claimed 75- or 87-operation universal-polynomial frontier: those are different, unbounded targets.

[write] Bearing on questions of *Canonical Diophantine certificates*. Its question cdc:q:acceleration asks, as a restricted target, for “a class with a canonical primitive-period factorization, together with a proof that the arithmetic witnesses preserve that factorization uniquely”. For one fixed universal Waterfall program, Theorem 32.1 supplies the forced macrostep factorization and Theorem 33.1 shows that the witnesses of F_k are exactly the macrostep data, uniquely; this realizes that restricted target for this program at each fixed horizon k , not the general question. It bears in the same way on that report’s Part XII questions “Canonical macro decompositions” and “A small explicit fixed interpreter”. The common-column class of Section 36 is a restricted unbounded class with horizon-free single-fold quadratic certificates (Proposition 36.2), an instance for question cdc:q:restricted; it is decidable, as Part XI requires of any such quadratic.

36 A decidable common-column class

This section isolates a setting in which the event history can genuinely be removed without leaving an external horizon. It is a companion result for a restricted class, not an alternative proof of universality.

There are n nonhalting clocks and one halt clock h . Suppose every nonhalt trigger has the form

$$M_{.i} = b_i \mathbf{1} + d_i \mathbf{e}_i, \quad b_i \geq 0, \quad d_i > 0, \quad M_{.h} = 0. \quad (79)$$

Let $a > 0$. After counts p_i , put $B(p) = \sum_i b_i p_i$. The deadlines are

$$d_i^{\text{abs}} = a_i + d_i p_i + B(p), \quad d_h^{\text{abs}} = a_h + B(p).$$

Subtracting $B(p)$ from every coordinate preserves all event comparisons. The relative event streams are the increasing arithmetic progressions

$$a_i + d_i \mathbb{N}, \quad (80)$$

with a fixed halt marker at a_h .

Theorem 36.1 (Complete progression criterion). *Under (79), the run reaches a legal halt if and only if no nonhalt stream hits a_h and no two nonhalt streams intersect below a_h . In that case*

$$p_i = \max \left(0, \left\lceil \frac{a_h - a_i}{d_i} \right\rceil \right), \quad \tau = a_h + \sum_i b_i p_i. \quad (81)$$

All these conditions and values are decidable by finite integer arithmetic without event-by-event replay.

Proof. Before any halt or tie, the run is exactly the sorted merge of the streams (80) below a_h : selecting stream i increments its index and leaves all other relative deadlines unchanged. Every stream has only finitely many elements below a_h . If two such elements coincide, their first common deadline is eventually exposed simultaneously, producing an undefined tie unless an even earlier tie has already stopped the run. If a stream hits a_h , it ties with the halt marker when that value is reached, again provided no earlier tie occurs. Conversely, in the absence of both obstructions, the finite merge is unique and ends with the uniquely minimal halt marker.

The number of indices $m \geq 0$ with $a_i + d_i m < a_h$ is exactly the quantity p_i in (81). Restoring the common shift gives the stated τ . A stream hits a_h exactly when $a_h \geq a_i$ and d_i divides $a_h - a_i$. Two streams intersect only if $\gcd(d_i, d_j)$ divides $a_j - a_i$; a congruence solution gives their first common value at least $\max(a_i, a_j)$, which can be compared with a_h . Thus finitely many greatest-common-divisor and congruence computations decide the run. $\square$

Strict minimum semantics matter. For example, streams $1 + 2\mathbb{N}$ and $3 + 4\mathbb{N}$, with halt marker 20, tie at 3 even though neither hits the halt marker. Checking only for a tie *at halt* would be unsound.

The condition $d_i > 0$ in this theorem is a condition on the *relative* diagonal increment, not merely on the physical self-reset $b_i + d_i$. If $d_i \leq 0$ and that clock becomes uniquely minimal below the halt marker, its relative deadline never increases and it remains uniquely minimal forever. This can happen with a positive physical self-reset. Such a lock is outside the positive-progression theorem.

36.1 A residue-separated horizon-free certificate

Fix $n \geq 1$, positive integers k_i , and nonnegative integers b_i . Put $N = n+1$ and supply $x_1, \dots, x_n, H \in \mathbb{N}$. Choose

$$a_i = Nx_i + i \ (1 \leq i \leq n), \quad a_h = N(H+1), \quad d_i = Nk_i. \quad (82)$$

The streams have distinct residues modulo N , including the halt residue 0. Consequently no tie is possible and the run always halts. With $W = H+1$, its counts and time are

$$p_i = \max \left(0, \left\lceil \frac{W - x_i}{k_i} \right\rceil \right), \quad T = NW + \sum_i b_i p_i. \quad (83)$$

The ceiling expression follows because $Nx_i + i + Nk_i m < NW$ is equivalent to $x_i + k_i m \leq W - 1$.

For each i introduce natural witnesses p_i, s_i, r_i, u_i and define

$$G = \left(T - NW - \sum_i b_i p_i \right)^2 + \sum_{i=1}^n \left[(x_i + k_i(p_i - s_i) - W - r_i)^2 + (r_i + u_i - (k_i - 1))^2 + p_i s_i \right]. \quad (84)$$

Proposition 36.2. *For fixed k_i, b_i , equation $G = 0$ is a horizon-free quadratic natural-domain certificate of the exact halt time in (83). For every supplied $x_1, \dots, x_n, H$, exactly one tuple $(T, \mathbf{p}, \mathbf{s}, \mathbf{r}, \mathbf{u})$ is a natural zero.*

Proof. Every summand in (84) is nonnegative on the natural domain, so each vanishes. The second residual implies $0 \leq r_i \leq k_i - 1$. The first then makes $z_i = p_i - s_i$ the unique integer with

$$x_i + k_i z_i = W + r_i, \quad 0 \leq r_i < k_i,$$

namely $z_i = \lceil (W - x_i)/k_i \rceil$. The product $p_i s_i = 0$ makes (p_i, s_i) the unique positive- and negative-part decomposition of z_i . Thus $p_i = \max(0, z_i)$, while r_i, u_i are uniquely forced by the two residuals. The time residual gives exactly (83). Conversely, these values make every summand zero. $\square$

Here the number of witnesses is $4n$, independent of the potentially very long run. This succeeds because the entire class is the merge of explicit increasing progressions with separated residues. If the k_i or b_i are promoted from fixed coefficients to input variables, formula (84) is still an explicit certificate on their stated domains, but generally has degree four; calling that parameterized polynomial quadratic would be incorrect.

36.2 A fully counted three-clock instance

For $n = 2$, $k_1 = k_2 = 1$, $b_1 = 2, b_2 = 3$, the remainder witnesses vanish and the certificate reduces to four natural witnesses p, q, r, s :

$$\begin{aligned} G_1 = & (p - r - H - 1 + x)^2 + pr \\ & + (q - s - H - 1 + y)^2 + qs \\ & + (T - 3(H + 1) - 2p - 3q)^2. \end{aligned} \tag{85}$$

The event counts are $p = \max(0, H + 1 - x)$ and $q = \max(0, H + 1 - y)$, and r, s are their corresponding negative parts. Reuse $W = H + 1$. Computing the first two linear residuals takes six additions/subtractions, the third takes three constant multiplications and three subtractions, and the five summands use three squarings and two products. Combining them takes four additions. Including W , the full straight-line evaluation therefore costs

$$8 \text{ multiplications} + 14 \text{ additions/subtractions} = 22 \text{ operations.}$$

Shifting the four natural witnesses to positive variables costs four more subtractions, giving 26 operations in this same literal ledger. These small counts belong to a decidable family; they are not counts for a universal Diophantine representation. A signed-domain substitution can invalidate the complementarity logic: for example $(x, y, H, T, p, q, r, s) = (0, 0, 0, 6, 0, 1, -1, 0)$ makes (85) vanish while using a negative witness.

37 Why endpoint counts do not replace a general history

It is tempting to take a proposed final count vector p , compute $d = a + Mp$, and require the halt deadline to lie between each nonhalt clock's current deadline and its immediately preceding self-reset deadline. This necessary-looking endpoint test is not sufficient for a general Waterfall matrix.

Consider

$$a = \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix}, \quad M = \begin{pmatrix} 5 & 3 & 0 \\ 3 & 6 & 0 \\ 2 & 3 & 0 \end{pmatrix}, \quad h = 3. \quad (86)$$

The actual successive prehalt events are clock 1 at time 1, clock 2 at time 5, and clock 1 at time 9. The deadline vectors are

$$(1, 2, 6) \rightarrow (6, 5, 8) \rightarrow (9, 11, 11) \rightarrow (14, 14, 13),$$

so the legal halt is at 13 with counts $(2, 1, 0)$. The intermediate equal coordinates at deadline 11 are not a minimum tie: the minimum there is the first coordinate 9.

Now propose the spurious counts $p = (2, 2, 0)$. They give $a + Mp = (17, 20, 16)$ and satisfy both endpoint inequalities

$$17 - 5 < 16 < 17, \quad 20 - 6 < 16 < 20.$$

Nevertheless, the second firing of clock 2 never occurs: the actual computation has already halted. Cross-increments from proposed future events have distorted the supposed final comparison. The common-column proof avoids this defect only because it removes a genuine common shift and reduces the ordering to independent progressions. The universal macrostep proof avoids it by checking every prefix position of its repeated blocks.

[write] Added 2 October 2026. The research note `waterfall_endpoint_alias.md` shows that the same shortcut also fails on the actual 46-clock matrix: a nonnegative nonhalting count vector v with $Mv = 194 \cdot \mathbf{1}$, 62 firings and ten control events, added to the seven-step halt of Section 34, preserves the canonical endpoint and the time equation (63) while changing (C, k, τ) from $(189, 7, 428)$ to $(251, 17, 622)$. This rejects only the endpoint-and-count bundle, not the ordered-history quadratic (78).

38 Reproducibility and evidence

The companion archive contains the editable LaTeX source, the rendered article, the two frozen mathematical source-data files, exact-integer replay programs, coefficient-level polynomial exports, and machine-readable receipts. It excludes source-paper PDFs and private working material. Its README gives one command for the full replay and individual commands for each claim.

The checks serve different roles. The complete matrix reconstruction checks that the compact definition really denotes the cited fixed matrix. The symbolic frontend checks strict-minimum inequalities on unbounded natural cones and all final coordinates. The independent frontend checks the same macrostep by a separate loop-domain parameterization. The grouped quadratic compiler exports every coefficient of every linear residual and complementarity factor; its fixtures and exact evaluations test the implementation of the proved construction. Event and countdown simulations check the worked example. The common-column replay checks formulas, endpoint obstructions, congruence intersections, and literal arithmetic ledgers.

These are exact integer computations, but executable checks are not automatically formal proofs. The logical reduction and natural-domain arguments are written above; the finite matrix-specific obligations are explicit and replayable. Neither peer review nor proof-assistant verification is claimed. Mutation tests and finite sample runs are implementation checks, not substitutes for the all-input proofs. The archive identifies upstream provenance and does not assign a blanket license to third-party material.

[write] In this report the replay programs and runner are `code/14-waterfall-*` (the eight `replay/*.py`, `run-replay.sh` and `build.sh`), the receipts and replay data are `data/14-waterfall-*`, and the two source-data files are `data/14-waterfall-UniversalTM15x2.tm.txt` and `data/14-waterfall-UniversalTM15x2.twm.txt`. Those two files are Iijil’s third-party machine data; they are *not* covered by the repository’s MIT-0 licence (see `14-waterfall-LICENSE-PROVENANCE.md` and the report’s README). The editable LaTeX source and rendered article named above are this Part (its gap table is Appendix G); they and the delivered README are not shipped. The runner rewrites twelve shipped JSON files; run it only on a copy in the delivered layout. At placement it passed (“All Waterfall replays passed.”) and reproduced every JSON file apart from line endings.

A The compact polynomial in the worked example

[write] Appendices A–C are source 08’s appendices A–C (Part I); Appendices D–E are source 12’s, Appendices F–G source 14’s, and Appendix H is the provenance of the merge.

Write $x = (x_A, x_B, x_C)$, $y = (y_A, y_B, y_C)$, and $r = (r_A, r_B, r_C)$. For each $i \in \{A, B\}$ let

$$D_i = (u_i + \bar{u}_i - 1)^2 + (r_i + v_i - u_i - w_i)^2 + u_i v_i + \bar{u}_i w_i.$$

The threshold is 1, so the term $a - 1$ is zero. The example polynomial is

$$\begin{aligned} Q = & (x_A - f_1 - f_2 - r_A)^2 + (x_B - f_1 - r_B)^2 + (x_C - r_C)^2 \\ & + (y_A - r_A)^2 + (y_B - f_2 - r_B)^2 + (y_C - f_1 - r_C)^2 \\ & + D_A + D_B + (u_A + u_B + s_1 - 1)^2 + (u_A + s_2)^2. \end{aligned}$$

There are 12 affine squares and 4 nonnegative products, in agreement with $2d + 2H + m = 12$ and $2H = 4$. At

$$x = (2, 1, 0), \quad f = (1, 1), \quad y = (0, 1, 1), \quad r = (0, 0, 0),$$

set both comparator tuples to $(0, 1, 0, 0)$ and $(s_1, s_2) = (1, 0)$. Every summand is zero. In the exported JSON, the comparator fields **b**, **c**, **v**, **w** correspond respectively to $u, \bar{u}, v, w$ here.

This example also shows why a single maximality slack is not enough by itself. The inequality $u_A + u_B \leq 1$ tests the cooperative rule, whereas $u_A = 0$ tests the competing unary rule. The shared u_A flag makes the two tests consistent without another comparison witness.

B A rank computation with independent pools

Take three input species with respectively two, one, and three competing rules. The consumption matrix is

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}.$$

Every cover must contain all three rows, whose rank is three. Thus $m = 6$, $\rho(A) = 3$, and the optimal exponent is three. At population (n, n, n) , maximality requires

$$f_1 + f_2 = n, \quad f_3 = n, \quad f_4 + f_5 + f_6 = n.$$

The exact number is

$$(n+1) \binom{n+2}{2} = \frac{(n+1)^2(n+2)}{2}.$$

Its degree attains the rank theorem. If each rule returns the pool object it consumes, this count repeats independently at each round and is raised to the T th power. The same phenomenon is tested in the executable validation.

C Source audit and scope of novelty

C.1 Repository identifiers

The repository tree inspection returned the commit

db3b377f0ae3ab8cb585df4eb4b859cbee5ba3c5.

The MRDP interface was also fetched explicitly at that commit. Its blob identifier was

74aea8c5e75923c7f8b0c041388b91e10e729d65.

The report catalogue consulted on the live branch was

SetTheory/Cardinals/docs/reports/hilbert-tenth-problem/canonical-diophantine-certificates/README.md.

Repository code search returned an older indexed revision in some results. No assertion that all indexed search snippets matched the tree snapshot was used in the proofs. No proof status was transferred from an existing Lean declaration to any new theorem in this manuscript.

[write] The README cited here summarizes Part XI of *Canonical Diophantine certificates*; Section 1.5 maps the results of this Part to that article’s statements. The MRDP file has the same blob at the time of this write.

C.2 What the external sources establish

The Alhazov–Verlan source supports established universality and the published 23-rule example. Pan–Păun–Song supplies the flat-maximal semantic background. Woods supplies the Presburger counting theorem and its rational-generating-function framework. Valiant supplies the counting complexity baseline; the relevant definition and theorem were inspected in the original paper. Minsky is the classical register-machine source. The repository sources establish the interfaces and prior-report claims being continued, not independent peer review of those reports.

C.3 What is proved in this manuscript

The shared-threshold formula and its exact counts, the detailed guarded and flat extensions, the resource-cover rank bound and its sharpness for every matrix, the resulting trace-degree estimate, the graph-to-network bijection, and the convex quadratic obstruction are supplied with proofs here.

The program-uniform quartic and its optimality combine the explicit construction with the credited lower-degree positivity lemma, for which a self-contained proof is included. None of these statements is labelled a verified first occurrence in the literature.

The source search was targeted rather than exhaustive. It did not certify a resolution of a named long-standing published conjecture, a new minimal universal-rule count, or a general finite-fold/single-fold theorem. The article is offered as a concrete research continuation with checkable mathematical assertions and clearly delimited uniformity.

D A complete 29-witness example in factored form

This appendix displays an entire example polynomial without printing its 126 expanded monomials. The expansion is supplied in results/locking_counterexample.json.

[write] Shipped as data/12-total-quadratic-locking_counterexample.json.

There are a seed s , sites p, c , and two local labels, 1 and 2. The rules are

$$s : 1 \rightarrow p : 1, \quad p : 1 \rightarrow c : 1, \quad c : 1 \rightarrow p : 2,$$

all with unit delay. Set $M = 3$ and $B = 3$. Allocate $t_p, t_c; \ell_{v,0}, \ell_{v,1}, \ell_{v,2}$ for $v = p, c$; (f_v, a_v, h_v) for the two used literals $v : 1$; three min-gate triples (z_i, α_i, β_i) for $0 \leq i \leq 2$; and winner gaps $g_{v,j}$ for $v = p, c$ and $0 \leq j \leq 2$. The total is $2 + 6 + 6 + 9 + 6 = 29$ variables.

Define the site keys and their offered costs by

$$\begin{aligned} K_p &= 3t_p + \ell_{p,1} + 2\ell_{p,2}, & (K_{p,0}, K_{p,1}, K_{p,2}) &= (9, 3z_0 + 1, 3z_1 + 2), \\ K_c &= 3t_c + \ell_{c,1} + 2\ell_{c,2}, & (K_{c,0}, K_{c,1}, K_{c,2}) &= (9, 3z_2 + 1, 11). \end{aligned}$$

Then the required single polynomial is

$$\begin{aligned} P = \sum_{v=p,c} & \left[(\ell_{v,0} + \ell_{v,1} + \ell_{v,2} - 1)^2 + (f_v - t_v - a_v)^2 + (3 - f_v - h_v)^2 \right. \\ & \left. + \ell_{v,1}a_v + (\ell_{v,0} + \ell_{v,2})h_v \right] \\ & + Q_{\min}(3, 1; z_0, \alpha_0, \beta_0) \\ & + Q_{\min}(3, f_c + 1; z_1, \alpha_1, \beta_1) \\ & + Q_{\min}(3, f_p + 1; z_2, \alpha_2, \beta_2) \\ & + \sum_{v=p,c} \sum_{j=0}^2 \left[(K_{v,j} - K_v - g_{v,j})^2 + \ell_{v,j}g_{v,j} \right]. \end{aligned}$$

Its unique natural zero is given by

$$\begin{aligned} (t_p, t_c) &= (1, 2), & (\ell_{v,0}, \ell_{v,1}, \ell_{v,2}) &= (0, 1, 0) \quad (v = p, c), \\ (f_p, a_p, h_p) &= (1, 0, 2), & (f_c, a_c, h_c) &= (2, 0, 1), \\ (z_0, \alpha_0, \beta_0) &= (1, 2, 0), & (z_1, \alpha_1, \beta_1) &= (3, 0, 0), & (z_2, \alpha_2, \beta_2) &= (2, 1, 0), \\ (g_{p,0}, g_{p,1}, g_{p,2}) &= (5, 0, 7), & (g_{c,0}, g_{c,1}, g_{c,2}) &= (2, 0, 4). \end{aligned}$$

Every summand vanishes separately. In particular, the two min-gate slacks are both zero at the cap tie $z_1 = 3$, so that tie creates no arbitrary auxiliary choice.

E Source and artifact ledger

E.1 Frozen repository sources

The following sources were read at `db3b377f0ae3ab8cb585df4eb4b859cbee5ba3c5`:

- The root README.md, giving the project map and trust-boundary conventions.
- The Hilbert-tenth README and its native-stream/queue research README, documenting the 75/87 universal bounds and the scopes of imported substrate work; their pinned links are references [8, 9].
- The README of the merged canonical-diophantine-certificates report under the Hilbert-tenth report category in the cardinal-research documentation, including its first-arrival self-assembly and history-free routing summaries [10].

The article makes no independent verification claim for the repository’s universal arithmetic counts or its reported degree-classification theorems. The relevant prior self-assembly manuscript is explicitly credited. The repository’s content continues to change; the above pin, not a later main branch, is the comparison point.

E.2 Package contents and reproduction

Path	Purpose
article.tex, article.pdf	This source and compiled paper.
code/quadratic_compiler.py	Exact simulator, general and Horn compilers, exporter.
code/test_compiler.py	Exhaustive small-instance and regression tests.
code/verify_exports.py	Independent exact verification of exported polynomials.
results/test_receipt.json	Actual machine-readable test counts.
results/*example.json and other example JSON files	Complete sparse polynomials and natural zero assignments.
README.md, SOURCE_AUDIT.md	Reproduction instructions and scope ledger.
build.sh	PDF compilation using pdfLaTeX.

The compiler and tests use Python’s standard library. The PDF uses a standard LaTeX installation with the packages named in the preamble. Neither external datasets nor the live GitHub repository are required to reproduce the supplied polynomial examples and tests.

[write] Shipped files: `code/` and `build.sh` are `code/12-total-quadratic-*`, the `results/` files are `data/12-total-quadratic-*`, and `SOURCE_AUDIT.md` is `12-total-quadratic-SOURCE_AUDIT.md`. Source 12’s `article.tex` is this Part; its `article.pdf` and `README.md` are not shipped (the report’s README replaces the latter). The shipped build script changes to its own directory and compiles an `article.tex` that is not there; it does not build this report.

F Complete compact definition of the matrix

This appendix defines every trigger increment without printing a 46×46 array. It also makes the matrix-specific finite checks independently reconstructible. All clocks are those listed in Section 31.

Fix a direction D and its opposite O . In each rule below, start with the stated default value at every destination coordinate, then apply the listed overrides. The resulting vector is the column for the clock named at left.

Table 6: Auxiliary trigger vectors. Defaults and overrides are increments.

Source clock	Default	Destination overrides
$DTape$	0	$DTape : 10; DMult, DDiv_0, DDiv_1 : 9$
$DTemp$	0	$DTemp : 8; DTrans : 7$
$DTrans$	2	$DTape : 4; DTemp : 0$
$DMult$	2	$DTape : 0; DTemp : 6$
$DDiv_s$	2	$DTape : 0; DTemp : 2 + 2s$; both directions' Div_s and all $Trans_{u,s} : 3$; both directions' Div_{1-s} and all $Trans_{u,1-s} : 1$
$DWrite_w$	9	$DWrite_w : 11; OMult : 0; DTape : 5 + 2w$; $DTemp, DTrans, DMult : 5; DDiv_0, DDiv_1 : 0$

For a defined source instruction $(q, s) \mapsto (w, D, q')$, put

$$\alpha_t = \mathbf{1}_{\{t=s\}} - \mathbf{1}_{\{t=0\}}, \quad t \in \{0, 1\}.$$

Its control-clock trigger $Trans_{q,s}$ is specified by

$$\begin{aligned} DTape : 0, & & DTemp : 2, & & DTrans : 3, & & DMult : 10, \\ OTape : 4, & & OTemp : 6, & & OTrans : 7, & & OMult : 5, \\ DDiv_t : 1 + \alpha_t, & & ODiv_t : 10 + \alpha_t, & & & & \\ DWrite_t : 10, & & OWrite_t : 10 - 2\mathbf{1}_{\{t=w\}}. & & & & \end{aligned}$$

For each of the thirty control destinations, its increment is

$$M_{Trans_{u,t}, Trans_{q,s}} = 10 - \mathbf{1}_{\{u=q'\}} + \mathbf{1}_{\{u=q\}} + \alpha_t. \quad (87)$$

The remaining source $(J, 1)$ has all increments zero. These rules define all entries, and all lie in $\{0, \dots, 12\}$. Every nonhalt self-reset is positive.

The exact source ordering is

$$\begin{aligned} <ape, LTemp, LTrans, LDiv_0, LDiv_1, LMult, LWrite_0, LWrite_1; \\ &Trans_{A,0}, Trans_{A,1}, \dots, Trans_{O,0}, Trans_{O,1}; \\ &RWrite_0, RWrite_1, RMult, RDiv_0, RDiv_1, RTrans, RTemp, RTape. \end{aligned}$$

The frozen serialized matrix has SHA-256

52cfed3f6cba671ed7b166c126288d555ed
687b74622ae5a9aaa5bee1345e46a.

This hash identifies the exact matrix bytes used by the replay; the mathematical definition is independent of formatting those bytes.

[write] Shipped as `data/14-waterfall-UniversalTM15x2.twm.txt` (third-party data, not MIT-0).

G Finite affine gap certificate

For each listed form, $(Q_{\min}, Y_{\min})$ is the active-domain lower bound. With $Q = Q_{\min} + q$, $Y = Y_{\min} + y$, the recorded coefficients give a gap $c + aq + by$. Every recorded c is at least 1 and both a, b are nonnegative. The multiplicity is the number of strict-minimum obligations represented by that row. Identical shifted forms with different active domains are kept separate so the original loop coverage is visible. The multiplicities sum to 43,065.

$Q_{\min}$	$Y_{\min}$	c	a	b	Count	$Q_{\min}$	$Y_{\min}$	c	a	b	Count
0	1	1	0	4	58	0	0	5	0	0	1073
1	0	1	4	0	29	0	1	5	0	0	2552
0	1	1	0	2	58	1	0	5	0	0	377
0	1	3	0	4	58	0	1	9	0	4	58
1	0	1	2	0	174	0	1	5	2	0	116
1	0	3	4	0	58	1	0	7	2	0	58
0	0	1	0	0	1914	0	0	6	0	0	1595
0	1	1	0	0	290	0	1	6	0	0	1856
1	0	1	0	0	435	1	0	6	0	0	348
0	0	1	0	2	58	0	0	6	0	4	58
1	0	1	0	2	116	0	0	6	2	0	58
0	0	1	2	0	29	0	0	7	0	0	1073
0	0	1	4	0	29	0	1	7	0	0	174
1	0	5	4	0	29	1	0	7	0	0	2610
0	0	2	0	0	2291	0	0	8	0	0	1305
0	1	2	0	0	232	1	0	8	0	0	2088
1	0	2	0	0	348	0	0	8	0	4	58
0	0	2	0	2	58	0	0	8	2	0	58
0	0	2	0	4	58	0	0	9	0	0	1566
0	0	2	2	0	116	1	0	9	0	0	4640
0	0	3	0	0	1102	0	0	10	0	0	1595
0	1	3	0	0	2552	1	0	10	0	0	3480
1	0	3	0	0	174	0	0	11	0	0	986
0	0	3	0	2	29	0	0	12	0	0	58
1	0	3	0	2	232	0	1	12	0	0	232
0	1	3	2	0	116	0	0	13	0	0	116
0	0	3	4	0	29	0	1	13	0	0	116
0	0	4	0	0	1363	0	0	14	0	0	58
0	1	4	0	0	1856	0	1	14	0	0	116
1	0	4	0	0	348	0	0	15	0	0	58
0	0	4	0	2	58	1	0	16	0	0	116
0	0	4	2	0	58	0	0	17	0	0	58

H Provenance of the merge

[write] This report was built from three manuscripts of batch 78 of ProveIt’s incoming reports, cluster H3. The archives arrived in the repository’s `docs/incoming/` directory and were placed in commit `41e7f1189`, which staged their programs, data and audits under the file

prefixes 08-maximal-parallel-, 12-total-quadratic- and 14-waterfall-, and retired the archives. Each archive survives in its arrival commit and can be recovered with `git show <arrival>:docs/incoming/<archive>`.

Source	Archive and title	Arrival	Repository pin	Printed as
08 (base)	Maximal_Parallel_Diophantine.zip; <i>Quadratic Certificates for Maximal Parallelism</i> (main file <code>article.tex</code> , 25-page PDF)	808b53ed8	db3b377f0 (search results also showed the older 4cccfa068)	Part I ; Appendices A–C
12	Total_Quadratic_Diophantine_Semantics.zip; <i>Total Quadratic Semantics for Irreversible Computation</i> (main file <code>article.tex</code> , 25-page PDF)	808b53ed8	db3b377f0	Part II ; Appendices D–E
14	Waterfall_Diophantine_Certificates.zip; <i>Quadratic Diophantine certificates for a fixed Waterfall universal machine</i> (main file <code>paper/waterfall-diophantine.tex</code> with <code>paper/affine-gap-table.tex</code> , 16-page PDF)	24a743255	none	Part III ; Appendices F–G

Author lines as delivered: source 08, “Research manuscript prepared for Vladimir Reshetnikov”, “Developed with ChatGPT”; source 12, “Research prepared for Vladimir Reshetnikov”, “Developed with ChatGPT”; source 14, “Research note prepared with AI assistance for private review”. All three are dated 2 October 2026.

What each source contributed. Source 08: Part [I](#) with its three appendices, its proof and source audits, its compiler and verifier, and its example and validation data. Source 12: Part [II](#) with its two appendices, its source audit, its compiler, tests and export verifier with build script, and its four exported example polynomials and two receipts. Source 14: Part [III](#) with its two appendices (the gap table inlined from `paper/affine-gap-table.tex`), its source and licence provenance notes, its eight replay programs, runner and build script, its seven replay data files and five receipts, and the two third-party machine-data files.

Where the merge had to choose.

- *Placement.* The three manuscripts form a new report rather than three more Parts of *Canonical Diophantine certificates*, which would have grown to nineteen Parts in this batch. Source 08 is the base by manuscript-number order and because it has the broadest degree framework. The alternative (that report’s Parts XVII–XIX) was recorded at placement.
- *Union.* Every result, proof, example, remark, question and limitation of the three sources is printed. Results that re-prove statements of *Canonical Diophantine certificates* are printed as second routes with [write] notes (Section [1.5](#)), not deleted, because the first proofs are in

another report. Small devices common to the sources keep their own statements (Section 1.4).

- *Labels and citations.* Every delivered label carries the prefix `qoc:mp:`, `qoc:ts:` or `qoc:wf:` plus its delivered name; bibliography keys carry `mp:`, `ts:` or `wf:`, and the credits added in the merge carry `w:`. The two entries keyed `repo-canonical` by sources 08 and 12 describe different consultations and are both kept.
- *Layout.* The three title pages are replaced by one; each Part opens with its source’s title, subtitle, author line, date, abstract and status statement. Fonts, page geometry, heading styles and theorem styles are unified (source 08 used the newtx fonts, A4 paper and coloured headings; source 12 printed remarks in the definition style). The appendices of all three sources are collected after Part III, in source order.
- *Text changes.* Source 12’s hard-coded “Section 7” became a cross-reference. Source 14’s author line is printed without “for private review”. No other source sentence was changed; every addition is marked [write].
- *Dated notes.* Improvements made in the repository after the manuscripts arrived (the reduced Waterfall polynomial, the endpoint counterexample on the universal matrix, the proposed reduction of Part II’s gates) are dated notes; the sources’ counts stay as delivered.

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References of source 08 (Part I)

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References of source 12 (Part II)

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- [9] Vladimir Reshetnikov, *WIP: native queue streams and research continuation*, ProveIt research README, same frozen commit, accessed 2 October 2026. [Pinned repository source](#).
- [10] *Canonical Diophantine certificates: witness-faithful polynomial representations of bounded discrete computation*, merged research report, ProveIt. README description of manuscript 09, *Order-Free Diophantine Self-Assembly and a Sharp Positivity-Restricted Degree Threshold*; same frozen commit, accessed 2 October 2026. [Pinned report README](#).
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